Elite Mathematics
Edexcel International A Level

IAL PURE 1 · WMA11

WMA11 Pure 1 Expertise With Answers

Questions 6 and above with worked-solution pages and cross-topic placements.

114	2019–2026	Questions with worked solutions
topic placements	19 exam sessions	checked by Dr Eslam



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TOPIC

Indices & Surds

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 MAY/JUNE 2023

Question 6

7 marks | Indices & Surds

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

(b) Express $\dfrac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

(Total for Question 6 is 7 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2023

1 **Expand the square**

$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2}$.

2 **Rationalise the denominator**

$\frac{1}{3+2\sqrt{2}} \times \frac{3-2\sqrt{2}}{3-2\sqrt{2}} = \frac{3-2\sqrt{2}}{9-8} = 3 - 2\sqrt{2}$.

3 **Let r be the square root**

Let $r = \sqrt{3+2\sqrt{2}}$. Then $r^2 = 3 + 2\sqrt{2}$ and $\frac{1}{r^2} = 3 - 2\sqrt{2}$.

4 **Use part (a)**

$\left(r - \frac{1}{r}\right)^2 = r^2 - 2 + \frac{1}{r^2} = (3 + 2\sqrt{2}) - 2 + (3 - 2\sqrt{2}) = 4$.

5 **Take the positive square root**

Since $r > 1$, $r - \frac{1}{r} > 0$, so $r - \frac{1}{r} = 2$.

FINAL ANSWER

$(a)\ r^2 - 2 + \frac{1}{r^2}$. $(b)\ 3 - 2\sqrt{2}$. (c) The expression equals 2.

TOPIC

Quadratics

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 OCTOBER 2019

Question 8

5 marks | Quadratics

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a, b and c are integers to be found.

(5)

(Total 5 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2019

1 Use a substitution

Let $u = \sqrt{x}$. Then $x = u^2$, so the equation becomes $u^2 - 6u + 4 = 0$.

2 Solve the quadratic

$$u = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}.$$

3 Return to x

Since $u = \sqrt{x}$, $x = u^2 = (3 \pm \sqrt{5})^2 = 9 + 5 \pm 6\sqrt{5} = 14 \pm 6\sqrt{5}$.

FINAL ANSWER

$x = 14 + 6\sqrt{5}$ or $x = 14 - 6\sqrt{5}$.

WMA11/01 MAY/JUNE 2022

Question 6

7 marks | Quadratics

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$
$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2022

1 **Rearrange the second equation**

From $y - x^3 + 6x = 0$, we get $y = x^3 - 6x$.

2 **Substitute into the first equation**

$2x(x^3 - 6x) - 3x^2 = 50$, so $2x^4 - 15x^2 - 50 = 0$.

3 **Solve the quadratic in x squared**

Let $u = x^2$. Then $2u^2 - 15u - 50 = 0$, giving $u = 10$ or $u = -\frac{5}{2}$.

4 **Reject the negative value**

Since x^2 cannot be negative, $x^2 = 10$, so $x = \pm\sqrt{10}$.

5 **Find y**

$y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$. Therefore $y = 4\sqrt{10}$ when $x = \sqrt{10}$ and $y = -4\sqrt{10}$ when $x = -\sqrt{10}$.

✓ **FINAL ANSWER**

$(x, y) = (\sqrt{10}, 4\sqrt{10})$ or $(-\sqrt{10}, -4\sqrt{10})$.

WMA11/01 JANUARY 2023

Question 8

8 marks | Quadratics

8.

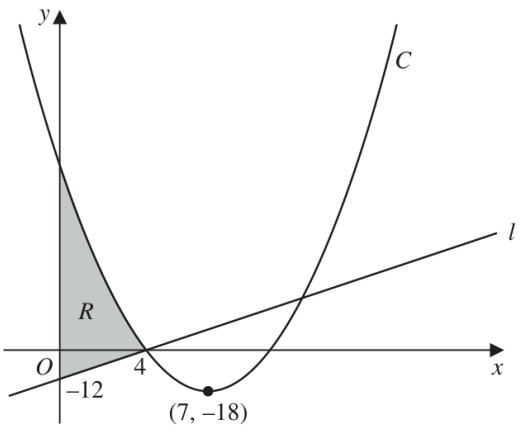


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C.

Given that l cuts the y-axis at -12 and cuts the x-axis at 4, as shown in Figure 2,

(a) find an equation for l, writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x-axis at 4 and at k, where k is a constant

(b) deduce the value of k,

(1)

(c) find $f(x)$.

(3)

The region R is shown shaded in Figure 2.

(d) Use inequalities to define R.

(2)

(Total for Question 8 is 8 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2023

1 Find the straight line

The line cuts the y-axis at -12 and the x-axis at 4, so its gradient is $\dfrac{0-(-12)}{4-0} = 3$. Hence l is $y = 3x - 12$.

2 Use symmetry of the quadratic

The minimum point has x-coordinate 7. The roots are 4 and k, so their midpoint is 7.

3 Find k

$\dfrac{4+k}{2} = 7$, so $k = 10$.

4 **Build the quadratic**

Let $f(x) = a(x-4)(x-10)$. Since the minimum point is $(7, -18)$, $-18 = a(3)(-3) = -9a$, so $a = 2$.

5 **Write f(x)**

$f(x) = 2(x-4)(x-10) = 2x^2 - 28x + 80$.

6 **Identify the shaded region**

The region lies between the y-axis and $x = 4$, above the line and below the curve: $0 \le x \le 4$ and $3x - 12 \le y \le 2x^2 - 28x + 80$.

✓ **FINAL ANSWER**

$(a)\ y = 3x - 12.$ $(b)\ k = 10.$ $(c)\ f(x) = 2(x-4)(x-10).$ $(d)\ 0 \le x \le 4,\ 3x - 12 \le y \le 2x^2 - 28x + 80.$

WMA11/01 OCTOBER 2023

Question 6

6 marks | Quadratics

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p-2x)=\frac{12+15p}{x+p} \qquad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2-10p-8>0$$

(3)

(b) Hence, using algebra, find the range of possible values of p

(3)

(Total for Question 6 is 6 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2023

1 Clear the denominator

$4(p-2x)=\dfrac{12+15p}{x+p}$ gives $4(p-2x)(x+p)=12+15p$.

2 Form a quadratic in x

Expanding gives $4p^2-4px-8x^2=12+15p$, so $8x^2+4px+12+15p-4p^2=0$.

3 Use the discriminant condition

For two distinct real roots, $b^2-4ac>0$.

4 Simplify the inequality

$(4p)^2-4(8)(12+15p-4p^2)>0$, so $144p^2-480p-384>0$.

5 Divide and factorise

Dividing by 48 gives $3p^2-10p-8>0$, and $3p^2-10p-8=(3p+2)(p-4)$.

6 State the range

The upward-opening quadratic is positive outside its roots, so $p<-\dfrac{2}{3}$ or $p>4$.

FINAL ANSWER

$p<-\dfrac{2}{3}$ or $p>4$.

Question 11

10 marks | Quadratics

11.

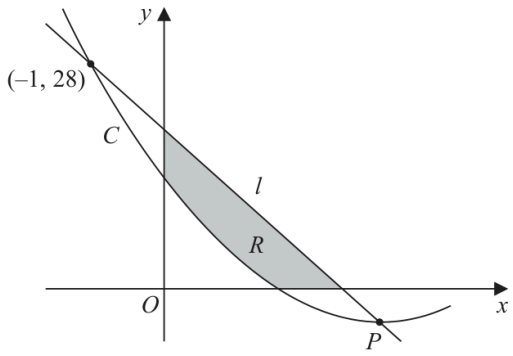


Figure 5

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

(a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

Given that C has a minimum at the point P

(b) state the coordinates of P

(1)

The line l intersects C at (–1, 28) and at P as shown in Figure 5.

(c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The finite region R, shown shaded in Figure 5, is bounded by the x-axis, l, the y-axis, and C.

(d) Use inequalities to define the region R.

(3)

(Total for Question 11 is 10 marks)

WORKED SOLUTION

Question 11 · WMA11/01 October 2023

1 Complete the square

$2x^2 - 12x + 14 = 2(x^2 - 6x) + 14 = 2\left((x-3)^2 - 9\right) + 14 = 2(x-3)^2 - 4$.

2 State the minimum point

The minimum of $2(x-3)^2 - 4$ is at $P = (3, -4)$.

3 Find the gradient of l

The line passes through $(-1, 28)$ and $P(3, -4)$, so $m = \dfrac{-4 - 28}{3 - (-1)} = \dfrac{-32}{4} = -8$.

4 **Find the equation of l**

Using $P(3,-4)$, $y+4=-8(x-3)$, so $y=-8x+20$.

5 **Use the axes as boundaries**

The region is to the right of the y-axis and above the x-axis, so $x \geq 0$ and $y \geq 0$.

6 **Use the line and curve as boundaries**

The region is below the line and above the curve, so $y \leq -8x+20$ and $y \geq 2x^2-12x+14$.

✓ **FINAL ANSWER**

$(a)\ 2(x-3)^2-4.\quad (b)\ P=(3,-4).\quad (c)\ y=-8x+20.\quad (d)\ x \geq 0,\ y \geq 0,\ y \leq -8x+20,\ y \geq 2x^2-12x+14.$

WMA11/01 JANUARY 2024

Question 9

10 marks | Quadratics

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes,

(2)

(b) find an expression for $f(x)$.

(3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P

(5)

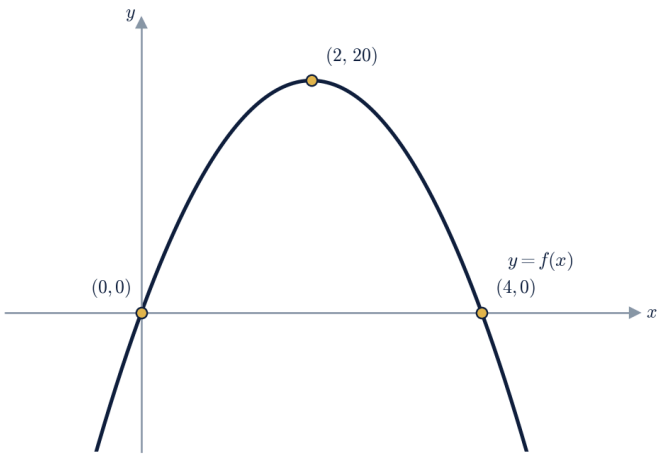
(Total for Question 9 is 10 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2024

1 **Use symmetry for the sketch**

The maximum is at $(2, 20)$ and the curve passes through $(0, 0)$. Since the axis of symmetry is $x = 2$, the other x-intercept is $(4, 0)$.



2 **Use completed-square form**
Let $f(x) = a(x-2)^2 + 20$.

3 **Use the origin**
$0 = a(0-2)^2 + 20$, so $4a = -20$ and $a = -5$.

4 **Write f(x)**
$f(x) = -5(x-2)^2 + 20 = -5x^2 + 20x = -5x(x-4)$.

5 **Equate the two curves**
$-5x^2 + 20x = x(x^2 - 4) = x^3 - 4x$.

6 **Factorise**
$x^3 + 5x^2 - 24x = 0$, so $x(x^2 + 5x - 24) = 0$ and $x(x+8)(x-3) = 0$.

7 **Use the negative x-coordinate**
The negative x-coordinate is $x = -8$.

8 **Find y**
$y = x(x^2 - 4) = (-8)(64 - 4) = -480$, so $P = (-8, -480)$.

✓ **FINAL ANSWER**

(a) downward parabola through $(0, 0)$ and $(4, 0)$ with maximum $(2, 20)$.

(b) $f(x) = -5x(x-4)$.

(c) $P = (-8, -480)$.

WMA11/01 MAY/JUNE 2025

Question 10

7 marks | Quadratics

10. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \qquad \text{where } k \text{ is a constant}$$

(a) Find the exact solutions to the given equation for $k = 4.5$

(3)

(b) Find the set of possible values of k for which the given equation has no real roots.

(4)

(Total for Question 10 is 7 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2025

1 Substitute k equals 4.5
The equation becomes $3.5x^6 + 4x^3 + 0.5 = 0$.

2 Clear decimals
Multiplying by 2 gives $7x^6 + 8x^3 + 1 = 0$.

3 Use a quadratic in x cubed
$7x^6 + 8x^3 + 1 = (x^3 + 1)(7x^3 + 1) = 0$.

4 Solve for x
$x^3 = -1$ or $x^3 = -\frac{1}{7}$, so $x = -1$ or $x = -\sqrt[3]{\frac{1}{7}}$.

5 View the equation as a quadratic
Let $u = x^3$. Then the equation is $(k-1)u^2 + 4u + (k-4) = 0$.

6 Use no real roots
Since every real value of u gives a real x, the original equation has no real roots exactly when this quadratic in u has no real roots.

7 Use the discriminant
Require $b^2 - 4ac < 0$: $16 - 4(k-1)(k-4) < 0$.

8 Simplify
$16 - 4(k^2 - 5k + 4) < 0$ gives $20k - 4k^2 < 0$, so $4k(5-k) < 0$.

9 Solve the inequality
$4k(5-k) < 0$ for $k < 0$ or $k > 5$.

FINAL ANSWER

$(a)\ x = -1,\ -\sqrt[3]{\frac{1}{7}}.\quad (b)\ k < 0 \text{ or } k > 5.$

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TOPIC

Simultaneous Equations

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 OCTOBER 2023

Question 8

7 marks | Simultaneous Equations

8. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \qquad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0$$

(2)

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ

(5)

(Total for Question 8 is 7 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2023

1 Write C1 in terms of y

$xy = \frac{15}{2} - 5x$, so $y = \frac{15}{2x} - 5$.

2 Equate the two curves

$\frac{15}{2x} - 5 = x^3 - \frac{7}{2}x - 5$.

3 Multiply by 2x

$15 = 2x^4 - 7x^2$, hence $2x^4 - 7x^2 - 15 = 0$.

4 Solve for x

Let $u = x^2$. Then $2u^2 - 7u - 15 = 0 = (2u + 3)(u - 5)$, so $x^2 = 5$ and $x = \pm\sqrt{5}$.

5 Find the y-coordinates

Using $y = \frac{15}{2x} - 5$, the points are $\left(\sqrt{5}, \frac{3\sqrt{5}}{2} - 5\right)$ and $\left(-\sqrt{5}, -\frac{3\sqrt{5}}{2} - 5\right)$.

6 **Use the distance formula**

$\Delta x = 2\sqrt{5}$ and $\Delta y = 3\sqrt{5}$, so $PQ = \sqrt{(2\sqrt{5})^2 + (3\sqrt{5})^2} = \sqrt{65}$.

FINAL ANSWER

$(a)\ 2x^4 - 7x^2 - 15 = 0.$ $(b)\ PQ = \sqrt{65}.$

TOPIC

Inequalities

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 JANUARY 2020

Question 8

6 marks | Inequalities

8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C.

(6)

(Total 6 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2020

1 Equate the line and curve

$k(2x - 1) = x^2 + 2x + 11$, so $x^2 + (2 - 2k)x + (11 + k) = 0$.

2 Use the no-intersection condition

For the line not to cross or touch the curve, the quadratic in x must have no real roots, so its discriminant is less than zero.

3 Form the inequality in k

$(2 - 2k)^2 - 4(1)(11 + k) < 0$. Dividing by 4 gives $(1 - k)^2 - (11 + k) < 0$, so $k^2 - 3k - 10 < 0$.

4 Solve the quadratic inequality

$(k - 5)(k + 2) < 0$, so k lies between the two roots: $-2 < k < 5$.

FINAL ANSWER

$-2 < k < 5$.

WMA11/01 MAY/JUNE 2023

Question 7

6 marks | Inequalities

7.

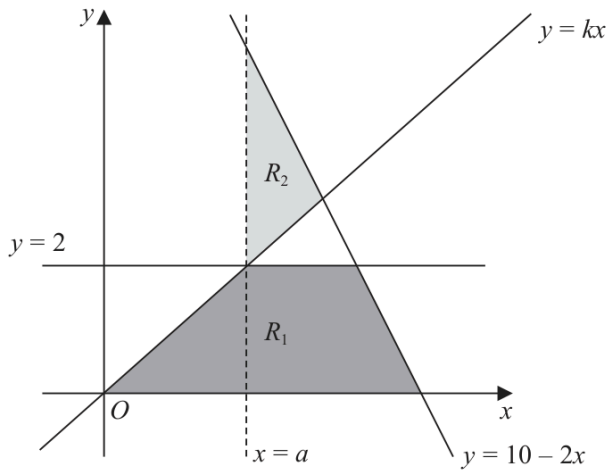


Figure 2

The region R_1, shown shaded in Figure 2, is defined by the inequalities

$$0 \leqslant y \leqslant 2 \qquad y \leqslant 10 - 2x \qquad y \leqslant kx$$

where k is a constant.

The line $x = a$, where a is a constant, passes through the intersection of the lines $y = 2$ and $y = kx$

Given that the area of R_1 is $\frac{27}{4}$ square units,

(a) find

(i) the value of a

(ii) the value of k

(4)

(b) Define the region R_2, also shown shaded in Figure 2, using inequalities.

(2)

(Total for Question 7 is 6 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2023

1 Find the area of R1

The bottom horizontal length is 5, from $x = 0$ to the intercept of $y = 10 - 2x$ at $x = 5$.

2 Use the top length

The top horizontal length is $4 - a$, because $y = 2$ meets $y = 10 - 2x$ at $x = 4$.

3 Solve for a

Area $R_1 = \frac{1}{2}(5 + 4 - a)(2) = 9 - a = \frac{27}{4}$, so $a = \frac{9}{4}$.

4 **Find k**

The line $x = a$ passes through the intersection of $y = 2$ and $y = kx$, so $2 = ka$. Hence $k = \dfrac{2}{9/4} = \dfrac{8}{9}$.

5 **Define R2**

Region R_2 is to the right of $x = a$, above $y = kx$, and below $y = 10 - 2x$.

✓ **FINAL ANSWER**

$(a)\ a = \dfrac{9}{4},\ k = \dfrac{8}{9}.$ $(b)\ x \geq \dfrac{9}{4},\ y \geq \dfrac{8}{9}x,\ y \leq 10 - 2x.$

WMA11/01 MAY/JUNE 2024

Question 6

7 marks | Inequalities

6.

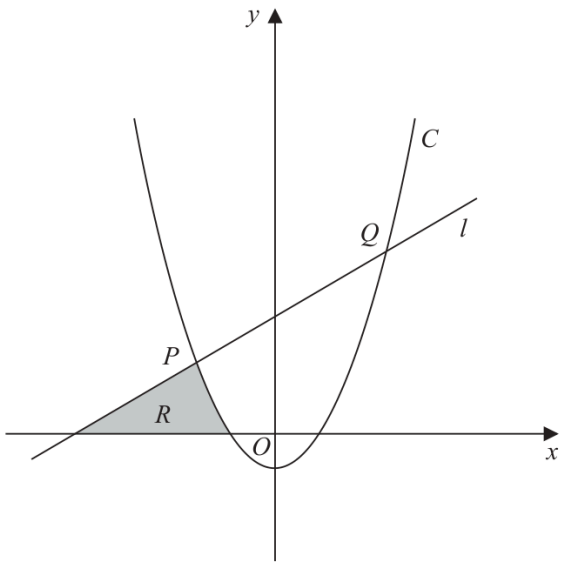


Figure 3

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 3 shows

- the line l with equation $y - 5x = 75$
- the curve C with equation $y = 2x^2 + x - 21$

The line l intersects the curve C at the points P and Q, as shown in Figure 3.

(a) Find, using algebra, the coordinates of P and the coordinates of Q.

(4)

The region R, shown shaded in Figure 3, is bounded by C, l and the x-axis.

(b) Use inequalities to define the region R.

(3)

(Total for Question 6 is 7 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2024

1 Write the line as y equals

$y - 5x = 75$, so $y = 5x + 75$.

2 Equate line and curve

$5x + 75 = 2x^2 + x - 21$, so $2x^2 - 4x - 96 = 0$.

3 Solve the quadratic

$x^2 - 2x - 48 = 0 = (x - 8)(x + 6)$, so $x = -6$ or $x = 8$.

4 **Find the y-coordinates**

Using $y = 5x + 75$, $x = -6$ gives $y = 45$ and $x = 8$ gives $y = 115$.

5 **State P and Q**

From the diagram, $P = (-6, 45)$ and $Q = (8, 115)$.

6 **Use the x-axis and left branch**

The shaded region is above the x-axis and on the left part of the diagram, so $y \geq 0$ and $x \leq 0$.

7 **Apply both boundaries**

The region is below both the line and the curve, so $y \leq 5x + 75$ and $y \leq 2x^2 + x - 21$.

✓ **FINAL ANSWER**

$(a)\ P = (-6, 45),\ Q = (8, 115).$ $(b)\ y \geq 0,\ y \leq 5x + 75,\ y \leq 2x^2 + x - 21,\ x \leq 0.$

TOPIC

Polynomials

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 MAY/JUNE 2021

Question 8

9 marks | Polynomials

8. The curve C_1 has equation

$$y = 3x^2 + 6x + 9$$

(a) Write $3x^2 + 6x + 9$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

The point P is the minimum point of C_1

(b) Deduce the coordinates of P.

(1)

A different curve C_2 has equation

$$y = Ax^3 + Bx^2 + Cx + D$$

where A, B, C and D are constants.

Given that C_2

- passes through P
- intersects the x-axis at -4, -2 and 3

(c) find, making your method clear, the values of A, B, C and D.

(5)

(Total 9 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2021

1 Complete the square

$3x^2 + 6x + 9 = 3(x^2 + 2x) + 9 = 3\left((x+1)^2 - 1\right) + 9 = 3(x+1)^2 + 6$.

2 Deduce the minimum point

The minimum occurs when $(x+1)^2 = 0$, so $x = -1$ and $y = 6$. Therefore $P = (-1, 6)$.

3 Use the x-intercepts of C2

Since C_2 intersects the x-axis at $-4, -2$ and 3, it has form $y = k(x+4)(x+2)(x-3)$.

4 **Use point P to find k**

Substitute $P=(-1,6)$: $6=k(3)(1)(-4)=-12k$, so $k=-\dfrac{1}{2}$.

5 **Expand the cubic**

$(x+4)(x+2)(x-3)=(x^2+6x+8)(x-3)=x^3+3x^2-10x-24$.

6 **Read off A, B, C and D**

Thus $y=-\dfrac{1}{2}x^3-\dfrac{3}{2}x^2+5x+12$, so $A=-\dfrac{1}{2}$, $B=-\dfrac{3}{2}$, $C=5$ and $D=12$.

✓ **FINAL ANSWER**

$(a)\ 3(x+1)^2+6.\quad (b)\ P=(-1,6).\quad (c)\ A=-\dfrac{1}{2},\ B=-\dfrac{3}{2},\ C=5,\ D=12.$

Question 6

9 marks | Polynomials

6.

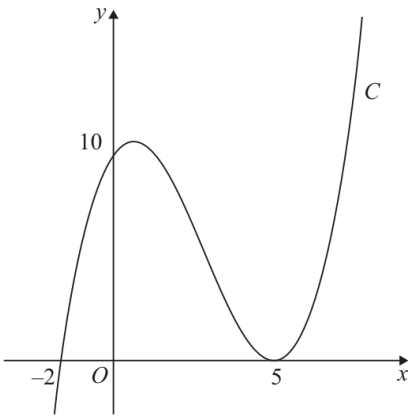


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = f(x)$ where $f(x)$ is a cubic function in x.

The curve C

- cuts the x-axis at $(-2, 0)$ and cuts the y-axis at $(0, 10)$
- touches the x-axis at $(5, 0)$

as shown in Figure 3.

(a) Deduce the roots of the equation

(i) $f\left(\frac{1}{3}x\right) = 0$

(ii) $f(x - 3) = 0$

(2)

(b) Find an expression for $f(x)$. You should leave your answer in factorised form.

(3)

The curve C intersects the straight line $y = 10(x + 2)$ at exactly three points.

(c) Use algebra to find the exact x coordinates of the three points of intersection.

(*Solutions based entirely on calculator technology are not acceptable.*)

(4)

(Total for Question 6 is 9 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2026

1 **Roots of f one third x**

The roots of $f(x) = 0$ are $x = -2$ and $x = 5$, with $x = 5$ repeated. For $f\left(\frac{1}{3}x\right) = 0$, set $\frac{1}{3}x = -2$ or 5, giving $x = -6$ or 15.

2 **Roots of f of x minus 3**

For $f(x - 3) = 0$, set $x - 3 = -2$ or 5, giving $x = 1$ or 8.

3 **Use the roots and repeated root**

Since the curve cuts at $(-2, 0)$ and touches at $(5, 0)$, write $f(x) = k(x + 2)(x - 5)^2$.

4 **Use the y-intercept**

The curve passes through $(0, 10)$, so $10 = k(2)(25) = 50k$.

5 **State f(x)**

$k = \frac{1}{5}$, hence $f(x) = \frac{1}{5}(x+2)(x-5)^2$.

6 **Set equal to the line**

$\frac{1}{5}(x+2)(x-5)^2 = 10(x+2)$.

7 **First solution**

The factor $x + 2$ gives one solution $x = -2$.

8 **Find the other solutions**

For $x \neq -2$, divide by $x + 2$: $\frac{1}{5}(x-5)^2 = 10$, so $(x-5)^2 = 50$.

9 **State all intersections**

$x = 5 \pm \sqrt{50} = 5 \pm 5\sqrt{2}$. Therefore the three x-coordinates are -2, $5 - 5\sqrt{2}$ and $5 + 5\sqrt{2}$.

FINAL ANSWER

$(a)(i)\ -6, 15.\quad (a)(ii)\ 1, 8.\quad (b)\ f(x) = \frac{1}{5}(x+2)(x-5)^2.\quad (c)\ x = -2,\ 5 \pm 5\sqrt{2}.$

WMA11/01 MAY/JUNE 2026

Question 6

7 marks | Polynomials

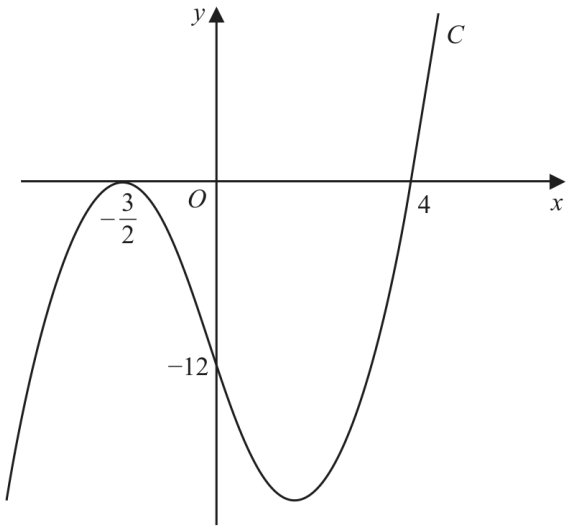


Figure 2

Figure 2 shows a sketch of the curve C with equation $y = f(x)$.

The curve C

- touches the x-axis at $-\frac{3}{2}$
- cuts the x-axis at 4
- cuts the y-axis at -12

Given that $f(x)$ is a cubic function,

(a) find $f(x)$. You may leave the answer in a factorised form.

(3)

The line with equation $y = 2x - 12$ cuts C at $(0, -12)$ and at two further points, P and Q.

(b) Use algebra to find the exact x coordinates of P and Q.
Give each answer in simplest form.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

(Total for Question 6 is 7 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2026

1 **(a) Build the cubic from its roots**

The curve touches the x-axis at $x = -\frac{3}{2}$, so this is a repeated root. It crosses the x-axis at $x = 4$, so this is a simple root. Hence

$$f(x) = k(2x + 3)^2(x - 4).$$

2 **(a) Use the y-intercept**

The curve passes through $(0, -12)$. Substituting $x = 0$ gives

$$-12 = k(3)^2(-4) = -36k,$$

so $k = \frac{1}{3}$. Therefore

$$f(x) = \frac{1}{3}(2x + 3)^2(x - 4).$$

---

3 **(b) Form the intersection equation**

At an intersection with the line $y = 2x - 12$,

$$\frac{1}{3}(2x + 3)^2(x - 4) = 2x - 12.$$

Multiplying by 3 and expanding gives

$$(4x^2 + 12x + 9)(x - 4) = 6x - 36,$$

so

$$4x^3 - 4x^2 - 45x = 0.$$

---

4 **(b) Remove the known intersection and solve**

Factor out the known root $x = 0$:

$$x(4x^2 - 4x - 45) = 0.$$

The two further intersections satisfy

$$4x^2 - 4x - 45 = 0.$$

Using the quadratic formula,

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(4)(-45)}}{8} = \frac{4 \pm 4\sqrt{46}}{8} = \frac{1 \pm \sqrt{46}}{2}.$$

FINAL ANSWER

$(a)\ f(x) = \frac{1}{3}(2x + 3)^2(x - 4).$ $(b)\ x = \frac{1 - \sqrt{46}}{2}$ or $x = \frac{1 + \sqrt{46}}{2}.$

TOPIC

Graphs of Functions

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 OCTOBER 2019

Question 6

8 marks | Graphs of Functions

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

(a) Sketch a graph of C, stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x-axis.

(3)

The line with equation $y = 10 - 2x$ is a tangent to C.

(b) Find the possible values for k.

(5)

(Total 8 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2019

1 Read the graph features

For $y = \frac{4}{x} + k$, the horizontal asymptote is $y = k$. Setting $y = 0$ gives $0 = \frac{4}{x} + k$, so the x-intercept is $x = -\frac{4}{k}$.

2 Equate line and curve

A tangent touches once, so solve $10 - 2x = \frac{4}{x} + k$. Multiplying by x gives $2x^2 + (k - 10)x + 4 = 0$.

3 Use the discriminant condition

For one repeated intersection, discriminant $= 0$: $(k - 10)^2 - 4(2)(4) = 0$. Hence $(k - 10)^2 = 32$.

4 Find k

$k - 10 = \pm 4\sqrt{2}$, so $k = 10 \pm 4\sqrt{2}$. Both are positive, so both satisfy the condition in the question.

FINAL ANSWER

Asymptote $y = k$, x-intercept $\left(-\frac{4}{k}, 0\right)$, and $k = 10 \pm 4\sqrt{2}$.

Question 10

10 marks | Graphs of Functions

10.

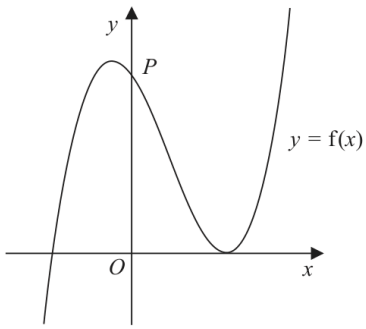


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

(a) Deduce the values of x for which $f(x) \leqslant 0$

(2)

The curve crosses the y-axis at the point P, as shown.

(b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)

(c) Hence, or otherwise, find

(i) the coordinates of P,

(ii) the gradient of the curve at P.

(2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

(d) (i) Find $g(x)$, giving your answer in a simplified factorised form.

(ii) Hence state the y intercept of the curve with equation $y = g(x)$.

(3)

(Total 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2019

1 Use the factorised form for the inequality

$f(x) = (2x + 5)(x - 3)^2$. Since $(x - 3)^2 \geq 0$, the sign is controlled by $2x + 5$, with an extra zero at $x = 3$. Hence $f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$.

2 Expand the cubic

$(x - 3)^2 = x^2 - 6x + 9$, so $f(x) = (2x + 5)(x^2 - 6x + 9) = 2x^3 - 7x^2 - 12x + 45$.

3 Find P and the gradient at P

At the y-axis, $x = 0$, so $P = (0, 45)$. Also $f'(x) = 6x^2 - 14x - 12$, so the gradient at P is $f'(0) = -12$.

4 **Translate the graph**

A translation 2 units in the positive x direction means $g(x) = f(x-2)$. Therefore $g(x) = (2(x-2)+5)((x-2)-3)^2 = (2x+1)(x-5)^2$.

5 **Find the new y-intercept**

Set $x = 0$: $g(0) = (1)(25) = 25$, so the y-intercept is 25.

✓ **FINAL ANSWER**

$f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$; $f(x) = 2x^3 - 7x^2 - 12x + 45$; $P = (0, 45)$; gradient $= -12$; $g(x) = (2x+1)(x-5)^2$; y-intercept 25.

Question 10

8 marks | Graphs of Functions

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

(a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

(b) Hence or otherwise

(i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

(ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

(c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

(ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

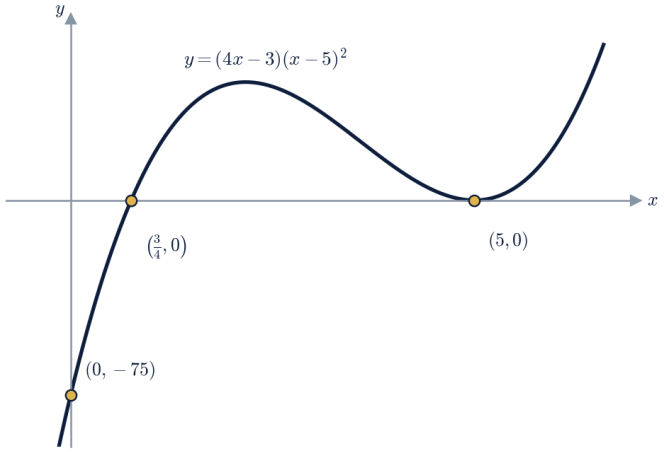
(Total 8 marks)

WORKED SOLUTION

Question 10 · WMA11/01 January 2020

1 **Use the factorised form for the sketch**

$f(x) = (4x - 3)(x - 5)^2$. The curve crosses the x-axis at $x = \frac{3}{4}$ and touches the x-axis at $x = 5$. The y-intercept is $f(0) = (-3)(25) = -75$.



2 **Solve f(x/4)=0**

For $f\left(\frac{1}{4}x\right) = 0$, either $4\left(\frac{x}{4}\right) - 3 = 0$ or $\frac{x}{4} - 5 = 0$. Therefore $x = 3$ or $x = 20$.

3 **Find p**

The curve $y = f(x) + p$ passes through the origin, so $0 = f(0) + p = -75 + p$. Hence $p = 75$.

4 **Find g(x)**

$g(x) = f(x+1) = (4(x+1) - 3)((x+1) - 5)^2 = (4x+1)(x-4)^2$.

5 **Find the y-intercept of C2**

At $x = 0$, $g(0) = (1)(16) = 16$. So the y-intercept is 16.

✓ **FINAL ANSWER**

Sketch: crosses at $\left(\frac{3}{4}, 0\right)$, touches at $(5, 0)$, crosses y-axis at $(0, -75)$. Also $x = 3, 20$, $p = 75$, $g(x) = (4x+1)(x-4)^2$, y-intercept 16.

WMA11/01 JANUARY 2021

Question 6

10 marks | Graphs of Functions

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \qquad k > 0 \qquad x \neq 0$$

(2)

(b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \qquad k > 0 \qquad x \neq 0$$

stating the coordinates of the point of intersection with the x-axis and, in terms of k, the equation of the horizontal asymptote.

(3)

(c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \qquad k > 0 \qquad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

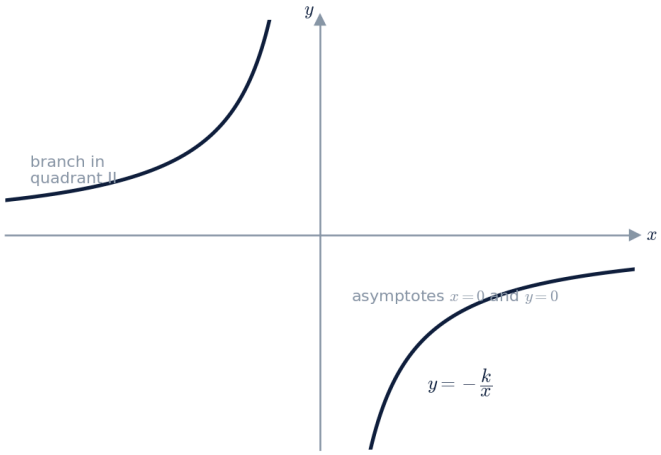
(Total 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2021

1 **Sketch y = -k/x**

Because $k > 0$, $y = -\frac{k}{x}$ has one branch in quadrant II and one branch in quadrant IV, with asymptotes $x = 0$ and $y = 0$.



2 **Translate the graph upward**

$y = -\frac{k}{x} + k$ is $y = -\frac{k}{x}$ translated up by k units. The vertical asymptote is still $x = 0$, and the horizontal asymptote is $y = k$

.

3 **Find the x-intercept**

Set $y = 0$: $0 = -\frac{k}{x} + k$. Since $k > 0$, $\frac{k}{x} = k$, so $x = 1$. The x-intercept is $(1, 0)$.

4 **Set up the intersection equation**

For intersections with $y = 3x + 4$, solve $-\frac{k}{x} + k = 3x + 4$. Multiplying by x gives $-k + kx = 3x^2 + 4x$.

5 **Use the discriminant**

This rearranges to $3x^2 + (4 - k)x + k = 0$. For no touch or intersection, the discriminant must be negative: $(4 - k)^2 - 12k < 0$.

6 **Solve the quadratic inequality in k**

$k^2 - 20k + 16 < 0$. The roots are $k = 10 \pm 2\sqrt{21}$, so the required range is $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$.

FINAL ANSWER

$(b)\ (1, 0),\ y = k.$ $\quad (c)\ 10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}.$

WMA11/01 JANUARY 2021

Question 8

9 marks | Graphs of Functions

8.

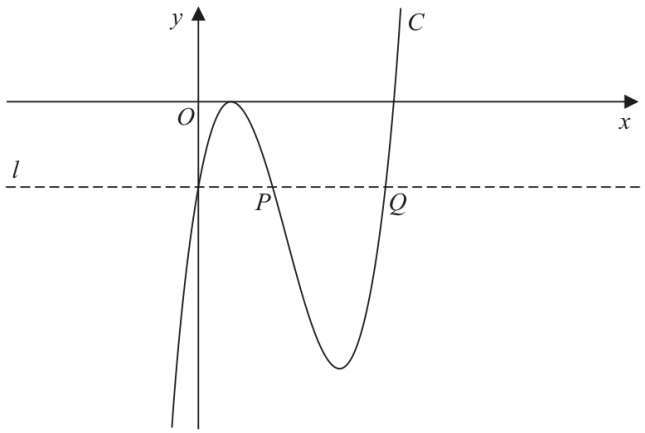


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

(a) Deduce the values of x for which $f(x) > 0$

(1)

(b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)

The line l, also shown in Figure 4, passes through the y intercept of C and is parallel to the x-axis.

The line l cuts C again at points P and Q, also shown in Figure 4.

(c) Using algebra and showing your working, find the length of line PQ. Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(5)

(Total 9 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2021

1 Use the factorised form

$f(x) = (3x - 2)^2(x - 4)$. The squared factor is never negative, and it is zero at $x = \frac{2}{3}$. Therefore the sign of $f(x)$ away from the repeated root is controlled by $x - 4$.

2 Find where f(x) is positive

For $f(x) > 0$, we need $x - 4 > 0$, so $x > 4$.

3 Expand f(x)

$(3x - 2)^2 = 9x^2 - 12x + 4$. Therefore $f(x) = (9x^2 - 12x + 4)(x - 4) = 9x^3 - 48x^2 + 52x - 16$.

4 **Find the line l**

The y-intercept of C is $f(0) = -16$, so the horizontal line l is $y = -16$.

5 **Find the other intersections with l**

Solve $f(x) = -16$: $9x^3 - 48x^2 + 52x - 16 = -16$, so $x(9x^2 - 48x + 52) = 0$. The root $x = 0$ is the y-intercept, and the two other roots give P and Q.

6 **Solve the quadratic**

$9x^2 - 48x + 52 = 0$. Hence $x = \dfrac{48 \pm \sqrt{48^2 - 4(9)(52)}}{18} = \dfrac{48 \pm 12\sqrt{3}}{18} = \dfrac{8 \pm 2\sqrt{3}}{3}$.

7 **Find PQ**

Because P and Q lie on the same horizontal line, PQ is the difference in their x-coordinates:
$PQ = \dfrac{8 + 2\sqrt{3}}{3} - \dfrac{8 - 2\sqrt{3}}{3} = \dfrac{4\sqrt{3}}{3}$.

FINAL ANSWER

$(a)\ x > 4.$ $(b)\ f(x) = 9x^3 - 48x^2 + 52x - 16.$ $(c)\ PQ = \dfrac{4\sqrt{3}}{3}$.

WMA11/01 OCTOBER 2021

Question 6

10 marks | Graphs of Functions

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C.

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $\mathrm{f}(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

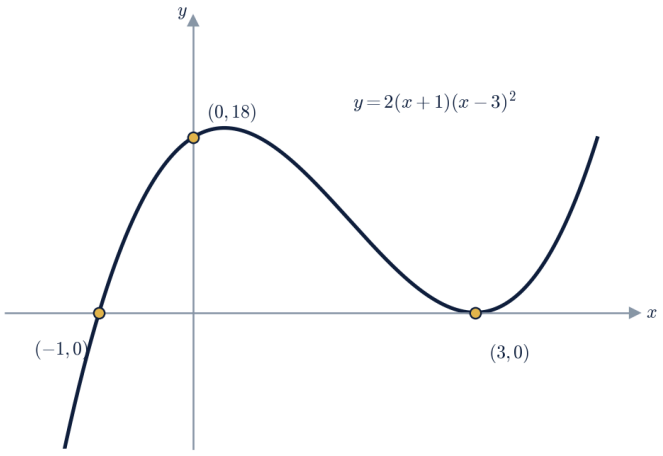
(Total 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2021

1 **Use the factorised form for the sketch**

$f(x) = 2(x+1)(x-3)^2$. The curve cuts the x-axis at $(-1, 0)$ and meets the x-axis at $(3, 0)$ because $x = 3$ is a repeated root.



2 **Find the y-intercept**

$f(0) = 2(1)(9) = 18$, so the y-intercept is $(0, 18)$.

3 **Describe the cubic shape**

The leading term is positive, so the cubic goes down to the left and up to the right. It crosses at $x = -1$ and touches at $x = 3$.

4 **Expand f(x)**

$(x+1)(x-3)^2 = (x+1)(x^2-6x+9) = x^3-5x^2+3x+9$, so $f(x) = 2x^3-10x^2+6x+18$.

5 **Differentiate**

$f'(x) = 6x^2-20x+6$.

6 **Find the gradient at x = 1/3**

$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{9}\right) - 20\left(\frac{1}{3}\right) + 6 = 0$, so the tangent is horizontal.

7 **Find the y-coordinate**

$f\left(\frac{1}{3}\right) = 2\left(\frac{4}{3}\right)\left(-\frac{8}{3}\right)^2 = \frac{512}{27}$.

8 **Write the tangent**

A horizontal tangent through this point has equation $y = \frac{512}{27}$.

✓ **FINAL ANSWER**

(a) x-intercepts $(-1,0),(3,0)$, y-intercept $(0,18)$. (b) $f(x) = 2x^3-10x^2+6x+18$. (c) $y = \frac{512}{27}$.

WMA11/01 JANUARY 2022

Question 10

9 marks | Graphs of Functions

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C.

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k.

(5)

(Total 9 marks)

WORKED SOLUTION

Question 10 · WMA11/01 January 2022

1 Find intercepts

For x-intercepts, set $\frac{1}{x^2} - 9 = 0$. Then $x^2 = \frac{1}{9}$, so $x = \pm\frac{1}{3}$. There is no y-intercept because $x = 0$ is not in the domain.

2 State asymptotes

The vertical asymptote is $x = 0$. As $|x|$ becomes large, $\frac{1}{x^2} \to 0$, so the horizontal asymptote is $y = -9$.

3 Set up intersections with D

For intersections with $y = kx^2$, solve $\frac{1}{x^2} - 9 = kx^2$.

4 Use u = x squared

Let $u = x^2$, where $u > 0$. Multiplying by x^2 gives $1 - 9u = ku^2$, so $ku^2 + 9u - 1 = 0$.

5 Require two positive u-values

Four distinct x-values require two distinct positive values of u, since each positive u gives $x = \pm\sqrt{u}$.

6 **Find the range of k**

For two positive roots, the product $-\frac{1}{k}$ and sum $-\frac{9}{k}$ must both be positive, so $k<0$. Also the discriminant must be positive: $81+4k>0$, so $k>-\frac{81}{4}$. Therefore $-\frac{81}{4}<k<0$.

FINAL ANSWER

(a) x-intercepts $\left(-\frac{1}{3},0\right),\left(\frac{1}{3},0\right)$; asymptotes $x=0,\ y=-9$. (b) $-\frac{81}{4}<k<0$.

WMA11/01 OCTOBER 2022

Question 6

6 marks | Graphs of Functions

6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \dfrac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k, where appropriate.

(5)

(b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

(Total for Question 6 is 6 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2022

1 Key points for y = (2x-k)(x+4)^2

The graph has x-intercepts $(-4, 0)$ and $\left(\frac{k}{2}, 0\right)$. The root at $x = -4$ is a repeated root, so the graph touches the x-axis there.

2 Use 0 < k < 4

Since $0 < k < 4$, the positive intercept $x = \frac{k}{2}$ lies between 0 and 2. The y-intercept is $(0, -16k)$.

3 Key features for y = k/x^2

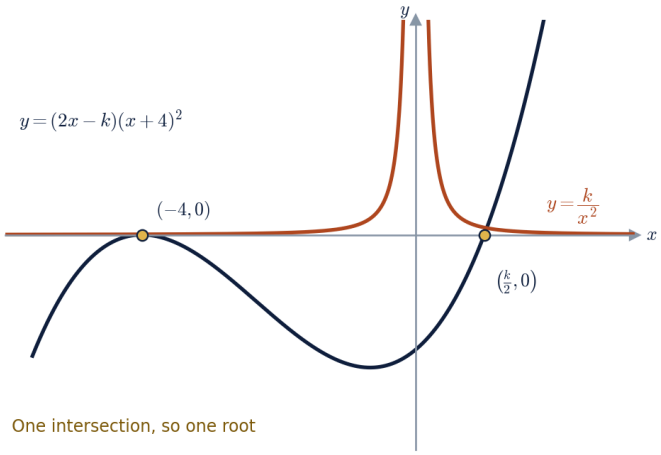
The graph is positive on both sides of the y-axis, with asymptotes $x = 0$ and $y = 0$. It has no intercepts with the coordinate axes.

4 Compare the two graphs

The equation $(2x - k)(x + 4)^2 = \frac{k}{x^2}$ represents intersections between the two graphs.

5 **State the number of roots**

From the sketches, the positive branch of $\dfrac{k}{x^2}$ meets the increasing positive branch of $(2x-k)(x+4)^2$ once only. Therefore there is one root.



FINAL ANSWER

The graphs intersect once, so the equation has one root.

WMA11/01 JANUARY 2024

Question 7

9 marks | Graphs of Functions

7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k.

(5)

(Total for Question 7 is 9 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2024

1 Identify the asymptote

For $y = \frac{4}{x - k}$, the vertical asymptote is where the denominator is zero, so $x = k$.

2 Find axis intercepts

At $x = 0$, $y = -\frac{4}{k}$, so the y-intercept is $(0, -\frac{4}{k})$. The graph has no x-intercept because the numerator is never zero.

3 Set up intersection with the line

If the line $y = 9 - x$ meets the curve, then $9 - x = \frac{4}{x - k}$.

4 Form a quadratic

$(9 - x)(x - k) = 4$, which simplifies to $x^2 - (9 + k)x + 9k + 4 = 0$.

5 Use no real intersection

For the line not to cross or touch the curve, the discriminant must be negative.

6 Solve the inequality

$(9 + k)^2 - 4(9k + 4) < 0$, so $k^2 - 18k + 65 < 0$. Since $k^2 - 18k + 65 = (k - 5)(k - 13)$, the range is $5 < k < 13$.

FINAL ANSWER

(a) y-intercept $\left(0, -\frac{4}{k}\right)$, no x-intercept, vertical asymptote $x = k$.

(b) $5 < k < 13$.

WMA11/01 MAY/JUNE 2024

Question 8

7 marks | Graphs of Functions

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

(a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

(b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0$$

(1)

(c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

(Total for Question 8 is 7 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2024

1 Find the intercepts of C1

$y = x(4 - x^2)$, so $y = 0$ when $x = 0$ or $x = \pm 2$. The curve crosses the axes at $(-2, 0)$, $(0, 0)$ and $(2, 0)$.

2 Use the cubic shape

$y = 4x - x^3$ has negative leading cubic term, so the left tail rises and the right tail falls.

3 Equate C1 and C2

$x(4 - x^2) = \frac{A}{x}$.

4 Multiply by x

$x^2(4 - x^2) = A$, so $4x^2 - x^4 = A$ and therefore $x^4 - 4x^2 + A = 0$.

5 Let u equal x squared

Let $u = x^2$. Then $u^2 - 4u + A = 0$.

6 Require two positive u-values

For 4 distinct x-values, the quadratic in u must have two distinct positive roots.

7 Find the range of A

The discriminant gives $16 - 4A > 0$, so $A < 4$. The product of the two roots is A, so $A > 0$. Therefore $0 < A < 4$.

FINAL ANSWER

(a) x-intercepts $(-2, 0)$, $(0, 0)$, $(2, 0)$.

(b) $x^4 - 4x^2 + A = 0$. (c) $0 < A < 4$.

WMA11/01 OCTOBER 2024

Question 6

9 marks | Graphs of Functions

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2 - x} \qquad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y-axis

(3)

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geqslant 0$$

(ii) find the range of possible values for k.

(6)

(Total for Question 6 is 9 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2024

1 **Identify the graph features**

$y = \dfrac{1}{2-x}$ has vertical asymptote $x = 2$ and horizontal asymptote $y = 0$.

2 **Find the y-intercept**

At $x = 0$, $y = \dfrac{1}{2}$, so the y-intercept is $(0, \dfrac{1}{2})$.

3 **Set up the intersection with the line**

If $y = kx - 4$ cuts C, then $kx - 4 = \dfrac{1}{2-x}$.

4 **Form the quadratic**

$(kx - 4)(2 - x) = 1$, so $kx^2 - (2k + 4)x + 9 = 0$.

5 **Use the discriminant**

For at least one intersection, the discriminant is non-negative.

6 **Derive the given inequality**

$(2k+4)^2 - 4(k)(9) \geq 0$, so $4(k^2 - 5k + 4) \geq 0$, hence $k^2 - 5k + 4 \geq 0$.

7 **Solve the inequality**

$k^2 - 5k + 4 = (k-1)(k-4)$, so $k \leq 1$ or $k \geq 4$.

✓ **FINAL ANSWER**

(a) vertical asymptote $x = 2$, y-intercept $(0, \frac{1}{2})$.

(b) $k \leq 1$ or $k \geq 4$.

WMA11/01 JANUARY 2025

Question 7

8 marks | Graphs of Functions

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

(a) Sketch the graph of C.

Show on your sketch

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

stating each in terms of k.

(3)

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

(b) find the range of possible values of k.

(5)

(Total for Question 7 is 8 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2025

1 **Identify the horizontal asymptote**

$y = \frac{2}{x} - k$ has horizontal asymptote $y = -k$.

2 **Find axis intersections**

Setting $y = 0$ gives $\frac{2}{x} = k$, so $x = \frac{2}{k}$. There is no y-intercept because $x = 0$ is not in the domain.

3 **Set up line-curve intersections**

$\frac{2}{x} - k = -kx - 6$.

4 **Form a quadratic**

Multiplying by x gives $2 - kx = -kx^2 - 6x$, so $kx^2 + (6 - k)x + 2 = 0$.

5 **Use two distinct intersections**

For two distinct points, the discriminant must be positive.

6 **Solve the inequality**

$(6-k)^2-8k>0$, so $k^2-20k+36>0$. Since $k^2-20k+36=(k-2)(k-18)$, $k<2$ or $k>18$.

7 **Use k positive**

As $k>0$, the range is $0<k<2$ or $k>18$.

✓ **FINAL ANSWER**

(a) x-intercept $(\frac{2}{k},0)$, no y-intercept, horizontal asymptote $y=-k$.

(b) $0<k<2$ or $k>18$.

WMA11/01 MAY/JUNE 2025

Question 7

10 marks | Graphs of Functions

7.

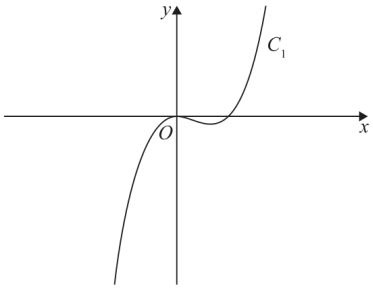


Figure 3

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x-axis at the origin and cuts the x-axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

The curve C_2 has equation $y = 1.2x(8 - x)$

On the following page there is a copy of Figure 3 called Diagram 1.

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect. Show each stage of your working.

(5)

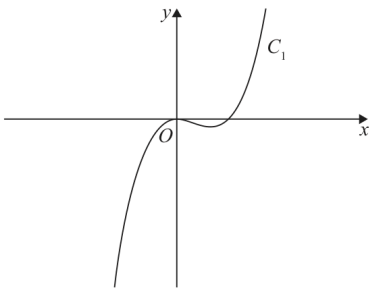


Diagram 1

(Total for Question 7 is 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2025

1 **Use the repeated root at the origin**

Since the cubic touches the x-axis at the origin and cuts it at $x = 4$, write $f(x) = \lambda x^2(x - 4)$.

2 **Use the point (10,120)**

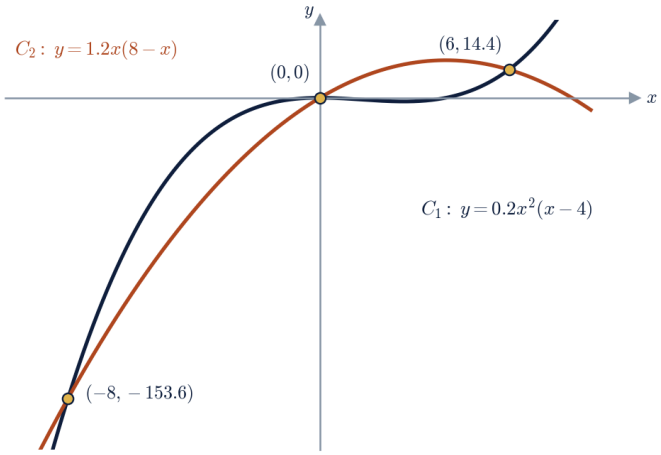
$120 = \lambda(10)^2(10 - 4) = 600\lambda$, so $\lambda = 0.2$.

3 **State f(x)**

$f(x) = 0.2x^2(x - 4)$.

4 **Sketch C2**

$C_2 : y = 1.2x(8 - x)$ is a downward-opening parabola through $x = 0$ and $x = 8$, with its maximum between them.



5 **Set intersections equal**

$1.2x(8 - x) = 0.2x^2(x - 4)$.

6 **Clear decimals**

Divide by 0.2 to get $6x(8 - x) = x^2(x - 4)$.

7 **Factor**

$48x - 6x^2 = x^3 - 4x^2$, so $x^3 + 2x^2 - 48x = 0$ and $x(x^2 + 2x - 48) = 0$.

8 **Solve for x**

$x = 0$, or $x^2 + 2x - 48 = 0 = (x + 8)(x - 6)$, so $x = -8$ or $x = 6$.

9 **Find y-values**

Using $y = 1.2x(8 - x)$ gives $(-8, -153.6)$, $(0, 0)$ and $(6, 14.4)$.

✓ **FINAL ANSWER**

$(a)\ f(x) = 0.2x^2(x - 4).$ $(c)\ (-8, -153.6),\ (0, 0),\ (6, 14.4).$

WMA11/01 OCTOBER 2025

Question 6

9 marks | Graphs of Functions

6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C,

(b) find the range of possible values of k.

(5)

(Total for Question 6 is 9 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2025

1 **Find coordinate-axis intersections**

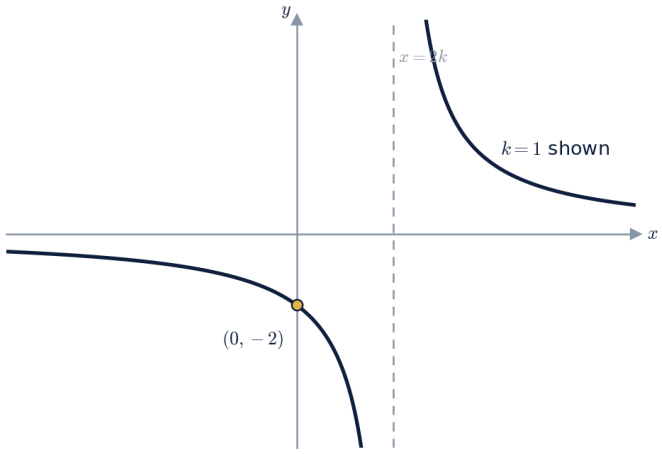
$y = \frac{4k}{x - 2k}$ has no x-intercept because the numerator is non-zero. At $x = 0$, $y = \frac{4k}{-2k} = -2$, so the y-intercept is $(0, -2)$.

2 **Find the vertical asymptote**

The denominator is zero when $x = 2k$, so the vertical asymptote is $x = 2k$.

3 **Sketch shape**

The curve has one branch in quadrant I to the right of $x = 2k$, and one branch passing through $(0, -2)$ and approaching the x-axis from below as $x \to -\infty$.



4 **Set line and curve equal**

$\dfrac{4k}{x - 2k} = 6 - 2x$.

5 **Form a quadratic in x**

$4k = (6 - 2x)(x - 2k) = 6x - 12k - 2x^2 + 4kx$, so $2x^2 - (4k + 6)x + 16k = 0$.

6 **Use at least one intersection**

For at least one real intersection, the discriminant must satisfy $b^2 - 4ac \geq 0$.

7 **Find the discriminant**

$(4k + 6)^2 - 4(2)(16k) \geq 0$, so $16k^2 - 80k + 36 \geq 0$.

8 **Solve in k**

$4k^2 - 20k + 9 \geq 0$, so $(2k - 1)(2k - 9) \geq 0$.

9 **Use k positive**

Therefore $k \leq \dfrac{1}{2}$ or $k \geq \dfrac{9}{2}$. Since k is positive, $0 < k \leq \dfrac{1}{2}$ or $k \geq \dfrac{9}{2}$.

✓ **FINAL ANSWER**

(a) y-intercept $(0, -2)$, vertical asymptote $x = 2k$.

(b) $0 < k \leq \dfrac{1}{2}$ or $k \geq \dfrac{9}{2}$.

WMA11/01 MAY/JUNE 2026

Question 10

11 marks | Graphs of Functions

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \qquad x \neq 2k$$

where k is a positive constant.

(a) Sketch a graph of C.
Show on your sketch the coordinates of the point where C crosses a coordinate axis.

(3)

The line l has equation $x + y = 12$

Given that l is a tangent to C,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

The line l meets C at a point P.

Given that P lies below the x-axis,

(c) find the coordinates of P.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

(Total for Question 10 is 11 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2026

1 **(a) Identify the graph features**

The curve

$$y = \frac{3k}{x - 2k}$$

is a rectangular hyperbola with vertical asymptote $x = 2k$ and horizontal asymptote $y = 0$. Since $k > 0$, the branch for $x > 2k$ lies above the x-axis. The other branch lies below the x-axis and passes through the y-axis.

2 **(a) Find the axis intercept**

At $x = 0$,

$$y = \frac{3k}{-2k} = -\frac{3}{2}.$$

So the curve crosses the y-axis at

$$\left(0, -\frac{3}{2}\right).$$

There is no x-intercept because the numerator $3k$ is non-zero.

---

3 **(b) Form the intersection quadratic**

The line is $y = 12 - x$. At an intersection,

$$\frac{3k}{x - 2k} = 12 - x.$$

Cross-multiplying gives

$$3k = (12 - x)(x - 2k) = 12x - x^2 - 24k + 2kx.$$

Hence

$$x^2 - (12 + 2k)x + 27k = 0.$$

---

4 **(b) Use the tangency condition**

A tangent gives a repeated solution for x, so the discriminant is zero:

$$(12 + 2k)^2 - 4(1)(27k) = 0.$$

Expanding and simplifying,

$$144 + 48k + 4k^2 - 108k = 0,$$

$$4k^2 - 60k + 144 = 0,$$

and therefore

$$k^2 - 15k + 36 = 0,$$

as required.

---

5 **(c) Find the possible values of k**

Factor the equation from part (b):

$$k^2 - 15k + 36 = (k - 3)(k - 12) = 0.$$

Thus $k = 3$ or $k = 12$.

---

6 **(c) Find and select the tangent point**

The repeated root of

$$x^2 - (12 + 2k)x + 27k = 0$$

is

$$x = \frac{12 + 2k}{2} = 6 + k.$$

Then $y = 12 - x = 6 - k$. For $k = 3$, the point is $(9, 3)$, which is above the x-axis. For $k = 12$, the point is $(18, -6)$, which is below the x-axis. Therefore

$$P = (18, -6).$$

FINAL ANSWER

(a) Rectangular hyperbola with asymptotes $x = 2k$ and $y = 0$, crossing the y-axis at $\left(0, -\frac{3}{2}\right)$.

(b) $k^2 - 15k + 36 = 0$.

(c) $P = (18, -6)$.

TOPIC

Transformations

Every past-paper question on this topic, October 2019 to May/June 2026.

Question 9

9 marks | Transformations

9. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

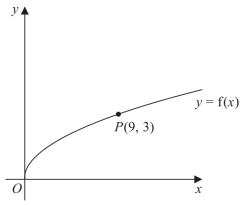


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \qquad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q.

(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3\left(\sqrt{2} + 1\right)$$

(3)

(c) Hence find, in simplest form, the coordinates of Q.

(3)

Question 9 continued

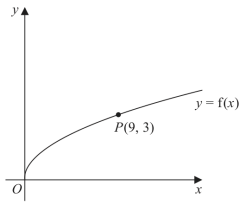
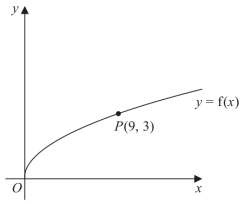


Diagram 1

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

Only use this copy if you need to redraw your graphs.



Copy of Diagram 1

(Total 9 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2021

1 **Transform P on y = f(2x)**

$y = f(2x)$ is a horizontal scale factor of $\frac{1}{2}$, so $P(9,3)$ maps to $\left(\frac{9}{2}, 3\right)$.

2 **Translate point P**

$y = f(x) + 3$ is a translation 3 units upward, so $P(9,3)$ maps to $(9,6)$.

3 **Set up the intersection equation**

At Q, $f(2x) = f(x) + 3$. Since $f(x) = \sqrt{x}$, this gives $\sqrt{2x} = \sqrt{x} + 3$.

4 **Show the required equation**

Let $s = \sqrt{x}$. Then $\sqrt{2x} = \sqrt{2}\,s$, so $\sqrt{2}s = s + 3$. Hence $(\sqrt{2}-1)s = 3$ and $s = \dfrac{3}{\sqrt{2}-1} = 3(\sqrt{2}+1)$.

5 **Find the x-coordinate**

$\sqrt{x} = 3(\sqrt{2}+1)$, so $x = 9(\sqrt{2}+1)^2 = 9(3+2\sqrt{2}) = 27+18\sqrt{2}$.

6 **Find the y-coordinate**

$y = \sqrt{x}+3 = 3(\sqrt{2}+1)+3 = 6+3\sqrt{2}$. Therefore $Q = (27+18\sqrt{2},\ 6+3\sqrt{2})$.

FINAL ANSWER

$(a)\ P \mapsto \left(\dfrac{9}{2}, 3\right)$ on $y = f(2x)$, $P \mapsto (9,6)$ on $y = f(x)+3$. $(c)\ Q = (27+18\sqrt{2},\ 6+3\sqrt{2})$.

WMA11/01 OCTOBER 2022

Question 7

8 marks | Transformations

7.

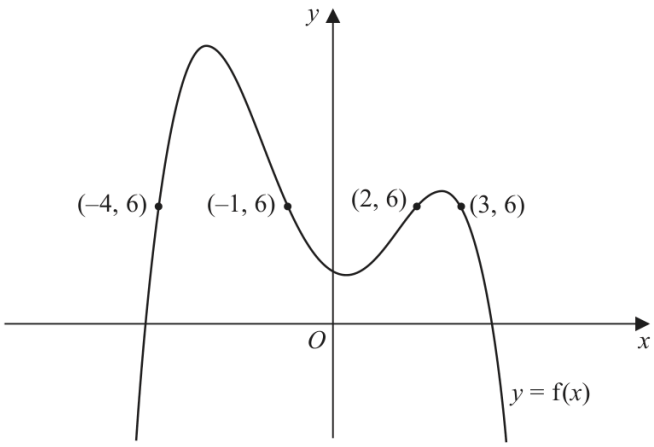


Figure 1

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6$$

(3)

(b) State the largest solution of the equation

$$f(2x) = 6$$

(1)

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P, Q, R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geqslant 6 \text{ and } x < 0$$

(4)

(Total for Question 7 is 8 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2022

1 Read f(x) < 6 from the graph

The curve is below the line $y = 6$ for $x < -4$, for $-1 < x < 2$, and for $x > 3$.

2 Solve f(2x) = 6

Since $f(u) = 6$ at $u = -4, -1, 2, 3$, we set $2x = -4, -1, 2, 3$. The solutions are $x = -2, -\frac{1}{2}, 1, \frac{3}{2}$.

3 State the largest solution

The largest solution is $x = \frac{3}{2}$.

4 **Reflect the points in the y-axis**

For $y = f(-x)$, each point (x, y) becomes $(-x, y)$.

5 **List the transformed points**

$P(-4, 6) \to (4, 6)$, $Q(-1, 6) \to (1, 6)$, $R(2, 6) \to (-2, 6)$, and $S(3, 6) \to (-3, 6)$.

6 **Use the reflected graph**

For $x < 0$, the part of the reflected graph on or above $y = 6$ is from $x = -3$ to $x = -2$.

✓ **FINAL ANSWER**

$(a)\ x < -4,\ -1 < x < 2,\ x > 3.\quad (b)\ \dfrac{3}{2}.\quad (c)(ii)\ -3 \le x \le -2.$

Question 7

10 marks | Transformations

7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \qquad x \neq 0$$

(2)

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \qquad x \neq 2$

(b) Fully describe this transformation.

(2)

The curve with equation

$$y = \frac{6}{x-2} \qquad x \neq 2$$

and the line with equation

$$y = kx + 7 \qquad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q.

Given that the x coordinate of point P is -4

(c) find the value of k,

(2)

(d) find, using algebra, the coordinates of point Q.

(*Solutions relying entirely on calculator technology are not acceptable.*)

(4)

Question 7 continued

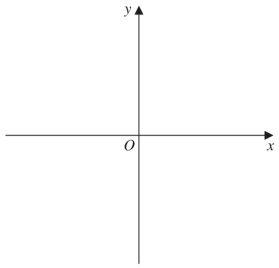
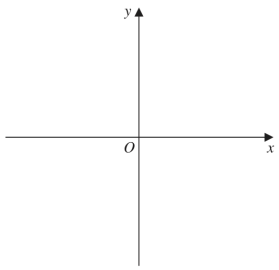


Diagram 1

Only use this copy of Diagram 1 if you need to redraw your graph.

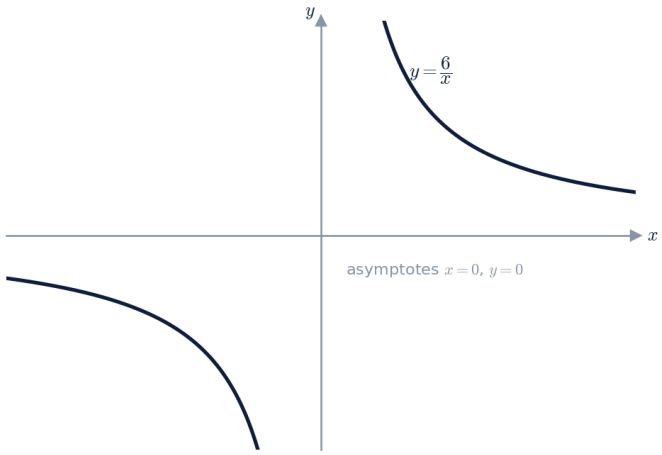


Copy of Diagram 1

(Total for Question 7 is 10 marks)

1 **Sketch y = 6/x**

The graph is a rectangular hyperbola with asymptotes $x = 0$ and $y = 0$, with branches in quadrants I and III.



2 **Describe the transformation**

$y = \dfrac{6}{x-2}$ is obtained from $y = \dfrac{6}{x}$ by a translation 2 units in the positive x-direction.

3 **Use P to find k**

At $x = -4$ on $y = \dfrac{6}{x-2}$, $y = \dfrac{6}{-6} = -1$.

4 **Substitute into the line**

Since P lies on $y = kx + 7$, $-1 = -4k + 7$, so $k = 2$.

5 **Find the second intersection**

Solve $\dfrac{6}{x-2} = 2x + 7$. Multiplying by $x - 2$ gives $6 = (2x + 7)(x - 2)$.

6 **Factor and choose Q**

$6 = 2x^2 + 3x - 14$, so $2x^2 + 3x - 20 = 0 = (2x - 5)(x + 4)$. The root $x = -4$ is P, so Q has $x = \dfrac{5}{2}$.

7 **Find y for Q**

$y = 2\left(\dfrac{5}{2}\right) + 7 = 12$, so $Q = \left(\dfrac{5}{2}, 12\right)$.

FINAL ANSWER

(b) Translation 2 units right.

(c) $k = 2$. (d) $Q = \left(\dfrac{5}{2}, 12\right)$.

WMA11/01 MAY/JUNE 2025

Question 9

10 marks | Transformations

9.

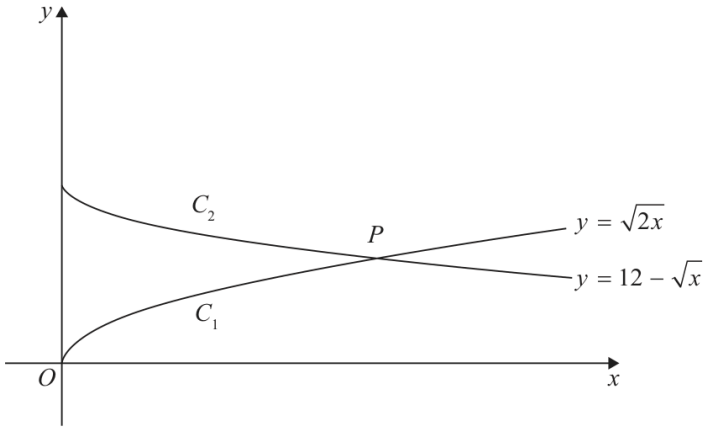


Figure 4

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

(i) the graph with equation $y = \sqrt{x}$ onto C_1

(ii) the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

The graphs C_1 and C_2 meet at the point P, as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12\left(\sqrt{2} - 1\right)$$

(ii) Hence find, in simplest form, the exact coordinates of P.

(6)

(Total for Question 9 is 10 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2025

1 Transform y equals root x to C1

C_1 has equation $y = \sqrt{2x}$. This is obtained from $y = \sqrt{x}$ by a stretch parallel to the x-axis with scale factor $\frac{1}{2}$.

2 Equivalent description

Equivalently, $y = \sqrt{2x} = \sqrt{2}\sqrt{x}$ is a stretch parallel to the y-axis with scale factor $\sqrt{2}$. The mark scheme also accepts the horizontal stretch description.

3 Translate the reflected curve

C_2 has equation $y = 12 - \sqrt{x} = -\sqrt{x} + 12$, so it is a translation of $y = -\sqrt{x}$ by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.

4 **Set the curves equal**

At P, $\sqrt{2x} = 12 - \sqrt{x}$.

5 **Factor out root x**

$\sqrt{2}\sqrt{x} + \sqrt{x} = 12$, so $(\sqrt{2}+1)\sqrt{x} = 12$.

6 **Rationalise**

$$\sqrt{x} = \frac{12}{\sqrt{2}+1} = \frac{12(\sqrt{2}-1)}{(\sqrt{2}+1)(\sqrt{2}-1)} = 12(\sqrt{2}-1).$$

7 **Find x**

$x = [12(\sqrt{2}-1)]^2 = 144(3-2\sqrt{2})$.

8 **Find y**

$y = \sqrt{2x} = \sqrt{2}\sqrt{x} = \sqrt{2}\cdot 12(\sqrt{2}-1) = 24 - 12\sqrt{2}$.

FINAL ANSWER

$(a)(i)$ stretch parallel to the x-axis scale factor $\frac{1}{2}$ (equivalently vertical stretch scale factor $\sqrt{2}$).

$(a)(ii)$ translation by $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$.

(b) $P = (144(3-2\sqrt{2}),\ 24-12\sqrt{2})$.

Question 9

6 marks | Transformations

9.

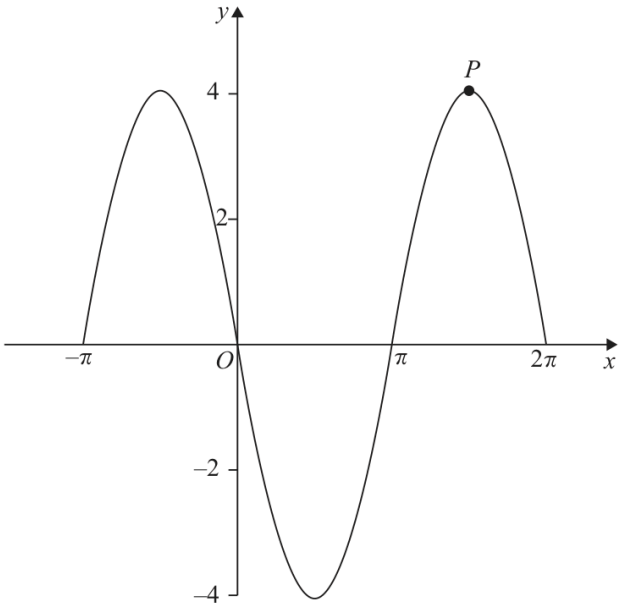


Figure 3

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = \mathrm{f}(x)$

(a) Write down an expression for $\mathrm{f}(x)$

(2)

The point P lies on $y = \mathrm{f}(x)$ and is shown in Figure 3.

(b) State the coordinates of the point to which P is transformed when the graph of $y = \mathrm{f}(x)$ is transformed to the graph with equation

(i) $y = \mathrm{f}\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}\mathrm{f}(x)$

(2)

(Total for Question 9 is 6 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2025

1 Identify amplitude and direction

The graph has amplitude 4, crosses the origin, and initially moves downwards, so $f(x) = -4\sin x$.

2 Identify P

The labelled maximum is at $P = \left(\frac{3\pi}{2}, 4\right)$.

3 Translate the graph right

$y = f\left(x - \frac{\pi}{6}\right)$ translates the graph right by $\frac{\pi}{6}$.

4 **New coordinates for part b(i)**

P becomes $\left(\frac{3\pi}{2}+\frac{\pi}{6}, 4\right)=\left(\frac{5\pi}{3}, 4\right)$.

5 **Transform y equals negative half f(x)**

$y=-\frac{1}{2}f(x)$ leaves x-coordinates unchanged and multiplies y-coordinates by $-\frac{1}{2}$.

6 **New coordinates for part b(ii)**

P becomes $\left(\frac{3\pi}{2}, -2\right)$.

✓ **FINAL ANSWER**

$(a)\ f(x)=-4\sin x.\quad (b)(i)\ \left(\frac{5\pi}{3}, 4\right).\quad (b)(ii)\ \left(\frac{3\pi}{2}, -2\right).$

TOPIC

Straight Line

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 JANUARY 2020

Question 6

8 marks | Straight Line

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x-axis at $R(8,0)$ and is parallel to l_1

(a) Find the equation of l_2, writing your answer in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

The line l_1 cuts the x-axis at P and the y-axis at Q.

Given that $PQRS$ is a parallelogram, find

(b) the area of $PQRS$,

(3)

(c) the coordinates of S.

(2)

(Total 8 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2020

1 Find the gradient of l1

$3x - 4y + 20 = 0$ gives $y = \frac{3}{4}x + 5$, so the gradient is $\frac{3}{4}$.

2 Find l2 through R

Line l_2 is parallel to l_1 and passes through $R(8,0)$, so $y = \frac{3}{4}(x - 8) = \frac{3}{4}x - 6$. Therefore $3x - 4y - 24 = 0$.

3 Find P and Q

For l_1, the x-intercept is $P\left(-\frac{20}{3}, 0\right)$ and the y-intercept is $Q = (0,5)$.

4 Find the area of PQRS

Base $PR = 8 - \left(-\frac{20}{3}\right) = \frac{44}{3}$ and height $OQ = 5$. Area $= \frac{44}{3} \cdot 5 = \frac{220}{3}$.

5 Find S using a vector

In parallelogram $PQRS$, $\vec{PS} = \vec{QR} = (8, -5)$. Hence $S = P + (8, -5) = \left(-\frac{20}{3} + 8, -5\right) = \left(\frac{4}{3}, -5\right)$.

FINAL ANSWER

$(a)\ 3x - 4y - 24 = 0.$ $(b)\ \frac{220}{3}.$ $(c)\ S = \left(\frac{4}{3}, -5\right).$

Question 8

10 marks | Straight Line

8.

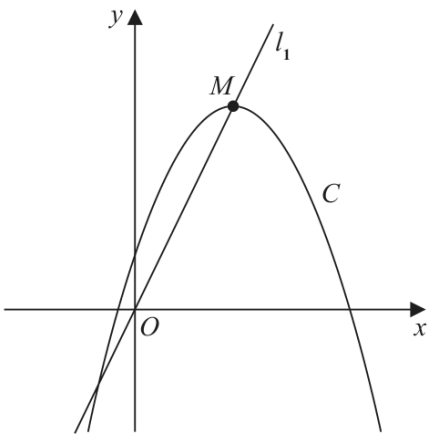


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C.

(a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a, b and c are constants to be found.

(ii) Hence, or otherwise, state the coordinates of M.

(5)

The line l_1 passes through O and M, as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

(b) Find an equation for l_2

(5)

(Total 10 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2021

1 Complete the square

$4 + 12x - 3x^2 = -3(x^2 - 4x) + 4 = -3\left((x-2)^2 - 4\right) + 4 = 16 - 3(x-2)^2$.

2 State the maximum point

The maximum occurs when $(x-2)^2 = 0$, so $M = (2, 16)$.

3 Find the gradient of l1

Line l_1 passes through $O(0,0)$ and $M(2,16)$, so its gradient is $\frac{16}{2} = 8$.

4 **Use the parallel tangent gradient**

Line l_2 is parallel to l_1, so its gradient is also 8. For $y = 4 + 12x - 3x^2$, $\dfrac{dy}{dx} = 12 - 6x$.

5 **Find the point of contact**

Set $12 - 6x = 8$, giving $x = \dfrac{2}{3}$. Then $y = 4 + 12\left(\dfrac{2}{3}\right) - 3\left(\dfrac{4}{9}\right) = \dfrac{32}{3}$.

6 **Write l2**

Using gradient 8 through $\left(\dfrac{2}{3}, \dfrac{32}{3}\right)$, $y - \dfrac{32}{3} = 8\left(x - \dfrac{2}{3}\right)$. Hence $y = 8x + \dfrac{16}{3}$.

✓ **FINAL ANSWER**

$(a)(i)\ 16 - 3(x-2)^2.$ $(a)(ii)\ M = (2, 16).$ $(b)\ y = 8x + \dfrac{16}{3}.$

Question 8

9 marks | Straight Line

8. The line l_1 has equation

$$2x - 5y + 7 = 0$$

(a) Find the gradient of l_1

(1)

Given that

- the point A has coordinates $(6, -2)$
- the line l_2 passes through A and is perpendicular to l_1

(b) find the equation of l_2 giving your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

The lines l_1 and l_2 intersect at the point M.

(c) Using algebra and showing all your working, find the coordinates of M.

(*Solutions relying on calculator technology are not acceptable.*)

(3)

Given that the diagonals of a square $ABCD$ meet at M,

(d) find the coordinates of the point C.

(2)

(Total 9 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2022

1 Find the gradient of l1

$2x - 5y + 7 = 0$ gives $y = \frac{2}{5}x + \frac{7}{5}$, so the gradient is $\frac{2}{5}$.

2 Find l2

The gradient perpendicular to $\frac{2}{5}$ is $-\frac{5}{2}$. Through $A(6, -2)$, $y + 2 = -\frac{5}{2}(x - 6)$, so $y = -\frac{5}{2}x + 13$.

3 Find M

At the intersection, $\frac{2}{5}x + \frac{7}{5} = -\frac{5}{2}x + 13$. Multiplying by 10 gives $4x + 14 = -25x + 130$, so $x = 4$.

4 **Find the y-coordinate of M**

$y = \frac{2}{5}(4) + \frac{7}{5} = 3$, so $M = (4, 3)$.

5 **Use midpoint of the diagonal**

In a square, the diagonals bisect each other, so M is the midpoint of AC. Thus $C = 2M - A = (8, 6) - (6, -2) = (2, 8)$.

✓ **FINAL ANSWER**

$(a)\ \frac{2}{5}.$ $(b)\ y = -\frac{5}{2}x + 13.$ $(c)\ M = (4, 3).$ $(d)\ C = (2, 8).$

WMA11/01 MAY/JUNE 2023

Question 10

10 marks | Straight Line

10.

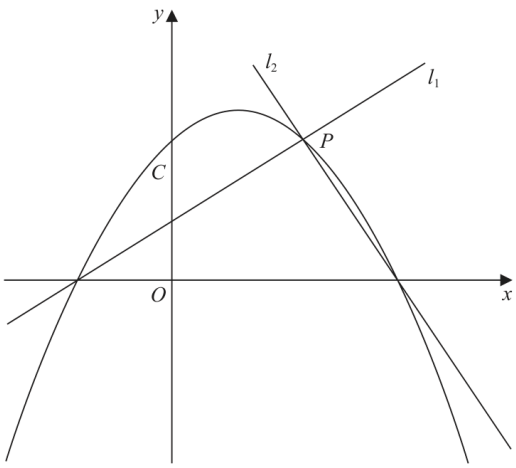


Figure 5

Figure 5 shows a sketch of the quadratic curve C with equation

$$y = -\frac{1}{4}(x+2)(x-b) \qquad \text{where } b \text{ is a positive constant}$$

The line l_1 also shown in Figure 5,

- has gradient $\frac{1}{2}$
- intersects C on the negative x-axis and at the point P

(a) (i) Write down an equation for l_1

(1)

(ii) Find, in terms of b, the coordinates of P

(3)

Given that the line l_2 is perpendicular to l_1 and intersects C on the positive x-axis,

(b) find, in terms of b, an equation for l_2

(2)

Given also that l_2 intersects C at the point P

(c) show that another equation for l_2 is

$$y = -2x + \frac{5b}{2} - 4$$

(2)

(d) Hence, or otherwise, find the value of b

(Total for Question 10 is 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2023

1 Find l1

Line l_1 has gradient $\frac{1}{2}$ and passes through the negative x-axis root $(-2, 0)$, so $y = \frac{1}{2}(x+2) = \frac{1}{2}x + 1$.

2 Find P

Set $-\frac{1}{4}(x+2)(x-b) = \frac{1}{2}(x+2)$.

3 Solve the intersection

Multiplying by -4 gives $(x+2)(x-b) = -2(x+2)$, so $(x+2)(x-b+2) = 0$.

4 **Choose the second point**

The root $x = -2$ is the x-axis point, so P has $x = b - 2$. Then $y = \frac{1}{2}(b-2) + 1 = \frac{b}{2}$.

5 **Find l2 from the positive root**

Since l_2 is perpendicular to l_1, its gradient is -2. It passes through the positive x-axis root $(b, 0)$, so $y = -2(x - b) = -2x + 2b$.

6 **Use P to write another l2**

Through $P\left(b-2, \frac{b}{2}\right)$ with gradient -2: $y - \frac{b}{2} = -2(x - (b-2))$, so $y = -2x + \frac{5b}{2} - 4$.

7 **Find b**

Both equations represent l_2, so $2b = \frac{5b}{2} - 4$. Hence $b = 8$.

✓ **FINAL ANSWER**

$(a)(i)\ y = \frac{1}{2}x + 1.$ $(a)(ii)\ P = \left(b-2, \frac{b}{2}\right).$ $(b)\ y = -2x + 2b.$ $(d)\ b = 8.$

WMA11/01 MAY/JUNE 2024

Question 9

9 marks | Straight Line

9. Given that

- the point A has coordinates $(4, 2)$
- the point B has coordinates $(15, 7)$
- the line l_1 passes through A and B

(a) find an equation for l_1, giving your answer in the form $px + qy + r = 0$ where p, q and r are integers to be found.

(3)

The line l_2 passes through A and is parallel to the x-axis.

The point C lies on l_2 so that the length of BC is $5\sqrt{5}$

(b) Find both possible pairs of coordinates of the point C.

(4)

(c) Hence find the minimum possible area of triangle ABC.

(2)

(Total for Question 9 is 9 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2024

1 Find the gradient of l1

$m = \dfrac{7-2}{15-4} = \dfrac{5}{11}$.

2 Use point A

$y - 2 = \dfrac{5}{11}(x - 4)$.

3 Rearrange

$11y - 22 = 5x - 20$, so $5x - 11y + 2 = 0$.

4 Use the horizontal line l2

Since l_2 passes through $A(4, 2)$ and is parallel to the x-axis, point C has coordinates $(c, 2)$.

5 Use BC equals 5 root 5

$(c - 15)^2 + (2 - 7)^2 = (5\sqrt{5})^2$.

6 Solve for c

$(c - 15)^2 + 25 = 125$, so $(c - 15)^2 = 100$ and $c = 5$ or 25.

7 **Find the smaller area**

The height from B to $y = 2$ is 5. For $C = (5, 2)$, $AC = 1$, so the minimum area is $\frac{1}{2}(1)(5) = \frac{5}{2}$.

FINAL ANSWER

$(a)\ 5x - 11y + 2 = 0.$ $(b)\ C = (5, 2)$ or $(25, 2).$ $(c)\ \frac{5}{2}.$

TOPIC

Radians

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 MAY/JUNE 2021

Question 7

10 marks | Radians

7.

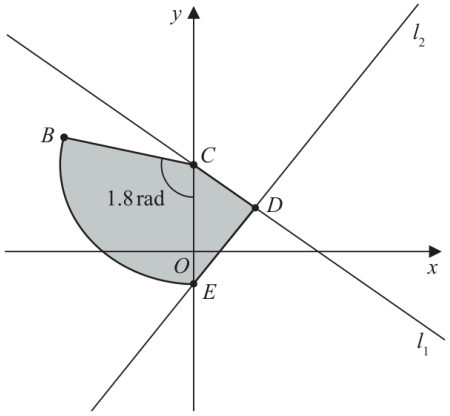


Figure 3

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y-axis at the point C, as shown in Figure 3.

(a) State the y coordinate of C.

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y-axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE.

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE.

(d) Calculate the exact area of the region $CBED$.

(3)

(Total 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2021

1 Find C

On the y-axis, $x = 0$. From $4y + 3x = 48$, $4y = 48$, so the y-coordinate of C is 12.

2 Find the gradient of l1

$4y + 3x = 48$ gives $y = -\frac{3}{4}x + 12$, so the gradient of l_1 is $-\frac{3}{4}$.

3 Use perpendicular gradients for l2

Since l_2 is perpendicular to l_1, its gradient is $\frac{4}{3}$. Through $D(8, 6)$, $y - 6 = \frac{4}{3}(x - 8)$.

4 **Show the y-coordinate of E**

$y = \frac{4}{3}x - \frac{32}{3} + 6 = \frac{4}{3}x - \frac{14}{3}$. Therefore when $x = 0$, $E_y = -\frac{14}{3}$.

5 **Find the radius of the sector**

$C = (0, 12)$ and $E = \left(0, -\frac{14}{3}\right)$, so $CE = 12 + \frac{14}{3} = \frac{50}{3}$.

6 **Find arc BE**

For sector BCE, $r = \frac{50}{3}$ and $\theta = 1.8 = \frac{9}{5}$. Hence arc $BE = r\theta = \frac{50}{3} \cdot \frac{9}{5} = 30$.

7 **Find the sector area**

Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{50}{3}\right)^2\left(\frac{9}{5}\right) = 250$.

8 **Find the triangle area and total area**

Triangle CDE has vertical base $CE = \frac{50}{3}$ and horizontal height 8, so its area is $\frac{1}{2} \cdot \frac{50}{3} \cdot 8 = \frac{200}{3}$. Total area $= 250 + \frac{200}{3} = \frac{950}{3}$.

FINAL ANSWER

$(a)\ 12.\quad (b)\ E_y = -\frac{14}{3}.\quad (c)\ 30.\quad (d)\ \frac{950}{3}.$

Question 7

10 marks | Radians

7.

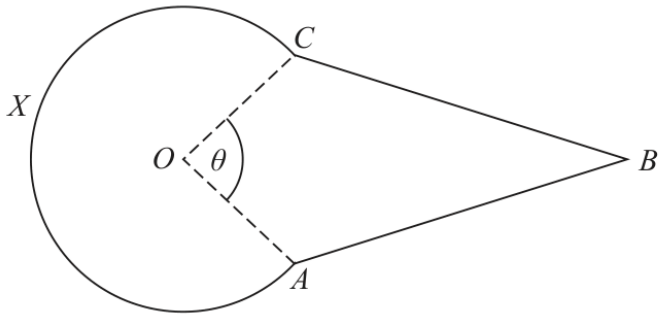


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O.

In the design

- $OA = OC = 0.6\,\text{m}$
- $AB = CB = 1.4\,\text{m}$
- Angle OAB = Angle OCB = 2 radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

(a) show that $\theta = 1.64$ radians to 3 significant figures,

(4)

(b) find the perimeter of the sign, in metres to 2 significant figures,

(2)

(c) find the area of the sign, in m^2 to 2 significant figures.

(4)

(Total 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2021

1 Work in triangle OAB

In triangle OAB, $OA = 0.6$, $AB = 1.4$, and $\angle OAB = 2$ radians.

2 Find OB

By the cosine rule, $OB^2 = 0.6^2 + 1.4^2 - 2(0.6)(1.4)\cos 2$, so $OB = 1.73756\ldots$.

3 Find half of theta

Using the cosine rule again, $\cos\angle AOB = \dfrac{0.6^2 + OB^2 - 1.4^2}{2(0.6)(OB)}$, giving $\angle AOB = 0.822199\ldots$ radians.

4 **Use symmetry of the kite**

The kite is symmetric, so $\theta = 2\angle AOB = 1.644398\ldots$, hence $\theta = 1.64$ radians to 3 significant figures.

5 **Find the outside arc length**

The sector $OCXA$ uses the major angle $2\pi - \theta$. Its arc length is $0.6(2\pi - 1.644398\ldots) = 2.78327\ldots$.

6 **Find the perimeter**

Perimeter $= AB + CB + \text{major arc} = 1.4 + 1.4 + 2.78327\ldots = 5.58327\ldots$, so the perimeter is 5.6 m.

7 **Find the kite area**

The kite consists of two congruent triangles like OAB, so area of kite $= 2\left(\frac{1}{2}(0.6)(1.4)\sin 2\right) = 0.76381\ldots$.

8 **Find the sector area and total area**

Major sector area $= \frac{1}{2}(0.6)^2(2\pi - 1.644398\ldots) = 0.83498\ldots$. Total area $= 0.76381\ldots + 0.83498\ldots = 1.59879\ldots$, so the area is $1.6\ \text{m}^2$.

FINAL ANSWER

$(a)\ \theta = 1.64\ \text{rad.}$ $(b)\ 5.6\ \text{m.}$ $(c)\ 1.6\ \text{m}^2$.

WMA11/01 MAY/JUNE 2022

Question 8

10 marks | Radians

8.

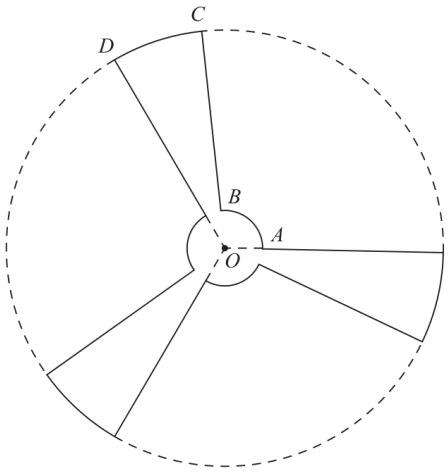


Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCDO$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

(a) show that the length of the arc CD is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)

(c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

(Total 10 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2022

1 Find the small sector angle

For arc AB, $s = r\theta$, so $15 = 9\theta$ and $\theta = \frac{5}{3}$ radians.

2 Find the outer arc angle

Each identical section has total angle $\frac{2\pi}{3}$, so the angle for arc CD is $\frac{2\pi}{3} - \frac{5}{3} = \frac{2\pi - 5}{3}$.

3 Find CD

$CD = 84\left(\frac{2\pi - 5}{3}\right) = 28(2\pi - 5) = 35.9\ \text{cm}$ to 3 significant figures.

4 **Find the perimeter**

There are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length $84 - 9 = 75$. Thus $P = 3(15) + 3(35.929\ldots) + 6(75) = 602.8\ldots \text{ cm}$.

5 **Find the area of one section**

One section has area $\frac{1}{2}(9^2)\left(\frac{5}{3}\right) + \frac{1}{2}(84^2)\left(\frac{2\pi - 5}{3}\right) = 1576.525\ldots \text{ cm}^2$.

6 **Find the total area**

The total area is $3(1576.525\ldots) = 4729.57\ldots \text{ cm}^2 = 4.73 \times 10^3 \text{ cm}^2$ to 3 significant figures.

✓ **FINAL ANSWER**

$(a)\ CD = 35.9 \text{ cm}.$ $(b)\ 603 \text{ cm}.$ $(c)\ 4.73 \times 10^3 \text{ cm}^2$.

Question 8

14 marks | Radians

8.

Diagram **NOT** to scale

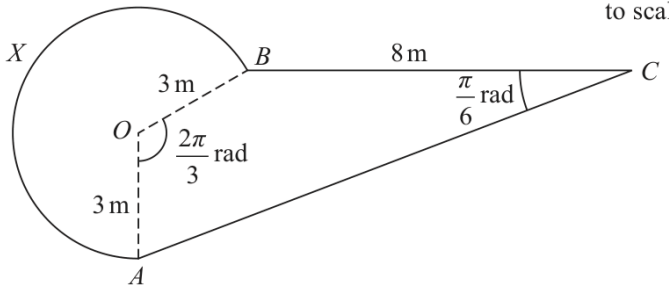


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8\,\text{m}$
- $OA = OB = 3\,\text{m}$
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB.

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2022

1 Find the angle of the sector AOBX

The sector AOBX is the major sector, so its angle is $2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$.

2 Find the sector area

$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}(3^2)\left(\frac{4\pi}{3}\right) = 6\pi\ \text{m}^2$.

3 Find the sector perimeter

The arc length is $r\theta = 3\left(\frac{4\pi}{3}\right) = 4\pi$. Adding the two radii gives $4\pi + 6$ m.

4 **Find the area of triangle AOB**

$\text{Area} = \frac{1}{2}(3)(3)\sin\left(\frac{2\pi}{3}\right) = \frac{9}{2}\cdot\frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4}\ \text{m}^2$.

5 **Show AB**

By the cosine rule, $AB^2 = 3^2 + 3^2 - 2(3)(3)\cos\left(\frac{2\pi}{3}\right) = 18 + 9 = 27$, so $AB = 3\sqrt{3}$.

6 **Find AC**

In triangle ABC, $AB^2 = AC^2 + 8^2 - 2(AC)(8)\cos\left(\frac{\pi}{6}\right)$, so $27 = AC^2 - 8\sqrt{3}\,AC + 64$.

7 **Choose the correct length**

Solving gives $AC = 4\sqrt{3} \pm \sqrt{11}$. From the diagram, AC is the longer value, so $AC = 4\sqrt{3} + \sqrt{11}$.

8 **Find the total area**

Area of triangle ABC is $\frac{1}{2}(8)(AC)\sin\left(\frac{\pi}{6}\right) = 2AC = 8\sqrt{3} + 2\sqrt{11}$. Total area is $6\pi + \frac{9\sqrt{3}}{4} + 8\sqrt{3} + 2\sqrt{11} = 43.236\ldots$, so $43\ \text{m}^2$ to 2 significant figures.

FINAL ANSWER

$(a)(i)\ 6\pi\ \text{m}^2,\ (ii)\ 4\pi + 6\ \text{m}.\quad (b)\ \frac{9\sqrt{3}}{4}\ \text{m}^2.\quad (c)\ AB = 3\sqrt{3}\ \text{m}.\quad (d)\ 43\ \text{m}^2$.

Question 6

10 marks | Radians

6.

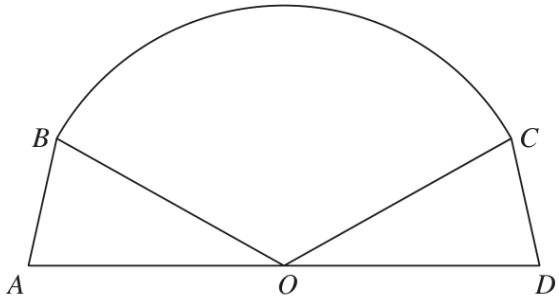


Diagram **NO**
accurately dr

Figure 1

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector OBC of a circle, with centre O, joined to two congruent triangles OAB and ODC.

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40\,\text{m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m, to 2 decimal places,

(2)

(b) find the size of angle AOB, in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2, to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m, to one decimal place.

(4)

(Total for Question 6 is 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2023

1 Find the radius

Area of sector BOC is $\frac{1}{2}r^2\theta$. So $40 = \frac{1}{2}r^2(2.4) = 1.2r^2$, giving $r = \sqrt{\frac{40}{1.2}} = 5.77$ m.

2 Find angle AOB

The triangles OAB and ODC are congruent, so $\angle AOB = \angle COD$. Since AOD is a straight line, $2\angle AOB + 2.4 = \pi$.

3 Calculate the angle

$\angle AOB = \frac{\pi - 2.4}{2} = 0.3708\ldots$, so 0.37 rad.

4 **Find the total area**

The two congruent triangle areas together are $2\left(\frac{1}{2}(6.25)(5.7735\ldots)\sin 0.3708\ldots\right) = 13.075\ldots$.

5 **Add the sector area**

Total area $= 40 + 13.075\ldots = 53.1\ \text{m}^2$ to one decimal place.

6 **Find AB**

Using the cosine rule, $AB^2 = 6.25^2 + r^2 - 2(6.25)(r)\cos 0.3708\ldots$, giving $AB = 2.265\ldots$ m.

7 **Find the perimeter**

Perimeter $= AD + AB + CD + \text{arc } BC = 12.5 + 2(2.265\ldots) + 5.7735\ldots(2.4) = 30.9$ m.

FINAL ANSWER

$(a)\ r = 5.77\ \text{m}.\quad (b)\ 0.37\ \text{rad}.\quad (c)\ 53.1\ \text{m}^2.\quad (d)\ 30.9\ \text{m}.$

Question 9

7 marks | Radians

9. Diagram **NO** accurately dr

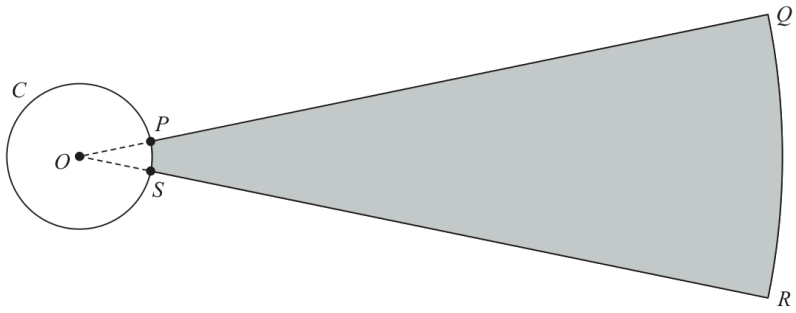


Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2\,\text{m}$

(1)

Given also that

- the target area, $PQRS$, is $90\,\text{m}^2$
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0$$

(3)

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2023

1 Use arc length

$s = r\theta$, so $0.72 = OP \times 0.6$. Hence $OP = 1.2\,\text{m}$.

2 Set up the target area

The outer sector has radius $x + 1.2$ and the inner sector has radius 1.2, with angle 0.6 radians.

3 Use sector area

$\frac{1}{2}(x+1.2)^2(0.6) - \frac{1}{2}(1.2)^2(0.6) = 90$.

4 **Simplify to the given quadratic**

$0.3\left((x+1.2)^2-1.44\right)=90$, so $0.3x^2+0.72x-90=0$. Multiplying by $\dfrac{50}{3}$ gives $5x^2+12x-1500=0$.

5 **Find the positive value of x**

$$x=\frac{-12+\sqrt{12^2+4(5)(1500)}}{10}=\frac{-6+4\sqrt{471}}{5}\approx 16.16.$$

6 **Find the perimeter**

$\text{Perimeter}=PQ+SR+\text{arc }QR+\text{arc }PS=x+x+0.6(x+1.2)+0.72\approx 43.46\text{ m}$.

7 **Round to the nearest metre**

The total perimeter is $\mathbf{43\text{ m}}$ to the nearest metre.

FINAL ANSWER

$(a)\ OP=1.2\text{ m}.\quad (b)\ 5x^2+12x-1500=0.\quad (c)\ 43\text{ m}.$

WMA11/01 JANUARY 2024

Question 8

9 marks | Radians

8.

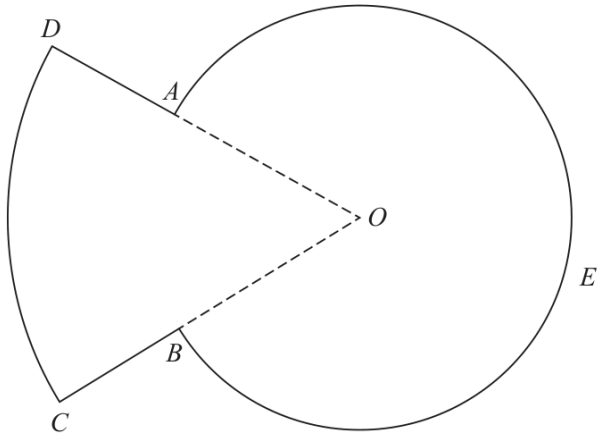


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O.

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9\,\text{m}$
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03\,\text{m}$ to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2024

1 Find the outer radius

For sector DOC, arc length $CD = 9$ and angle 0.8 radians, so $OD = \dfrac{9}{0.8} = 11.25\ \text{m}$.

2 Use the ratio DA to AO

$DA : AO = 3 : 5$, so $OD = DA + AO$ represents 8 parts. Hence $AO = \dfrac{5}{8}(11.25) = 7.03125\ \text{m} \approx 7.03\ \text{m}$.

3 Find the straight side lengths

$DA = CB = OD - AO = 11.25 - 7.03125 = 4.21875\ \text{m}$.

4 **Find the major arc**

The major sector $AOBEA$ has angle $2\pi - 0.8$, so arc $AEB = 7.03125(2\pi - 0.8)$.

5 **Calculate the perimeter**

$\text{Perimeter} = 9 + 2(4.21875) + 7.03125(2\pi - 0.8) \approx 56.0\ \text{m}$.

6 **Calculate the total area**

$\text{Area} = \frac{1}{2}(11.25)^2(0.8) + \frac{1}{2}(7.03125)^2(2\pi - 0.8) \approx 186\ \text{m}^2$ to the nearest whole number.

FINAL ANSWER

$(a)\ AO = 7.03\ \text{m}.$ $(b)\ 56.0\ \text{m}.$ $(c)\ 186\ \text{m}^2.$

Question 8

13 marks | Radians

8.

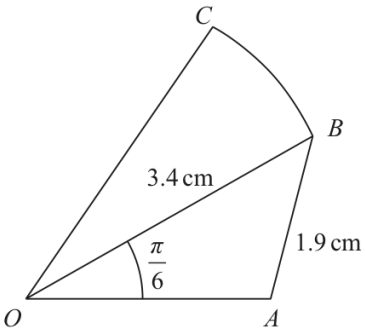


Figure 1

Figure 1 shows a sketch of a design for a badge.
The design consists of a triangle OAB joined to a sector OBC of a circle with centre O
In the design

- $OB = 3.4\,\text{cm}$
- $AB = 1.9\,\text{cm}$
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB, giving your answer in radians to 4 significant figures,

(3)

(b) find the area of triangle OAB, in cm^2, giving your answer to 3 significant figures.

(2)

Given that the ratio of the area of sector OBC to the area of triangle OAB is $3:2$

(c) show that angle BOC is 0.462 radians to 3 significant figures.

(3)

(d) Hence find the perimeter of the badge, in cm, to the nearest integer.

(5)

(Total for Question 8 is 13 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2025

1 Use the sine rule

In triangle OAB, $AB = 1.9$, $OB = 3.4$ and $\angle AOB = \frac{\pi}{6}$.

2 Find sine of angle OAB

$\frac{\sin \angle OAB}{3.4} = \frac{\sin(\pi/6)}{1.9}$, so $\sin \angle OAB = \frac{17}{19}$.

3 Choose the obtuse angle

Since $\angle OAB > \frac{\pi}{2}$, $\angle OAB = \pi - \sin^{-1}\left(\frac{17}{19}\right) \approx 2.034$ radians.

4 **Find OA**

Using the sine rule again, $OA \approx 2.096$ cm.

5 **Find the triangle area**

$\text{Area } OAB = \frac{1}{2}(OA)(OB)\sin(\pi/6) \approx 1.78 \text{ cm}^2$.

6 **Use the area ratio**

$\text{Area sector } OBC : \text{Area triangle } OAB = 3 : 2$, so sector area $= \frac{3}{2}(1.78156\ldots)$.

7 **Find angle BOC**

$\frac{1}{2}(3.4)^2\theta = \frac{3}{2}(1.78156\ldots)$, so $\theta \approx 0.462$ radians.

8 **Find the perimeter**

The perimeter is $OA + AB + OC + \text{arc } BC = 2.096 + 1.9 + 3.4 + 3.4(0.46234\ldots) \approx 8.968$ cm.

9 **Round**

The perimeter is 9 cm to the nearest integer.

FINAL ANSWER

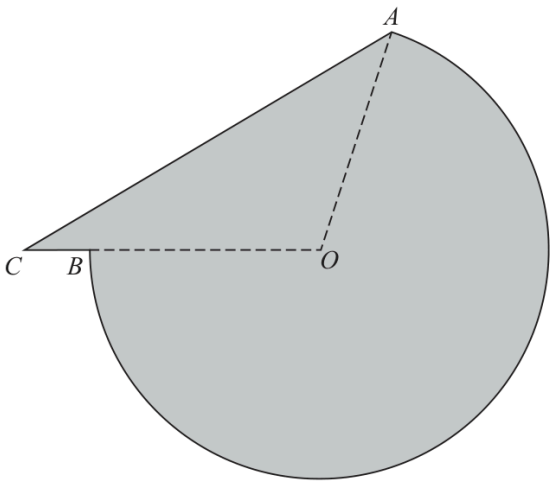
$(a)\ \angle OAB = 2.034 \text{ radians.}$ $(b)\ 1.78 \text{ cm}^2.$ $(c)\ \angle BOC = 0.462 \text{ radians.}$ $(d)\ 9 \text{ cm}.$

WMA11/01 MAY/JUNE 2025

Question 6

9 marks | Radians

6.



Not to scale

Figure 2

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC.

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)

(b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)

(c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)

(Total for Question 6 is 9 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2025

1 Set up triangle AOC

$AO = 15$, $OC = OB + BC = 17$, and $\angle ACO = 0.6$.

2 Use the sine rule

$\frac{\sin \angle CAO}{17} = \frac{\sin 0.6}{15}$, so $\angle CAO \approx 0.6944$ radians.

3 Show angle COA

$\angle COA = \pi - 0.6 - 0.6944... = 1.847...$, so $\angle COA = 1.847$ radians to 3 decimal places.

4 **Major sector angle**

The shaded sector is the major sector AOB, so its angle is $2\pi - 1.847\ldots$.

5 **Sector area**

$\text{Sector area} = \dfrac{1}{2}(15)^2(2\pi - 1.847\ldots) = 499.15\ldots$.

6 **Triangle area**

$\text{Area } AOC = \dfrac{1}{2}(15)(17)\sin(1.847\ldots) = 122.73\ldots$.

7 **Total area**

$499.15\ldots + 122.73\ldots = 621.88\ldots$, so the total area is $622\ \text{m}^2$ to 3 significant figures.

8 **Find AC**

By the cosine rule, $AC^2 = 15^2 + 17^2 - 2(15)(17)\cos(1.847\ldots)$, so $AC \approx 27.57$ m.

9 **Major arc length**

$\text{arc } AB = 15(2\pi - 1.847\ldots) = 66.54\ldots$ m.

10 **Perimeter**

$66.54\ldots + 27.57\ldots = 94.11\ldots$, so the perimeter is $94.1\ \text{m}$.

✓ **FINAL ANSWER**

$(a)\ \angle COA = 1.847\ \text{rad}.$ $(b)\ 622\ \text{m}^2.$ $(c)\ 94.1\ \text{m}.$

WMA11/01 OCTOBER 2025

Question 7

9 marks | Radians

7.

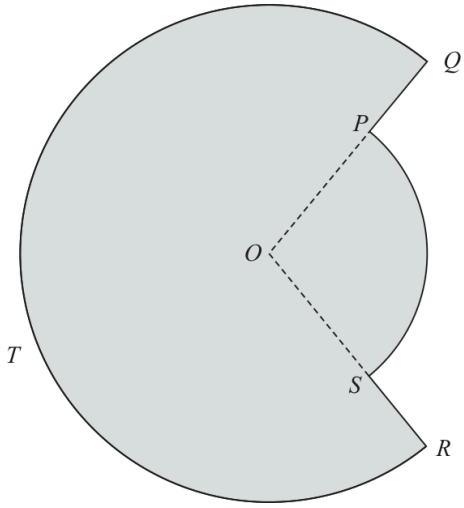


Figure 2

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O.

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8$ m

(a) show that, to 3 significant figures, $OQ = 8.83$ m

(3)

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

(Total for Question 7 is 9 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2025

1 Find OP from the small sector

Let $OP = r$. The area of sector POS is 30, so $\frac{1}{2}r^2(1.65) = 30$.

2 Calculate OP

$r^2 = \frac{60}{1.65}$, so $r = 6.030...$ m.

3 Show OQ

$OQ = OP + PQ = 6.030... + 2.8 = 8.830...$, so $OQ = 8.83$ m to 3 significant figures.

4 **Major sector angle**

The major sector $QORTQ$ has angle $2\pi - 1.65$ radians.

5 **Total area**

$\text{Area} = 30 + \frac{1}{2}(8.830\ldots)^2(2\pi - 1.65) = 210.6\ldots$.

6 **Round area**

The total surface area is $211\ \text{m}^2$ to the nearest integer.

7 **Arc lengths**

The inner arc PS has length $6.030\ldots(1.65) = 9.95\ldots$, and the outer major arc has length $8.830\ldots(2\pi - 1.65) = 40.91\ldots$.

8 **Add straight sides**

The straight sides PQ and SR each have length 2.8 m.

9 **Perimeter**

$9.95\ldots + 40.91\ldots + 2.8 + 2.8 = 56.46\ldots$, so the perimeter is $56\ \text{m}$ to 2 significant figures.

✓ **FINAL ANSWER**

$(a)\ OQ = 8.83\ \text{m}$. $(b)\ 211\ \text{m}^2$. $(c)\ 56\ \text{m}$.

WMA11/01 MAY/JUNE 2026

Question 7

7 marks | Radians

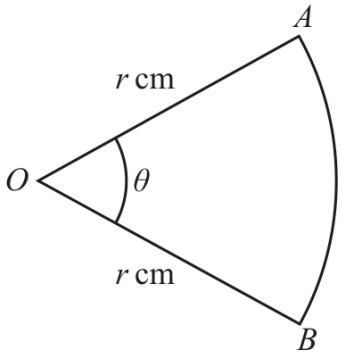


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a sector AOB of a circle, centre O.

The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is $1\,\text{cm}^2$
- the **perimeter** of sector AOB is $8\,\text{cm}$

(a) write down two different equations involving r and θ.

(2)

Given that $r > 2$

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ.
Write each answer in the form $a + b\sqrt{c}$ where a, b and c are integers.

(*Solutions relying on calculator technology are not acceptable.*)

(5)

(Total for Question 7 is 7 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2026

1 **(a) Translate the area and perimeter**

For a sector of radius r and angle θ radians,

$$\frac{1}{2}r^2\theta = 1.$$

Its perimeter is two radii plus the arc length, so

$$2r + r\theta = 8.$$

2 **(b) Eliminate the angle**

From the area equation,

$$\theta = \frac{2}{r^2}.$$

Substitute this into the perimeter equation:

$$2r + r\left(\frac{2}{r^2}\right) = 8.$$

Therefore

$$2r + \frac{2}{r} = 8,$$

and, since $r > 0$,

$$r^2 - 4r + 1 = 0.$$

---

3 **(b) Solve for the radius and apply the condition**

Complete the square:

$$(r-2)^2 = 3,$$

so

$$r = 2 \pm \sqrt{3}.$$

The condition $r > 2$ selects

$$r = 2 + \sqrt{3}.$$

---

4 **(b) Find the exact angle**

Using $\theta = \frac{2}{r^2}$,

$$\theta = \frac{2}{(2+\sqrt{3})^2}.$$

Since $\frac{1}{2+\sqrt{3}} = 2 - \sqrt{3}$,

$$\theta = 2(2-\sqrt{3})^2 = 2(7 - 4\sqrt{3}) = 14 - 8\sqrt{3}.$$

FINAL ANSWER

$(a)\ \frac{1}{2}r^2\theta = 1, \quad 2r + r\theta = 8. \qquad (b)\ r = 2 + \sqrt{3}\ \text{cm}, \quad \theta = 14 - 8\sqrt{3}\ \text{radians}.$

TOPIC

Trigonometric Functions

Every past-paper question on this topic, October 2019 to May/June 2026.

Question 9

4 marks | Trigonometric Functions

9.

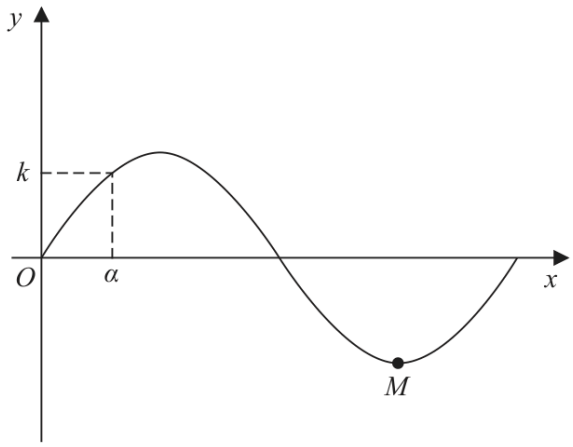


Figure 5

Figure 5 shows a sketch of part of the curve C with equation $y = \sin\left(\frac{x}{12}\right)$, where x is measured in radians. The point M shown in Figure 5 is a minimum point on C.

(a) State the period of C.

(1)

(b) State the coordinates of M.

(1)

The smallest positive solution of the equation $\sin\left(\frac{x}{12}\right) = k$, where k is a constant, is α.

Find, in terms of α,

(c) (i) the negative solution of the equation $\sin\left(\frac{x}{12}\right) = k$ that is closest to zero,

(ii) the smallest positive solution of the equation $\cos\left(\frac{x}{12}\right) = k$.

(2)

(Total 4 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2019

1 Use the scale factor inside the sine

For $y = \sin\left(\frac{x}{12}\right)$, one full sine cycle needs $\frac{x}{12}$ to increase by 2π, so the period is 24π.

2 Locate the minimum point

The sine function is -1 when its angle is $\frac{3\pi}{2}$. Therefore $\frac{x}{12} = \frac{3\pi}{2}$, so $x = 18\pi$ and $M = (18\pi, -1)$.

3 Use sine symmetry

If the smallest positive solution of $\sin\left(\frac{x}{12}\right) = k$ is α, then the closest negative solution is one half-period before the opposite sine position: $-12\pi - \alpha$.

4 **Convert to cosine**

Since $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$, the corresponding smallest positive cosine solution is $6\pi - \alpha$.

FINAL ANSWER

$(a)\ 24\pi.\quad (b)\ (18\pi, -1).\quad (c)(i)\ -12\pi - \alpha.\quad (ii)\ 6\pi - \alpha.$

WMA11/01 JANUARY 2020

Question 7

5 marks | Trigonometric Functions

7.

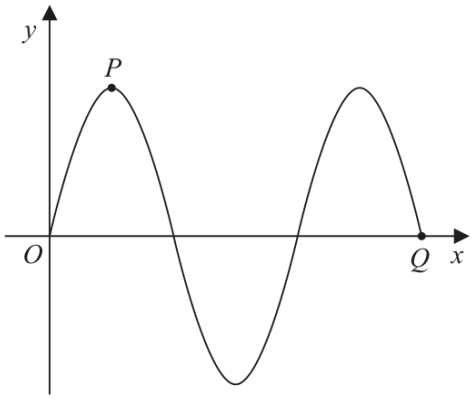


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3\sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

(i) the coordinates of P,

(ii) the coordinates of Q.

(3)

A different curve C_2 has equation $y = 3\sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R.

(2)

(Total 5 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2020

1 Read point P

The first maximum of $y = 3\sin x$ occurs at $x = 90°$, with $y = 3$. So $P = (90°, 3)$.

2 Read Q from the next shown zero

After the second maximum, the curve returns to the x-axis at $x = 540°$. So $Q = (540°, 0)$.

3 Find the vertical shift

For C_2, $y = 3\sin x + k$. Its maximum is $3 + k = 10$, so $k = 7$.

4 **Find the minimum point R**

The minimum value is $-3+7=4$. The smallest positive x-coordinate for a sine minimum is $270°$, so $R=(270°, 4)$.

FINAL ANSWER

$(a)(i)\ P=(90°, 3).$ $(a)(ii)\ Q=(540°, 0).$ $(b)\ R=(270°, 4).$

WMA11/01 MAY/JUNE 2021

Question 9

7 marks | Trigonometric Functions

9.

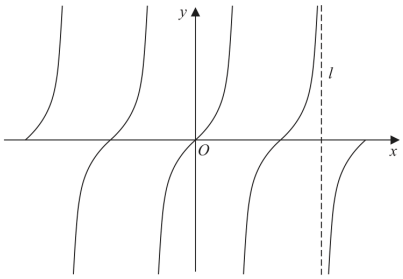


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

The line l, shown in Figure 4, is an asymptote to $y = \tan x$

(a) State an equation for l.

(1)

A copy of Figure 4, labelled Diagram 1, is shown on the next page.

(b) (i) On Diagram 1, sketch the curve with equation

$$y = \frac{1}{x} + 1 \qquad -2\pi \leqslant x \leqslant 2\pi$$

stating the equation of the horizontal asymptote of this curve.

(ii) Hence, **giving a reason**, state the number of solutions of the equation

$$\tan x = \frac{1}{x} + 1$$

in the region $-2\pi \leqslant x \leqslant 2\pi$

(4)

(c) State the number of solutions of the equation $\tan x = \frac{1}{x} + 1$ in the region

(i) $0 \leqslant x \leqslant 40\pi$

(ii) $-10\pi \leqslant x \leqslant \frac{5}{2}\pi$

(2)

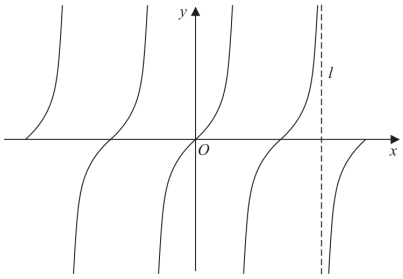


Diagram 1

(Total 7 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2021

1 **State the shown tangent asymptote**

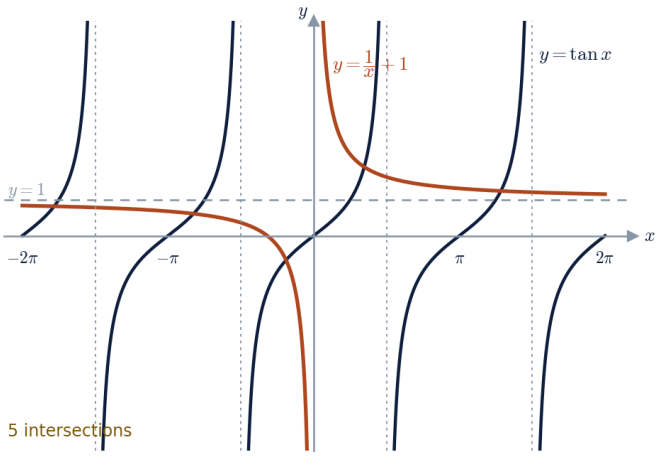
The shown dashed asymptote is the right-hand positive asymptote before 2π, so l has equation $x = \frac{3\pi}{2}$.

2 **Describe the reciprocal graph**

For $y = \frac{1}{x} + 1$, the vertical asymptote is $x = 0$ and the horizontal asymptote is $y = 1$. The right branch lies above $y = 1$ and the left branch lies below $y = 1$.

3 Count intersections on -2pi to 2pi

The number of solutions of $\tan x = \dfrac{1}{x} + 1$ is the number of intersections of the two graphs. From the sketch on $-2\pi \le x \le 2\pi$, there are 5 intersections.



4 Count first interval solutions

From the repeated tangent branches and the reciprocal curve, there is one intersection in each positive interval between consecutive tangent asymptotes. From 0 to 40π, this gives 40 solutions.

5 Count second interval solutions

Extending the same intersection pattern across the stated interval gives 14 intersections, so there are 14 solutions.

FINAL ANSWER

$(a)\ x = \dfrac{3\pi}{2}.$ $(b)(i)\ y = 1$ **is the horizontal asymptote.** $(b)(ii)\ 5.$ $(c)(i)\ 40.$ $(c)(ii)\ 14.$

WMA11/01 JANUARY 2022

Question 9

4 marks | Trigonometric Functions

9.

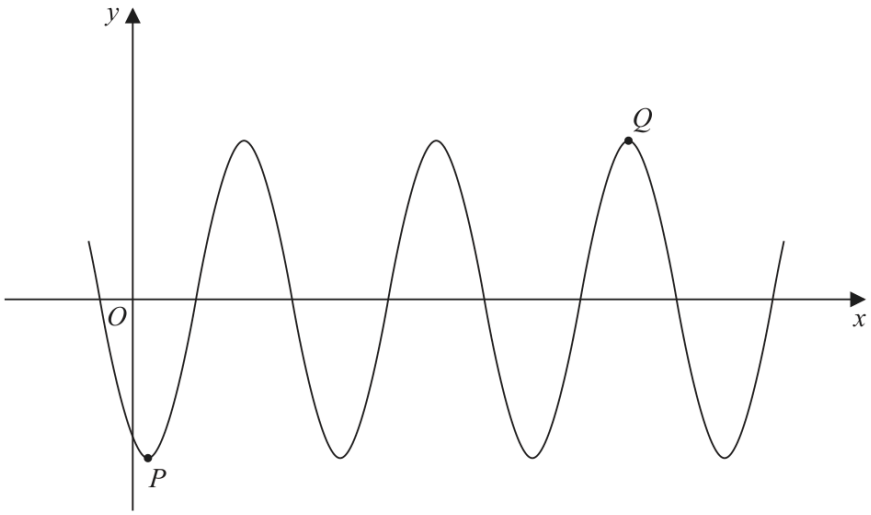


Figure 4

Figure 4 shows part of the curve with equation

$$y = A\cos(x - 30)°$$

where A is a constant.

The point P is a minimum point on the curve and has coordinates (30, –3) as shown in Figure 4.

(a) Write down the value of A.

(1)

The point Q is shown in Figure 4 and is a maximum point.

(b) Find the coordinates of Q.

(3)

(Total 4 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2022

1 Use the minimum point

At $P(30, -3)$, $x - 30 = 0$, so $\cos 0° = 1$. Therefore $A = -3$.

2 Find maximum points

Since $A = -3$, maximum points occur when $\cos(x - 30)° = -1$.

3 Use the shown maximum Q

$x - 30 = 180 + 360n$, so $x = 210 + 360n$. The maximum shown at Q is two full periods after the first maximum, so $x = 210 + 720 = 930$.

4 **Write Q**

The maximum y-value is 3, hence $Q = (930, 3)$.

FINAL ANSWER

$(a)\ A = -3. \quad (b)\ Q = (930, 3)$.

WMA11/01 MAY/JUNE 2022

Question 9

8 marks | Trigonometric Functions

9.

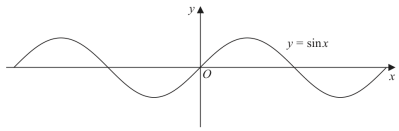


Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin\alpha = p$, where $0 < \alpha < 90°$

(a) state, in terms of p, the value of

(i) $2\sin(180° - \alpha)$

(ii) $\sin(\alpha - 180°)$

(iii) $3 + \sin(180° + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α, the x coordinates of any points in the interval $0 < x < 180°$ where

$$\sin 2x = p$$

(3)

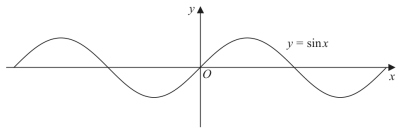


Diagram 1

(Total 8 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2022

1 **Use sine symmetry**

$\sin(180° - \alpha) = \sin\alpha = p$, so $2\sin(180° - \alpha) = 2p$.

2 **Use odd symmetry**

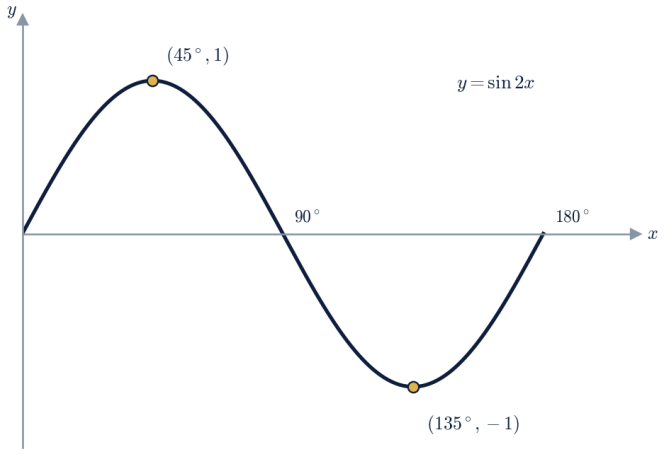
$\sin(\alpha - 180°) = -\sin(180° - \alpha) = -p$.

3 **Use the third quadrant identity**

$\sin(180° + \alpha) = -\sin\alpha = -p$, so $3 + \sin(180° + \alpha) = 3 - p$.

4 **Sketch y = sin 2x**

The graph has amplitude 1 and period $180°$. It crosses the x-axis at $0°, 90°, 180°$, has a maximum at $45°$, and a minimum at $135°$.



5 **Solve sin 2x = p**

Since $\sin\alpha = p$ and $0 < 2x < 360°$, the possible values are $2x = \alpha$ or $2x = 180° - \alpha$.

6 **Find x**

Therefore $x = \frac{\alpha}{2}$ or $x = 90° - \frac{\alpha}{2}$.

FINAL ANSWER

$(a)(i)\ 2p,\ (ii)\ -p,\ (iii)\ 3-p.\quad (c)\ x = \frac{\alpha}{2},\ 90° - \frac{\alpha}{2}.$

WMA11/01 JANUARY 2023

Question 9

4 marks | Trigonometric Functions

9.

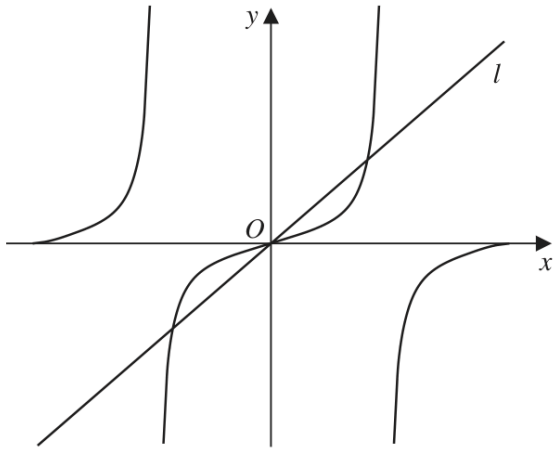


Figure 3

Figure 3 shows a sketch of

- the curve with equation $y = \tan x$
- the straight line l with equation $y = \pi x$

in the interval $-\pi < x < \pi$

(a) State the period of $\tan x$

(1)

(b) Write down the number of roots of the equation

(i) $\tan x = (\pi + 2)x$ in the interval $-\pi < x < \pi$

(1)

(ii) $\tan x = \pi x$ in the interval $-2\pi < x < 2\pi$

(1)

(iii) $\tan x = \pi x$ in the interval $-100\pi < x < 100\pi$

(1)

(Total for Question 9 is 4 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2023

1 State the period

The period of $\tan x$ is π.

2 Count roots for the steeper line

In $-\pi < x < \pi$, the line $y = (\pi + 2)x$ meets $y = \tan x$ once at the origin and once on each side of the origin, giving 3 roots.

3 Extend to -2pi < x < 2pi

For $y = \pi x$, the interval $-2\pi < x < 2\pi$ has the 3 central roots plus one further root on each neighbouring branch, giving 5 roots.

4 **Extend to -100pi < x < 100pi**

There are 99 positive outer roots, 99 negative outer roots, and 3 central roots, so the total is $99 + 99 + 3 = 201$.

✓ **FINAL ANSWER**

$(a)\ \pi.\quad (b)(i)\ 3,\ (ii)\ 5,\ (iii)\ 201.$

WMA11/01 MAY/JUNE 2023

Question 9

9 marks | Trigonometric Functions

9. (i)

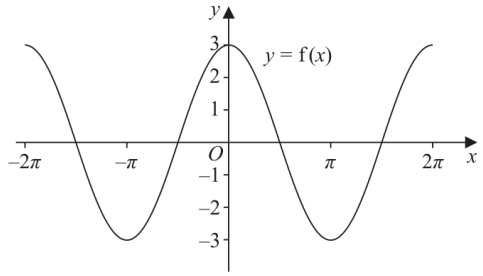


Figure 3

Figure 3 shows part of the graph of the trigonometric function with equation $y = f(x)$

(a) Write down an expression for $f(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = f\left(x + \frac{\pi}{4}\right)$

Show clearly the coordinates of all the points where the curve intersects the coordinate axes.

(3)

(ii)

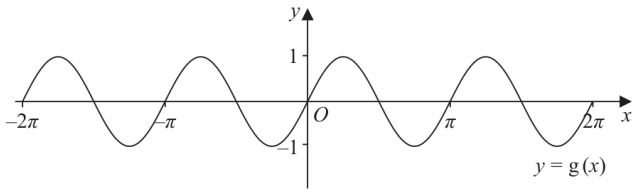


Figure 4

Figure 4 shows part of the graph of the trigonometric function with equation $y = g(x)$

(a) Write down an expression for $g(x)$

(2)

On a separate diagram,

(b) sketch, for $-2\pi < x < 2\pi$, the graph of the curve with equation $y = g(x) - 2$

Show clearly the coordinates of the y intercept.

(2)

(Total for Question 9 is 9 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2023

1 Identify f(x)

The first graph has amplitude 3, period 2π, and maximum at $x = 0$, so $f(x) = 3\cos x$.

2 Transform f

$y = f\left(x + \frac{\pi}{4}\right) = 3\cos\left(x + \frac{\pi}{4}\right)$, a shift left by $\frac{\pi}{4}$.

3 Find the x-intercepts

$\cos\left(x + \frac{\pi}{4}\right) = 0$ gives $x = \frac{\pi}{4} + n\pi$. In $-2\pi < x < 2\pi$, these are $-\frac{7\pi}{4}$, $-\frac{3\pi}{4}$, $\frac{\pi}{4}$, and $\frac{5\pi}{4}$.

4 **Find the y-intercept**

At $x = 0$, $y = 3\cos\frac{\pi}{4} = \frac{3\sqrt{2}}{2}$.

5 **Identify g(x)**

The second graph has amplitude 1, period π, and passes through the origin with positive gradient, so $g(x) = \sin 2x$.

6 **Transform g**

$y = g(x) - 2 = \sin 2x - 2$, so the graph is shifted down by 2. Its y-intercept is $(0, -2)$.

✓ **FINAL ANSWER**

$(i)(a)\ f(x) = 3\cos x.$ $(i)(b)$ x-intercepts $-\frac{7\pi}{4}, -\frac{3\pi}{4}, \frac{\pi}{4}, \frac{5\pi}{4}$ and y-intercept $\frac{3\sqrt{2}}{2}$. $(ii)(a)\ g(x) = \sin 2x.$ $(ii)(b)$ y-intercept $(0, -2)$.

Question 10

6 marks | Trigonometric Functions

10.

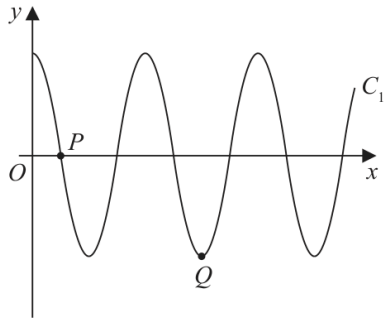


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 3\cos\left(\frac{x}{n}\right)^\circ \qquad x \geqslant 0$$

where n is a constant.

The curve C_1 cuts the positive x-axis for the first time at point $P(270, 0)$, as shown in Figure 4.

(a) (i) State the value of n

(ii) State the period of C_1

(2)

The point Q, shown in Figure 4, is a minimum point of C_1

(b) State the coordinates of Q.

(2)

The curve C_2 has equation $y = 2\sin x^\circ + k$, where k is a constant.

The point $R\left(a, \frac{12}{5}\right)$ and the point $S\left(-a, -\frac{3}{5}\right)$, both lie on C_2

Given that a is a constant less than 90

(c) find the value of k.

(2)

(Total for Question 10 is 6 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2023

1 Use the first positive x-intercept

For $y = 3\cos\left(\frac{x}{n}\right)^\circ$, the first positive zero occurs when $\frac{x}{n} = 90°$.

2 Find n

Since $P = (270, 0)$, $\frac{270}{n} = 90$, so $n = 3$.

3 Find the period

The period is $360n = 360(3) = 1080°$.

4 **Read the shown minimum point**

A minimum occurs when $\frac{x}{3} = 180° + 360°k$. The point Q shown is the next displayed minimum after one full period, so $x = 1620$ and $y = -3$.

5 **Use the two points on C2**

For $y = 2\sin x° + k$, the points give $2\sin a + k = \frac{12}{5}$ and $-2\sin a + k = -\frac{3}{5}$.

6 **Add the equations**

Adding gives $2k = \frac{12}{5} - \frac{3}{5} = \frac{9}{5}$, so $k = \frac{9}{10}$.

✓ **FINAL ANSWER**

$(a)(i)\ n = 3.$ $(a)(ii)\ 1080°.$ $(b)\ Q = (1620, -3).$ $(c)\ k = \frac{9}{10}.$

WMA11/01 JANUARY 2024

Question 6

6 marks | Trigonometric Functions

6.

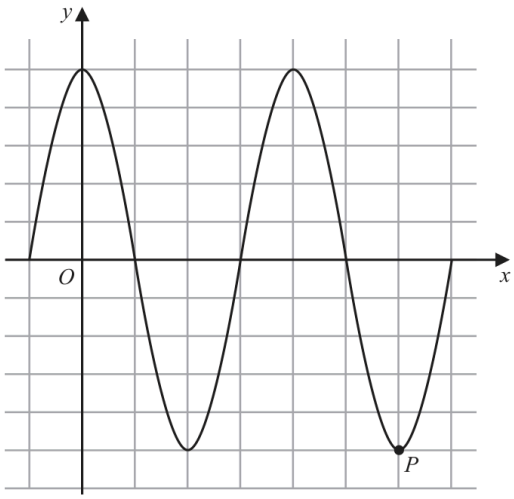


Figure 2

Figure 2 shows a plot of part of the curve C_1 with equation

$$y = 5\cos x$$

with x being measured in degrees.

The point P, shown in Figure 2, is a minimum point on C_1

(a) State the coordinates of P

(2)

The point Q lies on a different curve C_2

Given that point Q

- is a maximum point on the curve
- is the maximum point with the **smallest** x coordinate, $x > 0$

(b) find the coordinates of Q when

(i) C_2 has equation $y = 5\cos x - 2$

(ii) C_2 has equation $y = -5\cos x$

(4)

(Total for Question 6 is 6 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2024

1 Use the cosine graph

For $y = 5\cos x$, the minimum value is -5 and it occurs at $x = 180° + 360°n$.

2 Read P from the displayed minimum

The plotted point P is the displayed minimum at $x = 540°$, so $P = (540°, -5)$.

3 Translate the graph down

For $y = 5\cos x - 2$, the maximum value is $5 - 2 = 3$. The smallest positive x-coordinate for a maximum shown is $x = 360°$, so $Q = (360°, 3)$.

4 **Reflect in the x-axis**

For $y = -5\cos x$, a maximum occurs where $\cos x = -1$, first at $x = 180°$. The maximum value is 5, so $Q = (180°, 5)$.

FINAL ANSWER

$(a)\ P = (540°, -5).$ $(b)(i)\ Q = (360°, 3).$ $(b)(ii)\ Q = (180°, 5).$

WMA11/01 MAY/JUNE 2024

Question 11

6 marks | Trigonometric Functions

11.

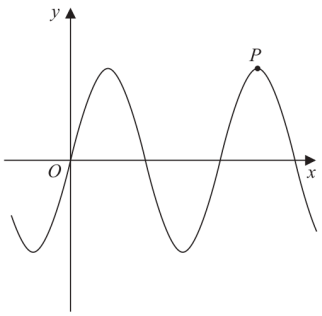


Figure 4

Figure 4 shows a sketch of part of the curve C_1 with equation

$$y = 12\sin x$$

where x is measured in radians.

The point P shown in Figure 4 is a maximum point on C_1

(a) Find the coordinates of P.

(2)

The curve C_2 has equation

$$y = 12\sin x + k$$

where k is a constant.

Given that the **maximum** value of y on C_2 is 3

(b) find the coordinates of the **minimum** point on C_2 which has the **smallest** positive x coordinate.

(2)

The curve C_3 has equation

$$y = 12\sin(x + B)$$

where B is a positive constant.

Given that $\left(\frac{\pi}{4}, A\right)$, where A is a constant, is the **minimum** point on C_3 which has the **smallest** positive x coordinate,

(c) find

(i) the value of A,

(ii) the smallest possible value of B.

(2)

(Total for Question 11 is 6 marks)

TOTAL FOR PAPER IS 75 MARKS

WORKED SOLUTION

Question 11 · WMA11/01 May/June 2024

1 Use maxima of sine

For $y = 12\sin x$, maxima occur when $x = \frac{\pi}{2} + 2\pi n$, with value 12.

2 Read point P

The maximum point P shown is the second positive maximum, so $P = (\frac{5\pi}{2}, 12)$.

3 Find k for C2

For $y = 12\sin x + k$, the maximum value is $12 + k$. Given the maximum is 3, $k = -9$.

4 **Find the smallest positive minimum on C2**

The minimum value is $-12-9=-21$, and it first occurs at $x=\dfrac{3\pi}{2}$. So the point is $(\dfrac{3\pi}{2},-21)$.

5 **Find A for C3**

For $y=12\sin(x+B)$, any minimum has value -12, so $A=-12$.

6 **Find the smallest positive B**

At the minimum $x=\dfrac{\pi}{4}$, so $\dfrac{\pi}{4}+B=\dfrac{3\pi}{2}$. Hence $B=\dfrac{5\pi}{4}$.

✓ **FINAL ANSWER**

$(a)\ P=(\dfrac{5\pi}{2},12).\quad (b)\ (\dfrac{3\pi}{2},-21).\quad (c)(i)\ A=-12.\quad (c)(ii)\ B=\dfrac{5\pi}{4}.$

WMA11/01 OCTOBER 2024

Question 7

8 marks | Trigonometric Functions

7.

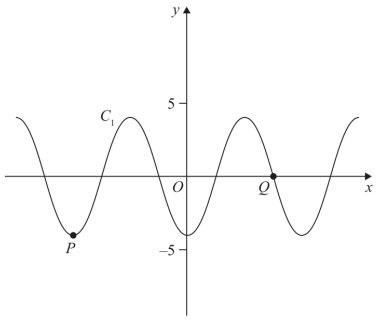


Figure 3

Figure 3 shows a plot of part of the curve C_1 with equation

$$y = -4\cos x$$

where x is measured in radians.

Points P and Q lie on the curve and are shown in Figure 3.

(a) State

(i) the coordinates of P

(ii) the coordinates of Q

(3)

The curve C_2 has equation $y = -4\cos x + k$ where x is measured in radians and k is a constant.

Given that C_2 has a maximum y value of 11

(b) (i) state the value of k

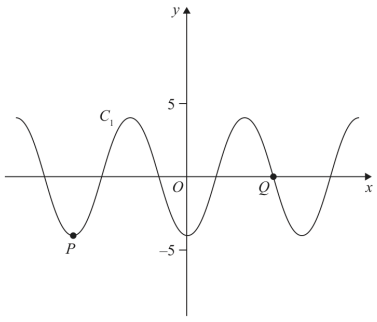
(ii) state the coordinates of the minimum point on C_2 with the smallest positive x coordinate.

(3)

On the opposite page there is a copy of Figure 3 labelled Diagram 1.

(c) Using Diagram 1, state the number of solutions of the equation

$$-4\cos x = 5 - \frac{10}{\pi}x$$



(Total for Question 7 is 8 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2024

1 **Use y equals negative 4 cos x**

For $y = -4\cos x$, minimum points occur at $x = 2\pi n$ with value -4.

2 **Read P**

The displayed minimum to the left of the y-axis is $P = (-2\pi, -4)$.

3 **Read Q**

The point Q is the positive x-intercept after the maximum at $x = \pi$, so $Q = (\frac{3\pi}{2}, 0)$.

4 **Find k for C2**

For $y = -4\cos x + k$, the maximum value is $4 + k$. Given the maximum is 11, $k = 7$.

5 **Find the smallest positive minimum on C2**

The minimum value is $-4 + 7 = 3$. The first positive minimum is at $x = 2\pi$, so the point is $(2\pi, 3)$.

6 **Interpret the equation graphically**

$-4\cos x = 5 - \frac{10}{\pi}x$ is the intersection of the cosine curve with the straight line $y = 5 - \frac{10}{\pi}x$.

7 **Count intersections**

The line passes through $(0, 5)$ and meets the displayed curve once, at the x-axis point, so there is 1 solution.

✓ **FINAL ANSWER**

$(a)(i)\ P = (-2\pi, -4).$ $(a)(ii)\ Q = (\frac{3\pi}{2}, 0).$ $(b)(i)\ k = 7.$ $(b)(ii)\ (2\pi, 3).$ $(c)\ 1 \text{ solution}.$

Question 9

13 marks | Trigonometric Functions

9.

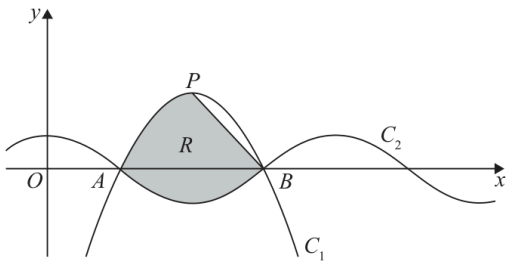


Figure 2

(a) Express $6x - \frac{27}{4} - x^2$ in the form $a + b(x + c)^2$ where a, b and c are constants to be found.

(3)

Figure 2 shows part of a sketch of curve C_1 with equation

$$y = 6x - \frac{27}{4} - x^2$$

Given that the point P is the maximum point on C_1

(b) state the coordinates of P

(2)

Figure 2 also shows part of a sketch of curve C_2 with equation

$$y = \cos(kx)$$

where k is a constant and x is measured in radians.

Given that C_1 and C_2 intersect on the x-axis at point A and at point B, as shown in Figure 2,

(c) (i) state the x coordinate of B

(ii) state the value of k

(iii) state the period of C_2

(3)

The line segment L joins P and B.
The region R, shown shaded in Figure 2, is bounded by L, C_1 and C_2

(d) Use inequalities to define R.

(5)

(Total for Question 9 is 13 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2025

1 **Complete the square**

$$6x - \frac{27}{4} - x^2 = -(x^2 - 6x) - \frac{27}{4} = -(x-3)^2 + 9 - \frac{27}{4}.$$

2 **State the completed-square form**

$$6x - \frac{27}{4} - x^2 = \frac{9}{4} - (x-3)^2.$$

3 **Find the maximum point**

The maximum point is $P = (3, \frac{9}{4})$.

4 **Find the x-intercepts of C1**

$\frac{9}{4} - (x-3)^2 = 0$ gives $x = 3 \pm \frac{3}{2}$. Hence A has $x = \frac{3}{2}$ and B has $x = \frac{9}{2}$.

5 **Use consecutive zeros of cosine**

The distance between consecutive zeros of $\cos(kx)$ is $\frac{\pi}{k}$.

6 **Find k and the period**

Since $x_B - x_A = \frac{9}{2} - \frac{3}{2} = 3$, $\frac{\pi}{k} = 3$, so $k = \frac{\pi}{3}$. The period is $\frac{2\pi}{k} = 6$.

7 **Find the line through P and B**

The line through $P(3, \frac{9}{4})$ and $B(\frac{9}{2}, 0)$ has gradient $-\frac{3}{2}$, so L is $y = -\frac{3}{2}x + \frac{27}{4}$.

8 **Define the x-range**

The region lies between A and B, so $\frac{3}{2} \le x \le \frac{9}{2}$.

9 **Define the lower boundary**

The lower boundary is C_2, so $y \ge \cos\left(\frac{\pi}{3}x\right)$.

10 **Define the upper boundaries**

The region is below both C_1 and the line segment L, so $y \le \frac{9}{4} - (x-3)^2$ and $y \le -\frac{3}{2}x + \frac{27}{4}$.

FINAL ANSWER

$(a)\ \frac{9}{4} - (x-3)^2$. $(b)\ P = (3, \frac{9}{4})$. $(c)(i)\ x_B = \frac{9}{2}$. $(c)(ii)\ k = \frac{\pi}{3}$. $(c)(iii)\ 6$. $(d)\ \frac{3}{2} \le x \le \frac{9}{2},\ y \ge \cos(\frac{\pi}{3}x),\ y \le \frac{9}{4} - (x-3)^2,\ y \le -\frac{3}{2}x + \frac{27}{4}$

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Question 9

9 marks | Trigonometric Functions

9.

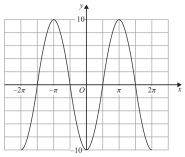


Figure 4

Figure 4 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$.

(a) Write down an expression for $f(x)$.

(2)

Copies of Figure 4 (labelled Diagram 1 and Diagram 2) can be found on the following pages.

(b) (i) On Diagram 1 sketch a graph of the curve with equation

$$y = 10 - x^2$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = 10 - x^2 \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

(3)

(c) (i) On Diagram 2 sketch a graph of the curve with equation

$$y = \tan x \qquad -2\pi \leqslant x \leqslant 2\pi$$

(ii) Hence find the **number** of solutions of the equation

$$f(x) = \tan x \quad \text{in the interval} \quad -100\pi \leqslant x \leqslant 100\pi$$

giving a reason for your answer.

(4)

(a) _____

(b)(i)

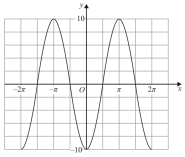
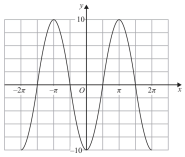


Diagram 1



Copy of Diagram 1

Only use this copy if you need to redraw your graph.

(b)(ii) _____

(c)(i)

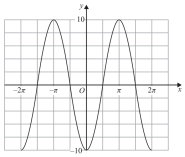
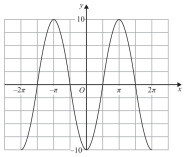


Diagram 2



Copy of Diagram 2

Only use this copy if you need to redraw your graph.

(c)(ii) _____

_____

(Total for Question 9 is 9 marks)

WORKED SOLUTION

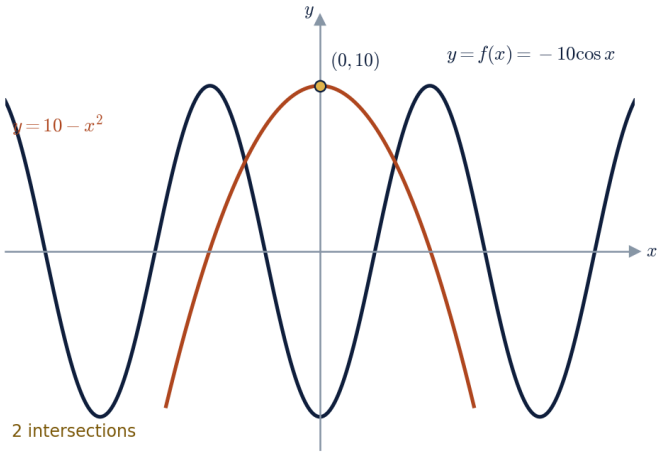
Question 9 · WMA11/01 January 2026

1 **Identify f(x)**

The graph has amplitude 10, period 2π, and value -10 at $x = 0$, so $f(x) = -10\cos x$.

2 **Sketch y equals 10 minus x squared**

The graph $y = 10 - x^2$ is a downward-opening parabola with vertex $(0, 10)$ and x-intercepts at $x = \pm\sqrt{10}$.



3 **Count intersections with the parabola**

On the given sketch, the parabola meets $f(x)$ twice. Outside the central part the parabola is below the range of $f(x)$, so there are 2 solutions in $-100\pi \le x \le 100\pi$.

4 **Sketch y equals tan x**

The graph $y = \tan x$ has zeros at integer multiples of π and vertical asymptotes at $x = \dfrac{\pi}{2} + n\pi$.

5 **Count in the reference interval**

In $-2\pi \le x \le 2\pi$, the graph $y = \tan x$ intersects $f(x)$ four times.

6 **Scale to the required interval**

The interval $-100\pi \le x \le 100\pi$ is 50 times as long as $-2\pi \le x \le 2\pi$, so the number of solutions is $50 \times 4 = 200$.

✓ **FINAL ANSWER**

$(a)\ f(x) = -10\cos x.\quad (b)(ii)\ 2.\quad (c)(ii)\ 200.$

Question 9

5 marks | Trigonometric Functions

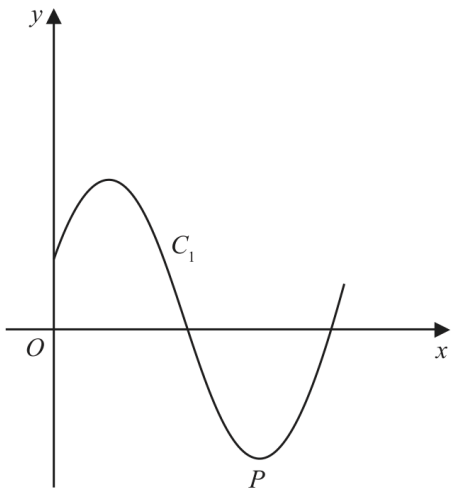


Figure 4

The curve C_1 has equation $y = \sin\left(x + \frac{\pi}{8}\right)$

Figure 4 shows a sketch of C_1 in the interval $0 \leqslant x < 2\pi$

The point P is a minimum turning point on C_1 and is shown in Figure 4.

(a) State the x coordinate of P.

(1)

(b) State the number of roots of the equation

$$\sin\left(x + \frac{\pi}{8}\right) = \sin\left(\frac{\pi}{8}\right)$$

in the interval $0 \leqslant x < 101\pi$, giving a reason for your answer.

(2)

The curve C_2 has equation $y = -5\sin\left(x + \frac{\pi}{8}\right)$

The point Q is the minimum turning point on C_2 in the interval $0 \leqslant x < 2\pi$

(c) State the coordinates of Q.

(2)

(Total for Question 9 is 5 marks)

WORKED SOLUTION

Question 9 · WMA11/01 May/June 2026

1 **(a) Locate the minimum of C1**

A sine curve reaches its minimum when its angle is $\frac{3\pi}{2}$. Therefore

$$x + \frac{\pi}{8} = \frac{3\pi}{2},$$

so

$$x = \frac{12\pi}{8} - \frac{\pi}{8} = \frac{11\pi}{8}.$$

2 **(b) Write both families of roots**

For $\sin u = \sin\left(\frac{\pi}{8}\right)$,

$$u = \frac{\pi}{8} + 2n\pi \quad \text{or} \quad u = \frac{7\pi}{8} + 2n\pi.$$

With $u = x + \frac{\pi}{8}$, this gives

$$x = 2n\pi \quad \text{or} \quad x = \frac{3\pi}{4} + 2n\pi.$$

---

3 **(b) Count the roots in the interval**

For $0 \le x < 101\pi$, each family has $n = 0, 1, \ldots, 50$, giving 51 roots from each family. Hence the total number of roots is

$$51 + 51 = 102.$$

---

4 **(c) Locate the minimum of C2**

Because $y = -5\sin\left(x + \frac{\pi}{8}\right)$, the minimum value -5 occurs when the sine value is 1. Thus

$$x + \frac{\pi}{8} = \frac{\pi}{2},$$

so $x = \frac{3\pi}{8}$. Therefore

$$Q = \left(\frac{3\pi}{8}, -5\right).$$

✓ **FINAL ANSWER**

$(a)\ x = \dfrac{11\pi}{8}.$ $(b)\ 102\ \text{roots}.$ $(c)\ Q = \left(\dfrac{3\pi}{8}, -5\right).$

TOPIC

Differentiation

Every past-paper question on this topic, October 2019 to May/June 2026.

Question 7

6 marks | Differentiation

7.

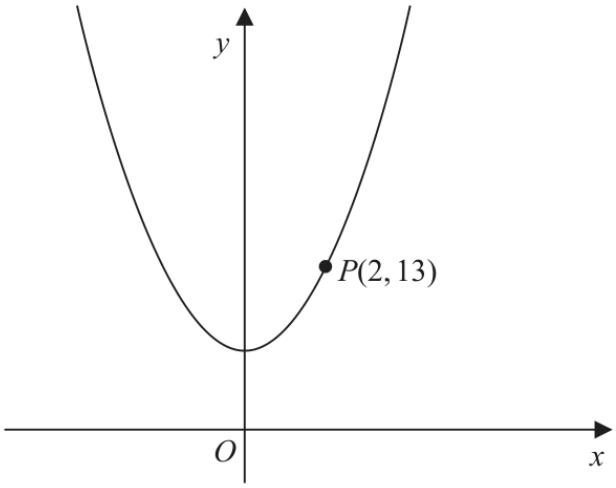


Figure 4

Figure 4 shows part of the curve with equation $y = 2x^2 + 5$

The point $P(2,13)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P.

(2)

The point Q with x coordinate $2 + h$ also lies on the curve.

(b) Find, in terms of h, the gradient of the line PQ. Give your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

(Total 6 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2019

1 Differentiate for the tangent gradient

For $y = 2x^2 + 5$, $\frac{dy}{dx} = 4x$. At P, $x = 2$, so the tangent gradient is 8.

2 Find the coordinates of Q

If $x_Q = 2 + h$, then $y_Q = 2(2 + h)^2 + 5 = 13 + 8h + 2h^2$.

3 Find the gradient of PQ

$$\text{gradient } PQ = \frac{(13 + 8h + 2h^2) - 13}{(2 + h) - 2} = \frac{8h + 2h^2}{h} = 8 + 2h.$$

4 **Explain the link**

As h gets closer to 0, point Q gets closer to P, and the chord gradient $8 + 2h$ tends to the tangent gradient 8.

FINAL ANSWER

$(a)\ 8. \quad (b)\ 8 + 2h. \quad (c)\ As\ h \to 0,\ 8 + 2h \to 8.$

WMA11/01 JANUARY 2020

Question 9

6 marks | Differentiation

9. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \qquad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

(Total 6 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2020

1 **Rewrite the function**

$y = \frac{4x^2+9}{2\sqrt{x}} = 2x^{3/2} + \frac{9}{2}x^{-1/2}$.

2 **Differentiate**

$\frac{dy}{dx} = 3x^{1/2} - \frac{9}{4}x^{-3/2}$.

3 **Set the derivative equal to zero**

$3x^{1/2} - \frac{9}{4}x^{-3/2} = 0$, so $3\sqrt{x} = \frac{9}{4x^{3/2}}$.

4 **Solve for x**

Multiply by $4x^{3/2}$: $12x^2 = 9$. Hence $x^2 = \frac{3}{4}$. Since $x > 0$, $x = \frac{\sqrt{3}}{2}$.

FINAL ANSWER

$x = \frac{\sqrt{3}}{2}$.

Question 7

9 marks | Differentiation

7. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \qquad x > 0$$

(a) Solve the equation

$$f(x) = 9$$

(4)

(b) Solve the equation

$$f''(x) = 6$$

(5)

(Total 9 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2021

1 **Solve f(x) = 9 without calculator methods**

$2x - 3\sqrt{x} - 5 = 9$, so $2x - 3\sqrt{x} - 14 = 0$.

2 **Use u = sqrt(x)**

Let $u = \sqrt{x}$, where $u > 0$. Then $x = u^2$, so $2u^2 - 3u - 14 = 0$.

3 **Factorise and reject the invalid root**

$2u^2 - 3u - 14 = (2u - 7)(u + 2)$. Since $u > 0$, $u = \frac{7}{2}$, so $x = u^2 = \frac{49}{4}$.

4 **Differentiate twice**

$f(x) = 2x - 3x^{1/2} - 5$, so $f'(x) = 2 - \frac{3}{2}x^{-1/2}$ and $f''(x) = \frac{3}{4}x^{-3/2}$.

5 **Solve f''(x) = 6**

$\frac{3}{4}x^{-3/2} = 6$, so $x^{-3/2} = 8$. Therefore $x^{3/2} = \frac{1}{8}$.

6 **Find x**

Since $x^{3/2} = (\sqrt{x})^3$, we get $\sqrt{x} = \frac{1}{2}$, hence $x = \frac{1}{4}$.

FINAL ANSWER

$(a)\ x = \frac{49}{4}$. $(b)\ x = \frac{1}{4}$.

Question 7

11 marks | Differentiation

7.

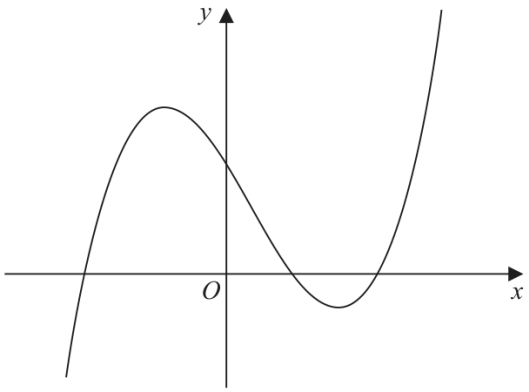


Figure 3

Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (x + 4)(x - 2)(2x - 9)$$

Given that the curve with equation $y = f(x) - p$ passes through the point with coordinates $(0, 50)$

(a) find the value of the constant p.

(2)

Given that the curve with equation $y = f(x + q)$ passes through the origin,

(b) write down the possible values of the constant q.

(2)

(c) Find $f'(x)$.

(4)

(d) Hence find the range of values of x for which the gradient of the curve with equation $y = f(x)$ is less than -18

(3)

(Total 11 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2022

1 Find p from the translated curve

$f(0) = (4)(-2)(-9) = 72$. Since $y = f(x) - p$ passes through $(0, 50)$, $72 - p = 50$, so $p = 22$.

2 Use roots for f(x + q)

If $y = f(x + q)$ passes through the origin, then $f(q) = 0$. From $f(x) = (x + 4)(x - 2)(2x - 9)$, the possible values are $q = -4, 2, \frac{9}{2}$.

3 Expand f(x)

$(x + 4)(x - 2)(2x - 9) = 2x^3 - 5x^2 - 34x + 72$.

4 **Differentiate**

$f'(x) = 6x^2 - 10x - 34$.

5 **Set the gradient less than -18**

$6x^2 - 10x - 34 < -18$, so $6x^2 - 10x - 16 < 0$, or $3x^2 - 5x - 8 < 0$.

6 **Solve the inequality**

$3x^2 - 5x - 8 = (3x - 8)(x + 1)$. Since the quadratic opens upward, it is negative between the roots: $-1 < x < \frac{8}{3}$.

FINAL ANSWER

$(a)\ p = 22.\quad (b)\ q = -4, 2, \frac{9}{2}.\quad (c)\ f'(x) = 6x^2 - 10x - 34.\quad (d)\ -1 < x < \frac{8}{3}.$

WMA11/01 MAY/JUNE 2022

Question 10

12 marks | Differentiation

10.

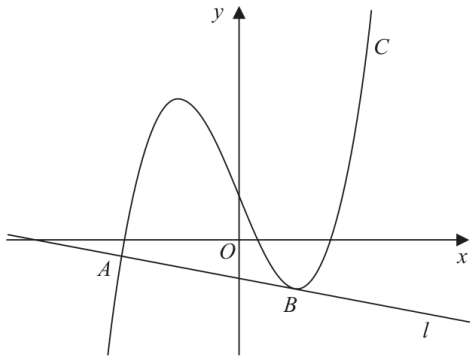


Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

(a) Find $\frac{dy}{dx}$

(2)

The line l, shown in Figure 5, is the normal to C at the point A with x coordinate $-\frac{7}{2}$

Given that l is also a tangent to C at the point B,

(b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)

(c) Hence find the x coordinate of B, justifying your answer.

(2)

Given that the y intercept of l is -1

(d) find the value of k.

(4)

(Total 12 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2022

1 Differentiate the curve

For $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$, $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$.

2 Find the tangent gradient at A

At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7}\left(\frac{49}{4}\right) + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$.

3 Find the gradient of l

Since l is the normal at A, its gradient is the negative reciprocal of 7, so $m_l = -\frac{1}{7}$.

4 **Use that l is tangent at B**

At B, the tangent gradient is also $-\frac{1}{7}$, so $\frac{6}{7}x^2+\frac{2}{7}x-\frac{5}{2}=-\frac{1}{7}$.

5 **Reach the given quadratic**

Multiplying by 14 gives $12x^2+4x-35=-2$, hence $12x^2+4x-33=0$.

6 **Solve for the x-coordinate of B**

$12x^2+4x-33=0$ gives $x=\frac{-4\pm 40}{24}$, so $x=\frac{3}{2}$ or $x=-\frac{11}{6}$. From the diagram, B is the point to the right of the y-axis, so $x_B=\frac{3}{2}$.

7 **Use the y-intercept of l**

Since the y-intercept is -1 and the gradient is $-\frac{1}{7}$, the line is $y=-\frac{1}{7}x-1$.

8 **Find the y-coordinate of A**

At $x=-\frac{7}{2}$ on the line, $y=-\frac{1}{7}\left(-\frac{7}{2}\right)-1=\frac{1}{2}-1=-\frac{1}{2}$.

9 **Substitute A into the curve**

$-\frac{1}{2}=\frac{2}{7}\left(-\frac{7}{2}\right)^3+\frac{1}{7}\left(-\frac{7}{2}\right)^2-\frac{5}{2}\left(-\frac{7}{2}\right)+k=-\frac{7}{4}+k$.

10 **Find k**

Therefore $k=\frac{5}{4}$.

FINAL ANSWER

$(a)\ \frac{dy}{dx}=\frac{6}{7}x^2+\frac{2}{7}x-\frac{5}{2}.\quad (c)\ x_B=\frac{3}{2}.\quad (d)\ k=\frac{5}{4}.$

Question 9

14 marks | Differentiation

9.

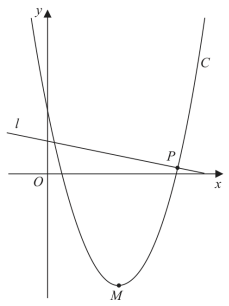


Figure 3

Figure 3 shows a sketch of the curve C with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

(a) Write $\frac{1}{2}x^2 - 10x + 22$ in the form

$$a(x + b)^2 + c$$

where a, b and c are constants to be found.

(3)

The point M is the minimum turning point of C, as shown in Figure 3.

(b) Deduce the coordinates of M

(2)

The line l is the normal to C at the point P, as shown in Figure 3.

Given that l has equation $y = k - \frac{1}{8}x$, where k is a constant,

(c) (i) find the coordinates of P

(ii) find the value of k

(6)

Question 9 continues on the next page

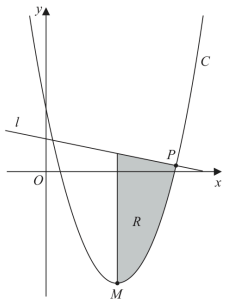


Figure 4

Figure 4 is a copy of Figure 3. The finite region R, shown shaded in Figure 4, is bounded by l, C and the line through M parallel to the y-axis.

(d) Identify the inequalities that define R.

(3)

(Total for Question 9 is 14 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2022

1 **Complete the square**

$$\frac{1}{2}x^2 - 10x + 22 = \frac{1}{2}(x^2 - 20x) + 22 = \frac{1}{2}\left((x-10)^2 - 100\right) + 22.$$

2 **Simplify the completed square form**

$$\frac{1}{2}\left((x-10)^2 - 100\right) + 22 = \frac{1}{2}(x-10)^2 - 28.$$

3 **Read the minimum point**

The minimum point is $M(10, -28)$.

4 **Use the normal gradient**

The line l has gradient $-\frac{1}{8}$, so the tangent gradient at P is 8.

5 **Differentiate the curve**

$\frac{dy}{dx} = x - 10$. At P, $x - 10 = 8$, so $x = 18$.

6 **Find P**

When $x = 18$, $y = \frac{1}{2}(18)^2 - 10(18) + 22 = 4$, so $P = (18, 4)$.

7 **Find k**

Since P lies on $y = k - \frac{1}{8}x$, $4 = k - \frac{1}{8}(18) = k - \frac{9}{4}$, so $k = \frac{25}{4}$.

8 **Write the upper boundary**

The line is $y = \frac{25}{4} - \frac{x}{8}$.

9 **Write the lower boundary**

The curve is $y = \frac{1}{2}x^2 - 10x + 22$.

10 **Identify the region**

The region lies to the right of the vertical line through M and between the curve and the normal line: $10 \le x \le 18$ and $\frac{1}{2}x^2 - 10x + 22 \le y \le \frac{25}{4} - \frac{x}{8}$.

✓ **FINAL ANSWER**

$(a)\ \frac{1}{2}(x-10)^2 - 28.\quad (b)\ M(10, -28).\quad (c)\ P(18,4),\ k = \frac{25}{4}.\quad (d)\ 10 \le x \le 18,\ \frac{1}{2}x^2 - 10x + 22 \le y \le \frac{25}{4} - \frac{x}{8}$

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WMA11/01 JANUARY 2023

Question 10

10 marks | Differentiation

10.

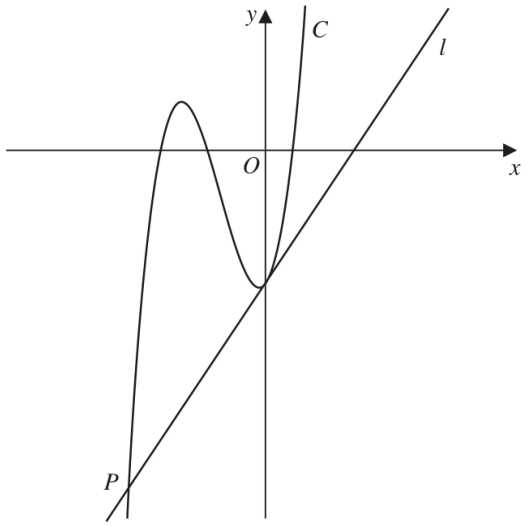


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x + 20)(x + 6)(2x - 3)$$

(a) Use the given information to state the values of x for which

$$f(x) > 0$$

(2)

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

(3)

The straight line l is the tangent to C at the point where C cuts the y-axis.

Given that l cuts C at the point P, as shown in Figure 4,

(c) find, using algebra, the x coordinate of P

(*Solutions based on calculator technology are not acceptable.*)

(5)

(Total for Question 10 is 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 January 2023

1 Use the roots

$f(x) = (3x + 20)(x + 6)(2x - 3)$ has roots $x = -\frac{20}{3}$, $x = -6$, and $x = \frac{3}{2}$.

2 State where f(x) is positive

Since the cubic has positive leading coefficient, $f(x) > 0$ for $-\frac{20}{3} < x < -6$ and for $x > \frac{3}{2}$.

3 Expand the first two brackets

$(3x + 20)(x + 6) = 3x^2 + 38x + 120$.

4 **Complete the expansion**

$(3x^2+38x+120)(2x-3)=6x^3+67x^2+126x-360$.

5 **Find the tangent at the y-axis**

$f'(x)=18x^2+134x+126$, so at $x=0$ the tangent gradient is 126. Also $f(0)=-360$.

6 **Write the tangent equation**

The tangent line is $y=126x-360$.

7 **Find the other intersection**

Set $6x^3+67x^2+126x-360=126x-360$. This gives $6x^3+67x^2=0$, so $x^2(6x+67)=0$.

8 **Choose point P**

$x=0$ is the tangent point on the y-axis, so the other intersection P has $x=-\dfrac{67}{6}$.

✓ **FINAL ANSWER**

$(a)\ -\dfrac{20}{3}<x<-6$ or $x>\dfrac{3}{2}$. $(b)\ 6x^3+67x^2+126x-360$. $(c)\ x_P=-\dfrac{67}{6}$.

WMA11/01 MAY/JUNE 2024

Question 7

8 marks | Differentiation

7. The curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2x^3 - kx^2 + 14x + 24$$

and k is a constant.

(a) Find, in simplest form,

(i) $\mathrm{f}'(x)$

(ii) $\mathrm{f}''(x)$

(3)

The curve with equation $y = \mathrm{f}'(x)$ intersects the curve with equation $y = \mathrm{f}''(x)$ at the points A and B.

Given that the x coordinate of A is 5

(b) find the value of k.

(2)

(c) Hence find the coordinates of B.

(3)

(Total for Question 7 is 8 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2024

1 Differentiate once

$f'(x) = 6x^2 - 2kx + 14$.

2 Differentiate again

$f''(x) = 12x - 2k$.

3 Use the intersection at A

At an intersection of $y = f'(x)$ and $y = f''(x)$, $f'(x) = f''(x)$. Given $x_A = 5$, substitute $x = 5$.

4 Find k

$6(5)^2 - 2k(5) + 14 = 12(5) - 2k$, so $164 - 10k = 60 - 2k$ and $k = 13$.

5 Find the other x-coordinate

With $k = 13$, $6x^2 - 26x + 14 = 12x - 26$, so $6x^2 - 38x + 40 = 0$. Hence $(3x - 4)(x - 5) = 0$.

6 Use the point B

The other point has $x = \frac{4}{3}$. Then $y = f''(\frac{4}{3}) = 12(\frac{4}{3}) - 26 = -10$.

FINAL ANSWER

$(a)(i)\ f'(x) = 6x^2 - 2kx + 14.$ $(a)(ii)\ f''(x) = 12x - 2k.$ $(b)\ k = 13.$ $(c)\ B = (\frac{4}{3}, -10).$

WMA11/01 OCTOBER 2024

Question 9

10 marks | Differentiation

9.

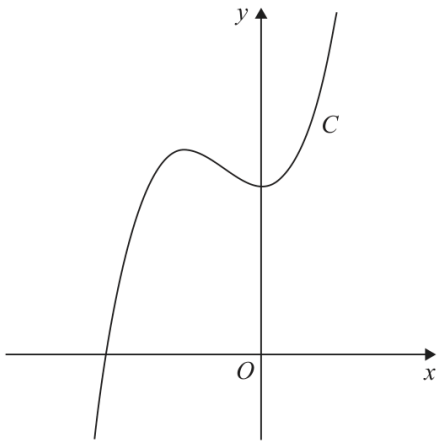


Figure 4

Figure 4 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (x + 5)(3x^2 - 4x + 20)$$

(a) Deduce the range of values of x for which $\mathrm{f}(x) \geqslant 0$

(1)

(b) Find $\mathrm{f}'(x)$ giving your answer in simplest form.

(3)

The point $R(-4, 84)$ lies on C.

Given that the tangent to C at the point P is parallel to the tangent to C at the point R

(c) find the x coordinate of P.

(4)

(d) Find the point to which R is transformed when the curve with equation $y = \mathrm{f}(x)$ is transformed to the curve with equation,

(i) $y = \mathrm{f}(x - 3)$

(ii) $y = 4\mathrm{f}(x)$

(2)

(Total for Question 9 is 10 marks)

WORKED SOLUTION

Question 9 · WMA11/01 October 2024

1 Use the positive quadratic factor

$3x^2 - 4x + 20$ has discriminant $(-4)^2 - 4(3)(20) < 0$ and positive leading coefficient, so it is always positive.

2 Deduce the sign of f(x)

The sign of $f(x) = (x + 5)(3x^2 - 4x + 20)$ is therefore the sign of $x + 5$. Hence $f(x) \geq 0$ for $x \geq -5$.

3 Expand f(x)

$f(x) = 3x^3 + 11x^2 + 100$.

4 **Differentiate**

$f'(x) = 9x^2 + 22x$.

5 **Find the gradient at R**

At $R(-4, 84)$, $f'(-4) = 9(16) + 22(-4) = 56$.

6 **Use parallel tangents**

For the tangent at P to be parallel, $f'(x) = 56$.

7 **Solve for x**

$9x^2 + 22x - 56 = 0$, so $x = \dfrac{-22 \pm 50}{18}$. The roots are $x = -4$ and $x = \dfrac{14}{9}$.

8 **Choose the point P**

$x = -4$ is the point R, so $x_P = \dfrac{14}{9}$.

9 **Translate point R**

$y = f(x - 3)$ translates the curve 3 units to the right, so $(-4, 84)$ maps to $(-1, 84)$.

10 **Transform R under y equals 4f(x)**

$y = 4f(x)$ stretches vertically by scale factor 4, so $(-4, 84)$ maps to $(-4, 336)$.

✓ **FINAL ANSWER**

$(a)\ x \geq -5.$ $(b)\ f'(x) = 9x^2 + 22x.$ $(c)\ x_P = \dfrac{14}{9}.$ $(d)(i)\ (-1, 84).$ $(d)(ii)\ (-4, 336).$

WMA11/01 OCTOBER 2025

Question 10

10 marks | Differentiation

10.

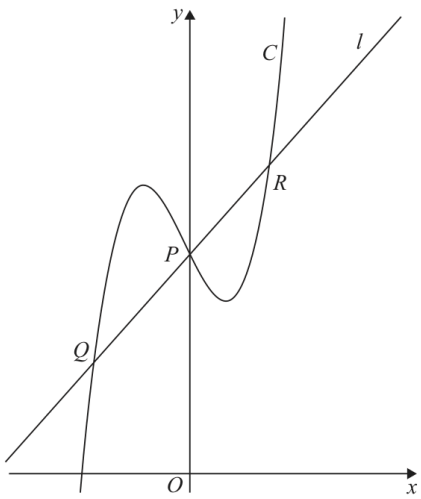


Figure 4

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 4 shows a sketch of the curve C with equation

$$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$$

The line l is the normal to C at the point P where $x = 0$

The line l also intersects C at points Q and R as shown in Figure 4.

(a) Find, using algebra, the x coordinate of point Q.

(6)

The point T lies on C.

Given that

- the tangent to C at T is parallel to l
- the x coordinate of T is positive

(b) find, using algebra, the exact x coordinate of T.

(4)

(Total for Question 10 is 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2025

1 Differentiate C

$y = 2x^3 + \frac{1}{2}x^2 - 2x + 5$, so $\frac{dy}{dx} = 6x^2 + x - 2$.

2 Find the tangent gradient at P

At P, $x = 0$, so the tangent gradient is -2.

3 Equation of the normal

The normal gradient is $\frac{1}{2}$ and $P = (0, 5)$, so $l : y = \frac{1}{2}x + 5$.

4 **Find intersections with the normal**

Set $2x^3 + \frac{1}{2}x^2 - 2x + 5 = \frac{1}{2}x + 5$.

5 **Form and factor the cubic**

$2x^3 + \frac{1}{2}x^2 - \frac{5}{2}x = 0$. Multiplying by 2 gives $4x^3 + x^2 - 5x = 0 = x(4x^2 + x - 5)$.

6 **Solve for x**

$x(4x + 5)(x - 1) = 0$, so the intersections have x-coordinates 0, $-\frac{5}{4}$ and 1.

7 **Select point Q**

The point Q is the left-hand intersection, so $x_Q = -\frac{5}{4}$.

8 **Parallel tangent condition**

The tangent at T is parallel to l, so its gradient is $\frac{1}{2}$.

9 **Set derivative equal to one half**

$6x^2 + x - 2 = \frac{1}{2}$, so $12x^2 + 2x - 5 = 0$.

10 **Use the positive solution**

$x = \frac{-2 \pm \sqrt{4 + 240}}{24} = \frac{-1 \pm \sqrt{61}}{12}$. Since x_T is positive, $x_T = \frac{-1 + \sqrt{61}}{12}$.

✓ **FINAL ANSWER**

$(a)\ x_Q = -\frac{5}{4}$. $(b)\ x_T = \frac{-1 + \sqrt{61}}{12}$.

WMA11/01 JANUARY 2026

Question 7

10 marks | Differentiation

7. The curve C has equation

$$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$$

(a) Show that $\frac{dy}{dx}$ can be written in the form

$$p(x+q)^2 + r$$

where p, q and r are constants to be found.

(5)

(b) Hence state

(i) the minimum value of $\frac{dy}{dx}$

(ii) the value of x at which this minimum value occurs.

(2)

Given that S is the point on C at which the gradient is a minimum,

(c) find the equation of the tangent to C at S, giving your answer in the form $y = mx + c$, where m and c are constants.

(3)

(Total for Question 7 is 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 January 2026

1 Differentiate

$y = \frac{2}{3}x^3 - 8x^2 + 43x - \frac{20}{3}$ gives $\frac{dy}{dx} = 2x^2 - 16x + 43$.

2 Complete the square

$2x^2 - 16x + 43 = 2(x^2 - 8x) + 43 = 2[(x-4)^2 - 16] + 43$.

3 Simplify

$\frac{dy}{dx} = 2(x-4)^2 + 11$.

4 Minimum gradient

Since $2(x-4)^2 \geq 0$, the minimum value of $\frac{dy}{dx}$ is 11.

5 **Where it occurs**

The minimum occurs when $x - 4 = 0$, so $x = 4$.

6 **Find the point S**

At $x = 4$, $y = \frac{2}{3}(4)^3 - 8(4)^2 + 43(4) - \frac{20}{3} = 80$.

7 **Tangent equation**

The gradient at S is 11, so $y - 80 = 11(x - 4)$.

8 **Simplify**

$y = 11x + 36$.

✓ **FINAL ANSWER**

$(a)\ \frac{dy}{dx} = 2(x-4)^2 + 11.$ $(b)\ \min \frac{dy}{dx} = 11$ at $x = 4.$ $(c)\ y = 11x + 36.$

WMA11/01 OCTOBER 2019

Question 10

10 marks | Differentiation | Main topic: Graphs of Functions

10.

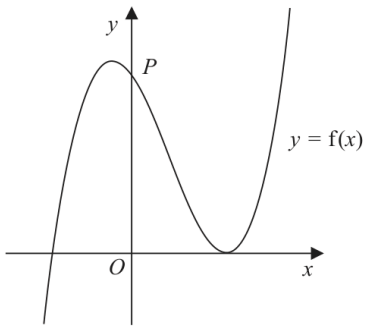


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

(a) Deduce the values of x for which $f(x) \leqslant 0$

(2)

The curve crosses the y-axis at the point P, as shown.

(b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)

(c) Hence, or otherwise, find

(i) the coordinates of P,

(ii) the gradient of the curve at P.

(2)

The curve with equation $y = f(x)$ is translated two units in the positive x direction to a curve with equation $y = g(x)$.

(d) (i) Find $g(x)$, giving your answer in a simplified factorised form.

(ii) Hence state the y intercept of the curve with equation $y = g(x)$.

(3)

(Total 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2019

1 Use the factorised form for the inequality

$f(x) = (2x + 5)(x - 3)^2$. Since $(x - 3)^2 \geq 0$, the sign is controlled by $2x + 5$, with an extra zero at $x = 3$. Hence $f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$.

2 Expand the cubic

$(x - 3)^2 = x^2 - 6x + 9$, so $f(x) = (2x + 5)(x^2 - 6x + 9) = 2x^3 - 7x^2 - 12x + 45$.

3 Find P and the gradient at P

At the y-axis, $x = 0$, so $P = (0, 45)$. Also $f'(x) = 6x^2 - 14x - 12$, so the gradient at P is $f'(0) = -12$.

4 **Translate the graph**

A translation 2 units in the positive x direction means $g(x) = f(x-2)$. Therefore $g(x) = (2(x-2)+5)((x-2)-3)^2 = (2x+1)(x-5)^2$.

5 **Find the new y-intercept**

Set $x = 0$: $g(0) = (1)(25) = 25$, so the y-intercept is 25.

✓ **FINAL ANSWER**

$f(x) \leq 0$ for $x \leq -\frac{5}{2}$ or $x = 3$; $f(x) = 2x^3 - 7x^2 - 12x + 45$; $P = (0, 45)$; gradient $= -12$; $g(x) = (2x+1)(x-5)^2$; y-intercept 25.

WMA11/01 OCTOBER 2019

Question 11

10 marks | Differentiation | Main topic: Integration

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P, writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

WORKED SOLUTION

Question 11 · WMA11/01 October 2019

1 Use the given gradient for the tangent

At $P\left(4, \frac{32}{3}\right)$, $f'(4) = 5$, so the tangent has gradient 5.

2 Find the tangent equation

$y - \frac{32}{3} = 5(x - 4)$. Hence $y = 5x - \frac{28}{3}$.

3 Integrate f'' to find f'

$f''(x) = 4x^{-1/2} - 3$, so $f'(x) = 8x^{1/2} - 3x + C$.

4 Use f'(4)=5

$5 = 8(2) - 12 + C = 4 + C$, so $C = 1$. Therefore $f'(x) = 8\sqrt{x} - 3x + 1$.

5 Integrate again

$f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x + D$.

6 Use the point on the curve

Substitute $x = 4$, $f(4) = \frac{32}{3}$: $\frac{32}{3} = \frac{16}{3}(8) - \frac{3}{2}(16) + 4 + D = \frac{68}{3} + D$, so $D = -12$.

FINAL ANSWER

$(a)\ y = 5x - \frac{28}{3}.\quad (b)\ f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x - 12.$

WMA11/01 JANUARY 2020

Question 11

11 marks | Differentiation | Main topic: Integration

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P,

(a) find the equation of the normal at P, writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

(b) find $f(x)$.

(8)

(Total 11 marks)

WORKED SOLUTION

Question 11 · WMA11/01 January 2020

1 Find the normal gradient

At P, $f'(4) = -4$, so the tangent gradient is -4. The normal gradient is therefore $\frac{1}{4}$.

2 Find the normal equation

Using $P(4, -50)$: $y + 50 = \frac{1}{4}(x - 4)$, so $y = \frac{1}{4}x - 51$.

3 Integrate f'' to find f'

$f''(x) = 6x^{-3/2} + x$. Therefore $f'(x) = -12x^{-1/2} + \frac{1}{2}x^2 + k$.

4 Use f'(4) = -4

$-4 = -\frac{12}{2} + \frac{1}{2}(16) + k = -6 + 8 + k = 2 + k$, so $k = -6$.

5 Integrate again

$f(x) = -24x^{1/2} + \frac{x^3}{6} - 6x + d$.

6 Use the point on the curve

Substitute $P(4, -50)$: $-50 = -24(2) + \frac{64}{6} - 24 + d = -72 + \frac{32}{3} + d = -\frac{184}{3} + d$. Hence $d = \frac{34}{3}$.

FINAL ANSWER

$(a)\ y = \frac{1}{4}x - 51. \quad (b)\ f(x) = -24\sqrt{x} + \frac{x^3}{6} - 6x + \frac{34}{3}.$

WMA11/01 JANUARY 2021

Question 9

11 marks | Differentiation | Main topic: Integration

9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}}\,\mathrm{d}x \qquad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = \mathrm{f}(x)$.

Given

- $\mathrm{f}'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3,-2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $\mathrm{f}(x)$.

(6)

(Total 11 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2021

1 **Expand and simplify the integrand**

$\frac{(3x+2)^2}{4\sqrt{x}} = \frac{9x^2+12x+4}{4x^{1/2}} = \frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}$.

2 **Integrate term by term**

$\int \left(\frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}\right)dx = \frac{9}{10}x^{5/2} + 2x^{3/2} + 2x^{1/2} + c$.

3 **Write in simplest form**

Factoring out $\sqrt{x}$ gives $\frac{\sqrt{x}(9x^2+20x+20)}{10} + c$.

4 **Use the gradient condition at P**

Since $f'(x) = x^2 + ax + b$ and the gradient at $P(3,-2)$ is 2, $f'(3) = 2$. Hence $9 + 3a + b = 2$, so $3a + b = -7$.

5 **Integrate f'(x)**

$f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + c$. The y-intercept is -8, so $c = -8$.

6 **Use P(3,-2)**

$-2 = \frac{27}{3} + \frac{9a}{2} + 3b - 8$, so $\frac{9a}{2} + 3b = -3$.

7 **Solve for a and b**

From $3a + b = -7$, $b = -7 - 3a$. Substitute into $\frac{9a}{2} + 3b = -3$ to get $a = -4$, and then $b = 5$.

8 **Write f(x)**

Therefore $f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8$.

FINAL ANSWER

$(i)\ \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c. \quad (ii)\ f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$

Question 6

8 marks | Differentiation | Main topic: Integration

6. The curve C has equation $y = f(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $f'(x) = \frac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(a) find the equation of the tangent to C at P. Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

(b) Find, in simplest form, $f(x)$.

(5)

(Total 8 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2021

1 Find the gradient at P

$f'(x) = \frac{32}{3x^2} + 3 - 2\sqrt[3]{x}$. At $x = 8$, $f'(8) = \frac{32}{3(64)} + 3 - 2(2) = \frac{1}{6} - 1 = -\frac{5}{6}$.

2 Write the tangent equation

Using $P(8, 2)$, $y - 2 = -\frac{5}{6}(x - 8)$, hence $y = -\frac{5}{6}x + \frac{26}{3}$.

3 Integrate f'(x)

$f'(x) = \frac{32}{3}x^{-2} + 3 - 2x^{1/3}$, so $f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + c$.

4 Use P(8,2) to find c

$2 = -\frac{32}{24} + 24 - \frac{3}{2}(16) + c = -\frac{4}{3} + c$, so $c = \frac{10}{3}$.

5 Write f(x)

$f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}$.

FINAL ANSWER

$(a)\ y = -\frac{5}{6}x + \frac{26}{3}$. $(b)\ f(x) = -\frac{32}{3x} + 3x - \frac{3}{2}x^{4/3} + \frac{10}{3}$.

WMA11/01 OCTOBER 2021

Question 6

10 marks | Differentiation | Main topic: Graphs of Functions

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C.

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $\mathrm{f}(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

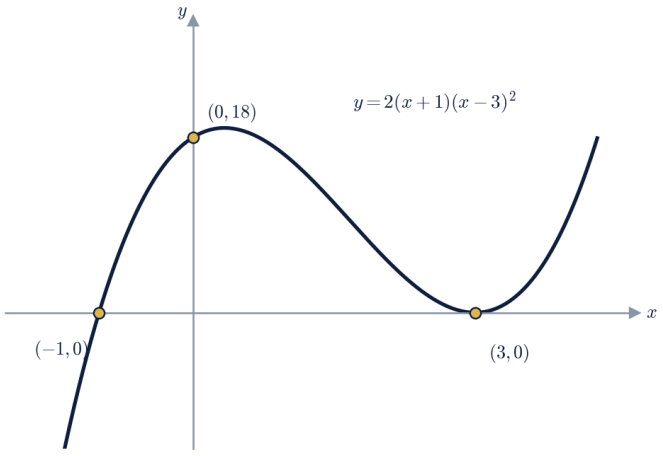
(Total 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 October 2021

1 **Use the factorised form for the sketch**

$f(x) = 2(x+1)(x-3)^2$. The curve cuts the x-axis at $(-1, 0)$ and meets the x-axis at $(3, 0)$ because $x = 3$ is a repeated root.



2 **Find the y-intercept**

$f(0) = 2(1)(9) = 18$, so the y-intercept is $(0, 18)$.

3 **Describe the cubic shape**

The leading term is positive, so the cubic goes down to the left and up to the right. It crosses at $x = -1$ and touches at $x = 3$.

4 **Expand f(x)**

$(x+1)(x-3)^2 = (x+1)(x^2-6x+9) = x^3-5x^2+3x+9$, so $f(x) = 2x^3-10x^2+6x+18$.

5 **Differentiate**

$f'(x) = 6x^2-20x+6$.

6 **Find the gradient at x = 1/3**

$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{9}\right) - 20\left(\frac{1}{3}\right) + 6 = 0$, so the tangent is horizontal.

7 **Find the y-coordinate**

$f\left(\frac{1}{3}\right) = 2\left(\frac{4}{3}\right)\left(-\frac{8}{3}\right)^2 = \frac{512}{27}$.

8 **Write the tangent**

A horizontal tangent through this point has equation $y = \frac{512}{27}$.

✓ **FINAL ANSWER**

(a) x-intercepts $(-1,0), (3,0)$, y-intercept $(0,18)$. (b) $f(x) = 2x^3-10x^2+6x+18$. (c) $y = \frac{512}{27}$.

WMA11/01 OCTOBER 2021

Question 8

10 marks | Differentiation | Main topic: Straight Line

8.

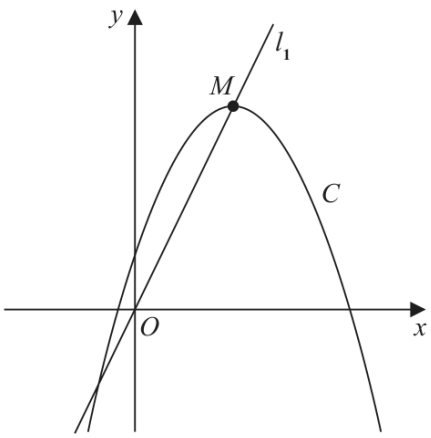


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C.

(a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a, b and c are constants to be found.

(ii) Hence, or otherwise, state the coordinates of M.

(5)

The line l_1 passes through O and M, as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

(b) Find an equation for l_2

(5)

(Total 10 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2021

1 Complete the square

$4 + 12x - 3x^2 = -3(x^2 - 4x) + 4 = -3\left((x-2)^2 - 4\right) + 4 = 16 - 3(x-2)^2$.

2 State the maximum point

The maximum occurs when $(x-2)^2 = 0$, so $M = (2, 16)$.

3 Find the gradient of l1

Line l_1 passes through $O(0,0)$ and $M(2,16)$, so its gradient is $\frac{16}{2} = 8$.

4 **Use the parallel tangent gradient**

Line l_2 is parallel to l_1, so its gradient is also 8. For $y = 4 + 12x - 3x^2$, $\dfrac{dy}{dx} = 12 - 6x$.

5 **Find the point of contact**

Set $12 - 6x = 8$, giving $x = \dfrac{2}{3}$. Then $y = 4 + 12\left(\dfrac{2}{3}\right) - 3\left(\dfrac{4}{9}\right) = \dfrac{32}{3}$.

6 **Write l2**

Using gradient 8 through $\left(\dfrac{2}{3}, \dfrac{32}{3}\right)$, $y - \dfrac{32}{3} = 8\left(x - \dfrac{2}{3}\right)$. Hence $y = 8x + \dfrac{16}{3}$.

✓ **FINAL ANSWER**

$(a)(i)\ 16 - 3(x-2)^2.$ $\quad (a)(ii)\ M = (2, 16).$ $\quad (b)\ y = 8x + \dfrac{16}{3}.$

WMA11/01 OCTOBER 2021

Question 10

7 marks | Differentiation | Main topic: Integration

10. A curve has equation $y = f(x), x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a.

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2021

1 Differentiate f'(x)

$f'(x) = ax - 12x^{1/3}$, so $f''(x) = a - 4x^{-2/3}$.

2 Use f''(27) = 0

Since $27^{2/3} = 9$, $0 = a - \frac{4}{9}$, so $a = \frac{4}{9}$.

3 Integrate f'(x)

With $a = \frac{4}{9}$, $f'(x) = \frac{4}{9}x - 12x^{1/3}$. Therefore $f(x) = \frac{2}{9}x^2 - 9x^{4/3} + c$.

4 Use the point (1, -8)

$-8 = \frac{2}{9} - 9 + c$, so $c = -8 + 9 - \frac{2}{9} = \frac{7}{9}$.

5 Write f(x)

$f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}$.

FINAL ANSWER

$(a)\ a = \frac{4}{9}$. $(b)\ f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}$.

WMA11/01 JANUARY 2022

Question 6

11 marks | Differentiation | Main topic: Integration

6. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$, simplifying your answer.

(7)

(Total 11 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2022

1 Simplify f'(x)

$f'(x) = \dfrac{(x+3)^2}{x\sqrt{x}} = \dfrac{x^2+6x+9}{x^{3/2}} = x^{1/2} + 6x^{-1/2} + 9x^{-3/2}$.

2 Find the gradient at P

At $x = 4$, $f'(4) = 2 + 3 + \dfrac{9}{8} = \dfrac{49}{8}$.

3 Find the tangent equation

Using $P(4, 20)$, $y - 20 = \dfrac{49}{8}(x-4)$. Multiplying out gives $49x - 8y - 36 = 0$.

4 Integrate f'(x)

$f(x) = \int \left(x^{1/2} + 6x^{-1/2} + 9x^{-3/2}\right)dx = \dfrac{2}{3}x^{3/2} + 12x^{1/2} - 18x^{-1/2} + c$.

5 Use P(4,20)

$20 = \dfrac{2}{3}(8) + 12(2) - \dfrac{18}{2} + c = \dfrac{61}{3} + c$, so $c = -\dfrac{1}{3}$.

6 Write f(x)

$f(x) = \dfrac{2}{3}x\sqrt{x} + 12\sqrt{x} - \dfrac{18}{\sqrt{x}} - \dfrac{1}{3}$.

FINAL ANSWER

$(a)(i)\ \dfrac{49}{8}.\quad (a)(ii)\ 49x-8y-36=0.\quad (b)\ f(x)=\dfrac{2}{3}x\sqrt{x}+12\sqrt{x}-\dfrac{18}{\sqrt{x}}-\dfrac{1}{3}.$

WMA11/01 MAY/JUNE 2022

Question 7

9 marks | Differentiation | Main topic: Integration

7. The curve C has equation $y = f(x),\ x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A.

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2022

1 Differentiate f'(x)

$f'(x) = 2x^{-1/2} + Ax^{-2} + 3$, so $f''(x) = -x^{-3/2} - 2Ax^{-3}$.

2 Use f''(4) = 0

$-4^{-3/2} - 2A(4^{-3}) = 0$, so $-\frac{1}{8} - \frac{A}{32} = 0$ and $A = -4$.

3 Substitute A into f'(x)

$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3$.

4 Integrate to find f(x)

$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c$.

5 Use f(12) = 8sqrt3

$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}$, so $c = -\frac{109}{3}$.

FINAL ANSWER

$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$.

WMA11/01 JANUARY 2023

Question 11

8 marks | Differentiation | Main topic: Integration

11. A curve C has equation $y = f(x), \quad x > 0$

Given that

- $f''(x) = 4x + \dfrac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

WORKED SOLUTION

Question 11 · WMA11/01 January 2023

1 Find point P

Since P has x-coordinate 4 and lies on the tangent $y = 3x + 4$, $P = (4, 16)$.

2 Find the normal

The tangent gradient is 3, so the normal gradient is $-\frac{1}{3}$. Hence the normal is $y - 16 = -\frac{1}{3}(x - 4)$.

3 Integrate f''(x)

$f''(x) = 4x + x^{-1/2}$, so $f'(x) = 2x^2 + 2\sqrt{x} + A$.

4 Use the tangent gradient

At $x = 4$, $f'(4) = 3$. Thus $2(16) + 2(2) + A = 3$, so $A = -33$.

5 Integrate again

$f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + B$.

6 Use P(4, 16)

$16 = \frac{2}{3}(64) + \frac{4}{3}(8) - 33(4) + B = \frac{160}{3} - 132 + B$.

7 Find B

$B = \frac{284}{3}$, so $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}$.

FINAL ANSWER

$(a)\ y - 16 = -\dfrac{1}{3}(x - 4).$ $(b)\ f(x) = \dfrac{2}{3}x^3 + \dfrac{4}{3}x^{3/2} - 33x + \dfrac{284}{3}.$

WMA11/01 MAY/JUNE 2023

Question 8

10 marks | Differentiation | Main topic: Integration

8. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find $f(x)$ giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2023

1 Differentiate the curve

For $y = \frac{1}{4}x^3 - 8x^{-1/2}$, $\frac{dy}{dx} = \frac{3}{4}x^2 + 4x^{-3/2}$.

2 Find the gradient at P

At $x = 4$, $\frac{dy}{dx} = \frac{3}{4}(16) + 4(4^{-3/2}) = 12 + \frac{1}{2} = \frac{25}{2}$.

3 Find the tangent

Using $P(4, 12)$, $y - 12 = \frac{25}{2}(x - 4)$.

4 Write in integer form

Multiplying by 2 gives $2y - 24 = 25x - 100$, hence $25x - 2y - 76 = 0$.

5 Integrate f'(x)

$f'(x) = \frac{1}{4}x^3 - 8x^{-1/2}$ gives $f(x) = \frac{x^4}{16} - 16\sqrt{x} + c$.

6 **Use P(4, 12)**

$12 = \frac{4^4}{16} - 16\sqrt{4} + c = 16 - 32 + c$, so $c = 28$.

FINAL ANSWER

$(a)\ 25x - 2y - 76 = 0. \quad (b)\ f(x) = \frac{x^4}{16} - 16\sqrt{x} + 28.$

WMA11/01 OCTOBER 2023

Question 7

10 marks | Differentiation | Main topic: Integration

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2023

1 Simplify f prime

$f'(x) = \dfrac{4x^2 + 10 - 7x^{1/2}}{4x^{1/2}} = x^{3/2} + \dfrac{5}{2}x^{-1/2} - \dfrac{7}{4}$.

2 Find the gradient at P

At $x = 4$, $f'(4) = 4^{3/2} + \dfrac{5}{2}(4^{-1/2}) - \dfrac{7}{4} = 8 + \dfrac{5}{4} - \dfrac{7}{4} = \dfrac{15}{2}$.

3 Find the normal gradient

The normal gradient is the negative reciprocal, so $m_n = -\dfrac{2}{15}$.

4 Use point P

Using $P(4, -1)$, $y + 1 = -\dfrac{2}{15}(x - 4)$.

5 Write the normal in integer form

$15y + 15 = -2x + 8$, hence $2x + 15y + 7 = 0$.

6 Integrate f prime

$f(x) = \int \left(x^{3/2} + \dfrac{5}{2}x^{-1/2} - \dfrac{7}{4}\right) dx = \dfrac{2}{5}x^{5/2} + 5x^{1/2} - \dfrac{7}{4}x + c$.

7 Use P to find c

$-1 = \dfrac{2}{5}(4^{5/2}) + 5(4^{1/2}) - \dfrac{7}{4}(4) + c = \dfrac{64}{5} + 10 - 7 + c$, so $c = -\dfrac{84}{5}$.

FINAL ANSWER

$(a)(i)\ \dfrac{15}{2}.\quad (a)(ii)\ 2x+15y+7=0.\quad (b)\ f(x)=\dfrac{2}{5}x^{5/2}+5x^{1/2}-\dfrac{7}{4}x-\dfrac{84}{5}.$

WMA11/01 JANUARY 2024

Question 10

8 marks | Differentiation | Main topic: Integration

10. **In this question you must show all stages of your working.**

The curve C has equation $y = f(x), x > 0$

Given that

- the point $P\left(2, 8\sqrt{2}\right)$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

First released on AP - Edexcel Discord
https://sites.google.com/view/ap-edexcel

(Total for Question 10 is 8 marks)

WORKED SOLUTION

Question 10 · WMA11/01 January 2024

1 **Differentiate f prime**

$f'(x) = 4x^{3/2} + kx^{-2}$, so $f''(x) = 6x^{1/2} - 2kx^{-3}$.

2 **Use f''(P) = 0**

At P, $x = 2$ and $f''(2) = 0$, so $6\sqrt{2} - 2k(2^{-3}) = 0$.

3 **Find k**

$6\sqrt{2} - \frac{k}{4} = 0$, hence $k = 24\sqrt{2}$.

4 **Integrate f prime**

$f(x) = \int \left(4x^{3/2} + 24\sqrt{2}x^{-2}\right)dx = \frac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c$.

5 **Use the point P**

$P(2, 8\sqrt{2})$ lies on C, so $8\sqrt{2} = \frac{8}{5}(2^{5/2}) - \frac{24\sqrt{2}}{2} + c$.

6 **Find c**

$8\sqrt{2} = \frac{32\sqrt{2}}{5} - 12\sqrt{2} + c$, so $c = \frac{68\sqrt{2}}{5}$.

FINAL ANSWER

$(a)\ k = 24\sqrt{2}.\quad (b)\ f(x) = \frac{8}{5}x^{5/2} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}.$

WMA11/01 MAY/JUNE 2024

Question 10

10 marks | Differentiation | Main topic: Integration

10. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = 6x - \dfrac{(2x-1)(3x+2)}{2\sqrt{x}}$
- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P, giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $\mathrm{f}(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2024

1 Expand the numerator

$(2x-1)(3x+2) = 6x^2 + x - 2$.

2 Simplify f prime

$f'(x) = 6x - \dfrac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \dfrac{1}{2}x^{1/2} + x^{-1/2}$.

3 Find the tangent gradient at P

$f'(4) = 24 - 3(8) - \dfrac{1}{2}(2) + \dfrac{1}{2} = -\dfrac{1}{2}$.

4 Use the normal gradient

The normal gradient is the negative reciprocal of $-\dfrac{1}{2}$, so it is 2.

5 Find the normal equation

Through $P(4, 12)$, $y - 12 = 2(x - 4)$, hence $y = 2x + 4$.

6 Integrate f prime

$f(x) = 3x^2 - \dfrac{6}{5}x^{5/2} - \dfrac{1}{3}x^{3/2} + 2x^{1/2} + c$.

7 Use P to find c

$12 = 3(4)^2 - \dfrac{6}{5}(4^{5/2}) - \dfrac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c$.

8 Simplify c

$12 = 48 - \dfrac{192}{5} - \dfrac{8}{3} + 4 + c = \dfrac{164}{15} + c$, so $c = \dfrac{16}{15}$.

FINAL ANSWER

$(a)\ y = 2x + 4. \quad (b)\ f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}.$

WMA11/01 OCTOBER 2024

Question 8

7 marks | Differentiation | Main topic: Integration

8. A curve C has equation $y = \mathrm{f}(x)$.

The point P with x coordinate 3 lies on C

Given

- $\mathrm{f}'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $\mathrm{f}(x)$.

(4)

(Total for Question 8 is 7 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2024

1 Find the y-coordinate of P

The normal is $y = -\frac{1}{24}x + 5$. At $x = 3$, $y = -\frac{3}{24} + 5 = \frac{39}{8}$.

2 Use the normal gradient

The normal gradient is $-\frac{1}{24}$, so the tangent gradient is 24.

3 Use f prime at x equals 3

$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k$.

4 Find k

$39 + 3k = 24$, so $k = -5$.

5 Integrate f prime

With $k = -5$, $f'(x) = 4x^2 - 5x + 3$, so $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c$.

6 Use point P

$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c$.

7 Find c

$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}$.

FINAL ANSWER

$k = -5$ and $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}$.

WMA11/01 JANUARY 2025

Question 6

10 marks | Differentiation | Main topic: Integration

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = f(x),\ x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $f'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24$$

(2)

Given also that $a + b = -9$

(b) find, in simplest form, $f(x)$

(7)

Curve C is transformed to the curve with equation $y = f(x - 3)$

Given that point P is transformed to the point Q,

(c) state the coordinates of Q.

(1)

(Total for Question 6 is 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2025

1 **Use the gradient at P**

At $x = 4$, $f'(4) = 7$. Since $f'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, we get $\dfrac{32 + 4a + b}{8} = 7$.

2 **Show the required equation**

$32 + 4a + b = 56$, so $4a + b = 24$.

3 **Solve for a and b**

Together with $a + b = -9$, subtracting gives $3a = 33$, so $a = 11$ and $b = -20$.

4 **Rewrite f prime**

$$f'(x) = \frac{2x^2 + 11x - 20}{4\sqrt{x}} = \frac{1}{2}x^{3/2} + \frac{11}{4}x^{1/2} - 5x^{-1/2}.$$

5 **Integrate**

$$f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} + c.$$

6 **Use P to find c**

$$-5 = \frac{1}{5}(4^{5/2}) + \frac{11}{6}(4^{3/2}) - 10(4^{1/2}) + c = \frac{16}{15} + c.$$

7 **State f(x)**

$c = -\frac{91}{15}$, so $f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}$.

8 **Transform P**

$y = f(x - 3)$ translates the curve 3 units to the right, so $P(4, -5)$ becomes $Q(7, -5)$.

✓ **FINAL ANSWER**

$(b)\ f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}.$ $(c)\ Q = (7, -5).$

WMA11/01 MAY/JUNE 2025

Question 8

11 marks | Differentiation | Main topic: Integration

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = f(x),\ x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P.

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2025

1 Use the tangent gradient

The tangent at P has equation $y = 10x + c$, so the gradient at P is 10.

2 Show k

$f'(4) = 3\sqrt{4} + k(4)^2 = 6 + 16k = 10$, so $16k = 4$ and $k = \frac{1}{4}$.

3 Find c

Substitute $P(4, 12)$ into $y = 10x + c$: $12 = 40 + c$, so $c = -28$.

4 Differentiate f prime

With $k = \frac{1}{4}$, $f'(x) = 3x^{1/2} + \frac{1}{4}x^2$.

5 Find f double prime

$f''(x) = \frac{3}{2}x^{-1/2} + \frac{1}{2}x$.

6 **Evaluate at P**

$f''(4) = \frac{3}{2} \cdot \frac{1}{2} + 2 = \frac{3}{4} + 2 = \frac{11}{4}.$

7 **Integrate f prime**

$f(x) = \int \left(3x^{1/2} + \frac{1}{4}x^2\right) dx = 2x^{3/2} + \frac{1}{12}x^3 + d.$

8 **Use P to find d**

$12 = 2(4^{3/2}) + \frac{1}{12}(4^3) + d = 16 + \frac{16}{3} + d.$

9 **State f(x)**

$d = -\frac{28}{3}$, so $f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$

✓ **FINAL ANSWER**

$(a)(i)\ k = \frac{1}{4}.$ $(a)(ii)\ c = -28.$ $(b)\ f''(P) = \frac{11}{4}.$ $(c)\ f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$

WMA11/01 OCTOBER 2025

Question 8

8 marks | Differentiation | Main topic: Integration

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = f(x), x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k.

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2025

1 Use tangent gradient

The tangent has equation $y = 5x - 3\sqrt{2}$, so the gradient at $x = \sqrt{2}$ is 5.

2 Find k

$f'(\sqrt{2}) = 2\sqrt{2} + \frac{8}{(\sqrt{2})^2} + k = 2\sqrt{2} + 4 + k = 5$, so $k = 1 - 2\sqrt{2}$.

3 Find the point on the curve

At $x = \sqrt{2}$, the tangent gives $y = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$.

4 Normal gradient

The tangent gradient is 5, so the normal gradient is $-\frac{1}{5}$.

5 Equation of the normal

$y - 2\sqrt{2} = -\frac{1}{5}(x - \sqrt{2})$, hence $y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}$.

6 Integrate f prime

$f(x) = \int \left(2x + \frac{8}{x^2} + k\right) dx = x^2 - \frac{8}{x} + kx + c$.

7 **Substitute k**

$f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + c$.

8 **Use the point**

$2\sqrt{2} = 2 - \dfrac{8}{\sqrt{2}} + (1 - 2\sqrt{2})\sqrt{2} + c = -2 - 3\sqrt{2} + c$.

9 **Find c**

$c = 2 + 5\sqrt{2}$, so $f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}$.

✓ **FINAL ANSWER**

$(a)\ k = 1 - 2\sqrt{2}.$ $(b)\ y = -\dfrac{1}{5}x + \dfrac{11\sqrt{2}}{5}.$ $(c)\ f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}$.

Question 8

12 marks | Differentiation | Main topic: Integration

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x), \quad x > 0$

A point P lies on C.

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P,

(ii) find, using algebra, the other possible x coordinate of P.

(6)

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P.

(2)

(c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2026

1 **Verify x equals root 3**

$f'(\sqrt{3}) = 4(\sqrt{3})^2 + \frac{6}{(\sqrt{3})^2} - 4 = 12 + 2 - 4 = 10$, which matches the tangent gradient.

2 **Find all possible x-coordinates**

Set $4x^2 + \frac{6}{x^2} - 4 = 10$.

3 **Form a quadratic in x squared**

Multiplying by x^2 gives $4x^4 - 14x^2 + 6 = 0$, so $2x^4 - 7x^2 + 3 = 0$.

4 **Factor**

$2x^4 - 7x^2 + 3 = (2x^2 - 1)(x^2 - 3) = 0$.

5 **Use x greater than zero**

$x = \sqrt{3}$ or $x = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$. The other possible x-coordinate is $\dfrac{\sqrt{2}}{2}$.

6 **Find the point P**

Given $x_P = \sqrt{3}$, the tangent gives $y_P = 10\sqrt{3} - 6\sqrt{3} = 4\sqrt{3}$.

7 **Normal equation**

The normal gradient is $-\dfrac{1}{10}$, so $y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3})$.

8 **Integrate f prime**

$f(x) = \int \left(4x^2 + \dfrac{6}{x^2} - 4\right) dx = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + c$.

9 **Use P to find c**

$4\sqrt{3} = \dfrac{4}{3}(\sqrt{3})^3 - \dfrac{6}{\sqrt{3}} - 4\sqrt{3} + c = 4\sqrt{3} - 2\sqrt{3} - 4\sqrt{3} + c$.

10 **State f(x)**

$c = 6\sqrt{3}$, so $f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}$.

✓ **FINAL ANSWER**

$(a)(i)$ verified.

$(a)(ii)\ x = \dfrac{\sqrt{2}}{2}$. $(b)\ y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3})$. $(c)\ f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}$.

TOPIC

Integration

Every past-paper question on this topic, October 2019 to May/June 2026.

WMA11/01 OCTOBER 2019

Question 11

10 marks | Integration

11. A curve has equation $y = f(x)$.

The point $P\left(4, \frac{32}{3}\right)$ lies on the curve.

Given that

- $f''(x) = \frac{4}{\sqrt{x}} - 3$
- $f'(x) = 5$ at P

find

(a) the equation of the tangent to the curve at P, writing your answer in the form $y = mx + c$, where m and c are constants to be found,

(2)

(b) $f(x)$.

(8)

(Total 10 marks)

WORKED SOLUTION

Question 11 · WMA11/01 October 2019

1 Use the given gradient for the tangent

At $P\left(4, \frac{32}{3}\right)$, $f'(4) = 5$, so the tangent has gradient 5.

2 Find the tangent equation

$y - \frac{32}{3} = 5(x - 4)$. Hence $y = 5x - \frac{28}{3}$.

3 Integrate f'' to find f'

$f''(x) = 4x^{-1/2} - 3$, so $f'(x) = 8x^{1/2} - 3x + C$.

4 Use f'(4)=5

$5 = 8(2) - 12 + C = 4 + C$, so $C = 1$. Therefore $f'(x) = 8\sqrt{x} - 3x + 1$.

5 Integrate again

$f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x + D$.

6 Use the point on the curve

Substitute $x = 4$, $f(4) = \frac{32}{3}$: $\frac{32}{3} = \frac{16}{3}(8) - \frac{3}{2}(16) + 4 + D = \frac{68}{3} + D$, so $D = -12$.

FINAL ANSWER

$(a)\ y = 5x - \frac{28}{3}.\quad (b)\ f(x) = \frac{16}{3}x^{3/2} - \frac{3}{2}x^2 + x - 12.$

WMA11/01 JANUARY 2020

Question 11

11 marks | Integration

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \qquad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P,

(a) find the equation of the normal at P, writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

(b) find $f(x)$.

(8)

(Total 11 marks)

WORKED SOLUTION

Question 11 · WMA11/01 January 2020

1 Find the normal gradient

At P, $f'(4) = -4$, so the tangent gradient is -4. The normal gradient is therefore $\frac{1}{4}$.

2 Find the normal equation

Using $P(4, -50)$: $y + 50 = \frac{1}{4}(x - 4)$, so $y = \frac{1}{4}x - 51$.

3 Integrate f'' to find f'

$f''(x) = 6x^{-3/2} + x$. Therefore $f'(x) = -12x^{-1/2} + \frac{1}{2}x^2 + k$.

4 Use f'(4) = -4

$-4 = -\frac{12}{2} + \frac{1}{2}(16) + k = -6 + 8 + k = 2 + k$, so $k = -6$.

5 Integrate again

$f(x) = -24x^{1/2} + \frac{x^3}{6} - 6x + d$.

6 Use the point on the curve

Substitute $P(4, -50)$: $-50 = -24(2) + \frac{64}{6} - 24 + d = -72 + \frac{32}{3} + d = -\frac{184}{3} + d$. Hence $d = \frac{34}{3}$.

FINAL ANSWER

$(a)\ y = \frac{1}{4}x - 51. \quad (b)\ f(x) = -24\sqrt{x} + \frac{x^3}{6} - 6x + \frac{34}{3}.$

WMA11/01 JANUARY 2021

Question 9

11 marks | Integration

9. (i) Find

$$\int \frac{(3x+2)^2}{4\sqrt{x}}\,\mathrm{d}x \qquad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = \mathrm{f}(x)$.

Given

- $\mathrm{f}'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3,-2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $\mathrm{f}(x)$.

(6)

(Total 11 marks)

WORKED SOLUTION

Question 9 · WMA11/01 January 2021

1 **Expand and simplify the integrand**

$\frac{(3x+2)^2}{4\sqrt{x}} = \frac{9x^2+12x+4}{4x^{1/2}} = \frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}$.

2 **Integrate term by term**

$\int \left(\frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}\right)dx = \frac{9}{10}x^{5/2} + 2x^{3/2} + 2x^{1/2} + c$.

3 **Write in simplest form**

Factoring out $\sqrt{x}$ gives $\frac{\sqrt{x}(9x^2+20x+20)}{10} + c$.

4 **Use the gradient condition at P**

Since $f'(x) = x^2 + ax + b$ and the gradient at $P(3,-2)$ is 2, $f'(3) = 2$. Hence $9 + 3a + b = 2$, so $3a + b = -7$.

5 **Integrate f'(x)**

$f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + c$. The y-intercept is -8, so $c = -8$.

6 **Use P(3,-2)**

$-2 = \frac{27}{3} + \frac{9a}{2} + 3b - 8$, so $\frac{9a}{2} + 3b = -3$.

7 **Solve for a and b**

From $3a + b = -7$, $b = -7 - 3a$. Substitute into $\frac{9a}{2} + 3b = -3$ to get $a = -4$, and then $b = 5$.

8 **Write f(x)**

Therefore $f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8$.

✓ **FINAL ANSWER**

$(i)\ \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c. \quad (ii)\ f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$

Question 6

8 marks | Integration

6. The curve C has equation $y = \mathrm{f}(x)$, $x > 0$

Given that

- C passes through the point $P(8, 2)$
- $\mathrm{f}'(x) = \dfrac{32}{3x^2} + 3 - 2\left(\sqrt[3]{x}\right)$

(a) find the equation of the tangent to C at P. Write your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

(b) Find, in simplest form, $\mathrm{f}(x)$.

(5)

(Total 8 marks)

WORKED SOLUTION

Question 6 · WMA11/01 May/June 2021

1 Find the gradient at P

$f'(x) = \dfrac{32}{3x^2} + 3 - 2\sqrt[3]{x}$. At $x = 8$, $f'(8) = \dfrac{32}{3(64)} + 3 - 2(2) = \dfrac{1}{6} - 1 = -\dfrac{5}{6}$.

2 Write the tangent equation

Using $P(8, 2)$, $y - 2 = -\dfrac{5}{6}(x - 8)$, hence $y = -\dfrac{5}{6}x + \dfrac{26}{3}$.

3 Integrate f'(x)

$f'(x) = \dfrac{32}{3}x^{-2} + 3 - 2x^{1/3}$, so $f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + c$.

4 Use P(8,2) to find c

$2 = -\dfrac{32}{24} + 24 - \dfrac{3}{2}(16) + c = -\dfrac{4}{3} + c$, so $c = \dfrac{10}{3}$.

5 Write f(x)

$f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + \dfrac{10}{3}$.

FINAL ANSWER

$(a)\ y = -\dfrac{5}{6}x + \dfrac{26}{3}$. $(b)\ f(x) = -\dfrac{32}{3x} + 3x - \dfrac{3}{2}x^{4/3} + \dfrac{10}{3}$.

WMA11/01 OCTOBER 2021

Question 10

7 marks | Integration

10. A curve has equation $y = f(x), x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a.

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

WORKED SOLUTION

Question 10 · WMA11/01 October 2021

1 Differentiate f'(x)

$f'(x) = ax - 12x^{1/3}$, so $f''(x) = a - 4x^{-2/3}$.

2 Use f''(27) = 0

Since $27^{2/3} = 9$, $0 = a - \frac{4}{9}$, so $a = \frac{4}{9}$.

3 Integrate f'(x)

With $a = \frac{4}{9}$, $f'(x) = \frac{4}{9}x - 12x^{1/3}$. Therefore $f(x) = \frac{2}{9}x^2 - 9x^{4/3} + c$.

4 Use the point (1, -8)

$-8 = \frac{2}{9} - 9 + c$, so $c = -8 + 9 - \frac{2}{9} = \frac{7}{9}$.

5 Write f(x)

$f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}$.

FINAL ANSWER

$(a)\ a = \frac{4}{9}$. $(b)\ f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}$.

WMA11/01 JANUARY 2022

Question 6

11 marks | Integration

6. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{(x+3)^2}{x\sqrt{x}}$
- the point $P(4, 20)$ lies on C

(a) (i) find the value of the gradient at P

(ii) Hence find the equation of the tangent to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $\mathrm{f}(x)$, simplifying your answer.

(7)

(Total 11 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2022

1 Simplify f'(x)

$f'(x) = \dfrac{(x+3)^2}{x\sqrt{x}} = \dfrac{x^2+6x+9}{x^{3/2}} = x^{1/2} + 6x^{-1/2} + 9x^{-3/2}$.

2 Find the gradient at P

At $x = 4$, $f'(4) = 2 + 3 + \dfrac{9}{8} = \dfrac{49}{8}$.

3 Find the tangent equation

Using $P(4, 20)$, $y - 20 = \dfrac{49}{8}(x-4)$. Multiplying out gives $49x - 8y - 36 = 0$.

4 Integrate f'(x)

$f(x) = \int \left(x^{1/2} + 6x^{-1/2} + 9x^{-3/2}\right)dx = \dfrac{2}{3}x^{3/2} + 12x^{1/2} - 18x^{-1/2} + c$.

5 Use P(4,20)

$20 = \dfrac{2}{3}(8) + 12(2) - \dfrac{18}{2} + c = \dfrac{61}{3} + c$, so $c = -\dfrac{1}{3}$.

6 Write f(x)

$f(x) = \dfrac{2}{3}x\sqrt{x} + 12\sqrt{x} - \dfrac{18}{\sqrt{x}} - \dfrac{1}{3}$.

FINAL ANSWER

$(a)(i)\ \dfrac{49}{8}.\quad (a)(ii)\ 49x-8y-36=0.\quad (b)\ f(x)=\dfrac{2}{3}x\sqrt{x}+12\sqrt{x}-\dfrac{18}{\sqrt{x}}-\dfrac{1}{3}.$

WMA11/01 MAY/JUNE 2022

Question 7

9 marks | Integration

7. The curve C has equation $y = f(x),\ x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$, where A is a constant
- $f''(x) = 0$ when $x = 4$

(a) find the value of A.

(4)

Given also that

- $f(x) = 8\sqrt{3}$, when $x = 12$

(b) find $f(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

WORKED SOLUTION

Question 7 · WMA11/01 May/June 2022

1 Differentiate f'(x)

$f'(x) = 2x^{-1/2} + Ax^{-2} + 3$, so $f''(x) = -x^{-3/2} - 2Ax^{-3}$.

2 Use f''(4) = 0

$-4^{-3/2} - 2A(4^{-3}) = 0$, so $-\frac{1}{8} - \frac{A}{32} = 0$ and $A = -4$.

3 Substitute A into f'(x)

$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3$.

4 Integrate to find f(x)

$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c$.

5 Use f(12) = 8sqrt3

$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}$, so $c = -\frac{109}{3}$.

FINAL ANSWER

$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}$.

WMA11/01 JANUARY 2023

Question 11

8 marks | Integration

11. A curve C has equation $y = f(x), \quad x > 0$

Given that

- $f''(x) = 4x + \frac{1}{\sqrt{x}}$
- the point P has x coordinate 4 and lies on C
- the tangent to C at P has equation $y = 3x + 4$

(a) find an equation of the normal to C at P

(2)

(b) find $f(x)$, writing your answer in simplest form.

(6)

(Total for Question 11 is 8 marks)

WORKED SOLUTION

Question 11 · WMA11/01 January 2023

1 Find point P

Since P has x-coordinate 4 and lies on the tangent $y = 3x + 4$, $P = (4, 16)$.

2 Find the normal

The tangent gradient is 3, so the normal gradient is $-\frac{1}{3}$. Hence the normal is $y - 16 = -\frac{1}{3}(x - 4)$.

3 Integrate f''(x)

$f''(x) = 4x + x^{-1/2}$, so $f'(x) = 2x^2 + 2\sqrt{x} + A$.

4 Use the tangent gradient

At $x = 4$, $f'(4) = 3$. Thus $2(16) + 2(2) + A = 3$, so $A = -33$.

5 Integrate again

$f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + B$.

6 Use P(4, 16)

$16 = \frac{2}{3}(64) + \frac{4}{3}(8) - 33(4) + B = \frac{160}{3} - 132 + B$.

7 Find B

$B = \frac{284}{3}$, so $f(x) = \frac{2}{3}x^3 + \frac{4}{3}x^{3/2} - 33x + \frac{284}{3}$.

FINAL ANSWER

$(a)\ y - 16 = -\dfrac{1}{3}(x - 4).\quad (b)\ f(x) = \dfrac{2}{3}x^3 + \dfrac{4}{3}x^{3/2} - 33x + \dfrac{284}{3}.$

WMA11/01 MAY/JUNE 2023

Question 8

10 marks | Integration

8. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Find the equation of the tangent to the curve with equation

$$y = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

at the point $P(4, 12)$

Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(5)

The curve with equation $y = f(x)$ also passes through the point $P(4, 12)$

Given that

$$f'(x) = \frac{1}{4}x^3 - 8x^{-\frac{1}{2}}$$

(b) find $f(x)$ giving the coefficients in simplest form.

(5)

(Total for Question 8 is 10 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2023

1 Differentiate the curve

For $y = \frac{1}{4}x^3 - 8x^{-1/2}$, $\frac{dy}{dx} = \frac{3}{4}x^2 + 4x^{-3/2}$.

2 Find the gradient at P

At $x = 4$, $\frac{dy}{dx} = \frac{3}{4}(16) + 4(4^{-3/2}) = 12 + \frac{1}{2} = \frac{25}{2}$.

3 Find the tangent

Using $P(4, 12)$, $y - 12 = \frac{25}{2}(x - 4)$.

4 Write in integer form

Multiplying by 2 gives $2y - 24 = 25x - 100$, hence $25x - 2y - 76 = 0$.

5 Integrate f'(x)

$f'(x) = \frac{1}{4}x^3 - 8x^{-1/2}$ gives $f(x) = \frac{x^4}{16} - 16\sqrt{x} + c$.

6 **Use P(4, 12)**

$12 = \dfrac{4^4}{16} - 16\sqrt{4} + c = 16 - 32 + c$, so $c = 28$.

FINAL ANSWER

$(a)\ 25x - 2y - 76 = 0.\quad (b)\ f(x) = \dfrac{x^4}{16} - 16\sqrt{x} + 28.$

WMA11/01 OCTOBER 2023

Question 7

10 marks | Integration

7. The curve C has equation $y = f(x)$ where $x > 0$

Given that

- $f'(x) = \dfrac{4x^2 + 10 - 7x^{\frac{1}{2}}}{4x^{\frac{1}{2}}}$
- the point $P(4, -1)$ lies on C

(a) (i) find the value of the gradient of C at P

(ii) Hence find the equation of the normal to C at P, giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

(b) Find $f(x)$.

(6)

(Total for Question 7 is 10 marks)

WORKED SOLUTION

Question 7 · WMA11/01 October 2023

1 Simplify f prime

$f'(x) = \dfrac{4x^2 + 10 - 7x^{1/2}}{4x^{1/2}} = x^{3/2} + \dfrac{5}{2}x^{-1/2} - \dfrac{7}{4}$.

2 Find the gradient at P

At $x = 4$, $f'(4) = 4^{3/2} + \dfrac{5}{2}(4^{-1/2}) - \dfrac{7}{4} = 8 + \dfrac{5}{4} - \dfrac{7}{4} = \dfrac{15}{2}$.

3 Find the normal gradient

The normal gradient is the negative reciprocal, so $m_n = -\dfrac{2}{15}$.

4 Use point P

Using $P(4, -1)$, $y + 1 = -\dfrac{2}{15}(x - 4)$.

5 Write the normal in integer form

$15y + 15 = -2x + 8$, hence $2x + 15y + 7 = 0$.

6 Integrate f prime

$f(x) = \int \left(x^{3/2} + \dfrac{5}{2}x^{-1/2} - \dfrac{7}{4}\right) dx = \dfrac{2}{5}x^{5/2} + 5x^{1/2} - \dfrac{7}{4}x + c$.

7 Use P to find c

$-1 = \dfrac{2}{5}(4^{5/2}) + 5(4^{1/2}) - \dfrac{7}{4}(4) + c = \dfrac{64}{5} + 10 - 7 + c$, so $c = -\dfrac{84}{5}$.

FINAL ANSWER

$(a)(i)\ \dfrac{15}{2}.\quad (a)(ii)\ 2x+15y+7=0.\quad (b)\ f(x)=\dfrac{2}{5}x^{5/2}+5x^{1/2}-\dfrac{7}{4}x-\dfrac{84}{5}.$

WMA11/01 JANUARY 2024

Question 10

8 marks | Integration

10. **In this question you must show all stages of your working.**

The curve C has equation $y = f(x), x > 0$

Given that

- the point $P\left(2, 8\sqrt{2}\right)$ lies on C
- $f'(x) = 4\sqrt{x^3} + \frac{k}{x^2}$ where k is a constant
- $f''(x) = 0$ at P

(a) find the exact value of k,

(4)

(b) find $f(x)$, giving your answer in simplest form.

(4)

First released on AP - Edexcel Discord
https://sites.google.com/view/ap-edexcel

(Total for Question 10 is 8 marks)

WORKED SOLUTION

Question 10 · WMA11/01 January 2024

1 Differentiate f prime

$f'(x) = 4x^{3/2} + kx^{-2}$, so $f''(x) = 6x^{1/2} - 2kx^{-3}$.

2 Use f''(P) = 0

At P, $x = 2$ and $f''(2) = 0$, so $6\sqrt{2} - 2k(2^{-3}) = 0$.

3 Find k

$6\sqrt{2} - \frac{k}{4} = 0$, hence $k = 24\sqrt{2}$.

4 Integrate f prime

$f(x) = \int \left(4x^{3/2} + 24\sqrt{2}x^{-2}\right)dx = \frac{8}{5}x^{5/2} - 24\sqrt{2}x^{-1} + c$.

5 Use the point P

$P(2, 8\sqrt{2})$ lies on C, so $8\sqrt{2} = \frac{8}{5}(2^{5/2}) - \frac{24\sqrt{2}}{2} + c$.

6 Find c

$8\sqrt{2} = \frac{32\sqrt{2}}{5} - 12\sqrt{2} + c$, so $c = \frac{68\sqrt{2}}{5}$.

FINAL ANSWER

$(a)\ k = 24\sqrt{2}.\quad (b)\ f(x) = \frac{8}{5}x^{5/2} - \frac{24\sqrt{2}}{x} + \frac{68\sqrt{2}}{5}.$

WMA11/01 MAY/JUNE 2024

Question 10

10 marks | Integration

10. The curve C has equation $y = \mathrm{f}(x)$ where $x > 0$

Given that

- $\mathrm{f}'(x) = 6x - \dfrac{(2x-1)(3x+2)}{2\sqrt{x}}$
- the point $P(4, 12)$ lies on C

(a) find the equation of the normal to C at P, giving your answer in the form $y = mx + c$ where m and c are integers to be found,

(4)

(b) find $\mathrm{f}(x)$, giving each term in simplest form.

(6)

(Total for Question 10 is 10 marks)

WORKED SOLUTION

Question 10 · WMA11/01 May/June 2024

1 Expand the numerator

$(2x-1)(3x+2) = 6x^2 + x - 2$.

2 Simplify f prime

$f'(x) = 6x - \dfrac{6x^2 + x - 2}{2\sqrt{x}} = 6x - 3x^{3/2} - \dfrac{1}{2}x^{1/2} + x^{-1/2}$.

3 Find the tangent gradient at P

$f'(4) = 24 - 3(8) - \dfrac{1}{2}(2) + \dfrac{1}{2} = -\dfrac{1}{2}$.

4 Use the normal gradient

The normal gradient is the negative reciprocal of $-\dfrac{1}{2}$, so it is 2.

5 Find the normal equation

Through $P(4, 12)$, $y - 12 = 2(x - 4)$, hence $y = 2x + 4$.

6 Integrate f prime

$f(x) = 3x^2 - \dfrac{6}{5}x^{5/2} - \dfrac{1}{3}x^{3/2} + 2x^{1/2} + c$.

7 Use P to find c

$12 = 3(4)^2 - \dfrac{6}{5}(4^{5/2}) - \dfrac{1}{3}(4^{3/2}) + 2(4^{1/2}) + c$.

8 Simplify c

$12 = 48 - \dfrac{192}{5} - \dfrac{8}{3} + 4 + c = \dfrac{164}{15} + c$, so $c = \dfrac{16}{15}$.

FINAL ANSWER

$(a)\ y = 2x + 4. \quad (b)\ f(x) = 3x^2 - \frac{6}{5}x^{5/2} - \frac{1}{3}x^{3/2} + 2x^{1/2} + \frac{16}{15}.$

WMA11/01 OCTOBER 2024

Question 8

7 marks | Integration

8. A curve C has equation $y = \mathrm{f}(x)$.

The point P with x coordinate 3 lies on C

Given

- $\mathrm{f}'(x) = 4x^2 + kx + 3$ where k is a constant
- the normal to C at P has equation $y = -\frac{1}{24}x + 5$

(a) show that $k = -5$

(3)

(b) Hence find $\mathrm{f}(x)$.

(4)

(Total for Question 8 is 7 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2024

1 Find the y-coordinate of P

The normal is $y = -\frac{1}{24}x + 5$. At $x = 3$, $y = -\frac{3}{24} + 5 = \frac{39}{8}$.

2 Use the normal gradient

The normal gradient is $-\frac{1}{24}$, so the tangent gradient is 24.

3 Use f prime at x equals 3

$f'(3) = 4(3)^2 + 3k + 3 = 39 + 3k$.

4 Find k

$39 + 3k = 24$, so $k = -5$.

5 Integrate f prime

With $k = -5$, $f'(x) = 4x^2 - 5x + 3$, so $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x + c$.

6 Use point P

$\frac{39}{8} = \frac{4}{3}(27) - \frac{5}{2}(9) + 3(3) + c = \frac{45}{2} + c$.

7 Find c

$c = \frac{39}{8} - \frac{45}{2} = -\frac{141}{8}$.

FINAL ANSWER

$k = -5$ and $f(x) = \frac{4}{3}x^3 - \frac{5}{2}x^2 + 3x - \frac{141}{8}$.

WMA11/01 JANUARY 2025

Question 6

10 marks | Integration

6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation $y = \mathrm{f}(x),\ x > 0$

Given that

- the point $P(4, -5)$ lies on C
- $\mathrm{f}'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, where a and b are constants
- the gradient of the tangent to C at P is 7

(a) show that

$$4a + b = 24$$

(2)

Given also that $a + b = -9$

(b) find, in simplest form, $\mathrm{f}(x)$

(7)

Curve C is transformed to the curve with equation $y = \mathrm{f}(x - 3)$

Given that point P is transformed to the point Q,

(c) state the coordinates of Q.

(1)

(Total for Question 6 is 10 marks)

WORKED SOLUTION

Question 6 · WMA11/01 January 2025

1 **Use the gradient at P**

At $x = 4$, $f'(4) = 7$. Since $f'(x) = \dfrac{2x^2 + ax + b}{4\sqrt{x}}$, we get $\dfrac{32 + 4a + b}{8} = 7$.

2 **Show the required equation**

$32 + 4a + b = 56$, so $4a + b = 24$.

3 **Solve for a and b**

Together with $a + b = -9$, subtracting gives $3a = 33$, so $a = 11$ and $b = -20$.

4 **Rewrite f prime**

$$f'(x) = \frac{2x^2 + 11x - 20}{4\sqrt{x}} = \frac{1}{2}x^{3/2} + \frac{11}{4}x^{1/2} - 5x^{-1/2}.$$

5 **Integrate**

$$f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} + c.$$

6 **Use P to find c**

$$-5 = \frac{1}{5}(4^{5/2}) + \frac{11}{6}(4^{3/2}) - 10(4^{1/2}) + c = \frac{16}{15} + c.$$

7 **State f(x)**

$c = -\frac{91}{15}$, so $f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}$.

8 **Transform P**

$y = f(x - 3)$ translates the curve 3 units to the right, so $P(4, -5)$ becomes $Q(7, -5)$.

✓ **FINAL ANSWER**

$(b)\ f(x) = \frac{1}{5}x^{5/2} + \frac{11}{6}x^{3/2} - 10x^{1/2} - \frac{91}{15}.$ $(c)\ Q = (7, -5).$

Question 8

11 marks | Integration

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

A curve has equation $y = f(x),\ x > 0$

The point $P(4, 12)$ lies on the curve.

Given that

- $f'(x) = 3\sqrt{x} + kx^2$ where k is a constant
- the equation of the tangent to the curve at P has equation $y = 10x + c$ where c is a constant

(a) (i) show that $k = \frac{1}{4}$

(ii) find the value of c

(4)

(b) Hence find the value of $f''(x)$ at P.

(3)

(c) Find $f(x)$.

(4)

(Total for Question 8 is 11 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2025

1 Use the tangent gradient

The tangent at P has equation $y = 10x + c$, so the gradient at P is 10.

2 Show k

$f'(4) = 3\sqrt{4} + k(4)^2 = 6 + 16k = 10$, so $16k = 4$ and $k = \frac{1}{4}$.

3 Find c

Substitute $P(4, 12)$ into $y = 10x + c$: $12 = 40 + c$, so $c = -28$.

4 Differentiate f prime

With $k = \frac{1}{4}$, $f'(x) = 3x^{1/2} + \frac{1}{4}x^2$.

5 Find f double prime

$f''(x) = \frac{3}{2}x^{-1/2} + \frac{1}{2}x$.

6 **Evaluate at P**

$f''(4) = \frac{3}{2} \cdot \frac{1}{2} + 2 = \frac{3}{4} + 2 = \frac{11}{4}.$

7 **Integrate f prime**

$f(x) = \int \left(3x^{1/2} + \frac{1}{4}x^2\right) dx = 2x^{3/2} + \frac{1}{12}x^3 + d.$

8 **Use P to find d**

$12 = 2(4^{3/2}) + \frac{1}{12}(4^3) + d = 16 + \frac{16}{3} + d.$

9 **State f(x)**

$d = -\frac{28}{3}$, so $f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$

✓ **FINAL ANSWER**

$(a)(i)\ k = \frac{1}{4}.$ $(a)(ii)\ c = -28.$ $(b)\ f''(P) = \frac{11}{4}.$ $(c)\ f(x) = 2x^{3/2} + \frac{1}{12}x^3 - \frac{28}{3}.$

WMA11/01 OCTOBER 2025

Question 8

8 marks | Integration

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve has equation $y = f(x), x > 0$

Given that

- $f'(x) = 2x + \frac{8}{x^2} + k$, where k is a constant
- the equation of the tangent to the curve at $x = \sqrt{2}$ is $y = 5x - 3\sqrt{2}$

(a) find the exact value of k.

(2)

(b) Find an equation of the normal to the curve at $x = \sqrt{2}$

(2)

(c) Find $f(x)$, writing your answer in simplest form.

(4)

(Total for Question 8 is 8 marks)

WORKED SOLUTION

Question 8 · WMA11/01 October 2025

1 Use tangent gradient

The tangent has equation $y = 5x - 3\sqrt{2}$, so the gradient at $x = \sqrt{2}$ is 5.

2 Find k

$f'(\sqrt{2}) = 2\sqrt{2} + \frac{8}{(\sqrt{2})^2} + k = 2\sqrt{2} + 4 + k = 5$, so $k = 1 - 2\sqrt{2}$.

3 Find the point on the curve

At $x = \sqrt{2}$, the tangent gives $y = 5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$.

4 Normal gradient

The tangent gradient is 5, so the normal gradient is $-\frac{1}{5}$.

5 Equation of the normal

$y - 2\sqrt{2} = -\frac{1}{5}(x - \sqrt{2})$, hence $y = -\frac{1}{5}x + \frac{11\sqrt{2}}{5}$.

6 Integrate f prime

$f(x) = \int \left(2x + \frac{8}{x^2} + k\right) dx = x^2 - \frac{8}{x} + kx + c$

7 **Substitute k**

$f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + c$.

8 **Use the point**

$2\sqrt{2} = 2 - \dfrac{8}{\sqrt{2}} + (1 - 2\sqrt{2})\sqrt{2} + c = -2 - 3\sqrt{2} + c$.

9 **Find c**

$c = 2 + 5\sqrt{2}$, so $f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}$.

✓ **FINAL ANSWER**

$(a)\ k = 1 - 2\sqrt{2}.$ $(b)\ y = -\dfrac{1}{5}x + \dfrac{11\sqrt{2}}{5}.$ $(c)\ f(x) = x^2 - \dfrac{8}{x} + (1 - 2\sqrt{2})x + 2 + 5\sqrt{2}$.

Question 8

12 marks | Integration

8. **In this question you must show all stages of your working.**
Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x), \quad x > 0$

A point P lies on C.

Given that

- $f'(x) = 4x^2 + \frac{6}{x^2} - 4$
- the equation of the tangent to C at P is $y = 10x - 6\sqrt{3}$

(a) (i) verify that $\sqrt{3}$ is a possible x coordinate of P,

(ii) find, using algebra, the other possible x coordinate of P.

(6)

Given that the x coordinate of P is $\sqrt{3}$

(b) find an equation of the normal to C at P.

(2)

(c) Find $f(x)$, writing your answer in simplest form.
You must show each stage of your working.

(4)

(Total for Question 8 is 12 marks)

WORKED SOLUTION

Question 8 · WMA11/01 January 2026

1 **Verify x equals root 3**

$f'(\sqrt{3}) = 4(\sqrt{3})^2 + \frac{6}{(\sqrt{3})^2} - 4 = 12 + 2 - 4 = 10$, which matches the tangent gradient.

2 **Find all possible x-coordinates**

Set $4x^2 + \frac{6}{x^2} - 4 = 10$.

3 **Form a quadratic in x squared**

Multiplying by x^2 gives $4x^4 - 14x^2 + 6 = 0$, so $2x^4 - 7x^2 + 3 = 0$.

4 **Factor**

$2x^4 - 7x^2 + 3 = (2x^2 - 1)(x^2 - 3) = 0$.

5 **Use x greater than zero**

$x = \sqrt{3}$ or $x = \dfrac{1}{\sqrt{2}} = \dfrac{\sqrt{2}}{2}$. The other possible x-coordinate is $\dfrac{\sqrt{2}}{2}$.

6 **Find the point P**

Given $x_P = \sqrt{3}$, the tangent gives $y_P = 10\sqrt{3} - 6\sqrt{3} = 4\sqrt{3}$.

7 **Normal equation**

The normal gradient is $-\dfrac{1}{10}$, so $y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3})$.

8 **Integrate f prime**

$f(x) = \displaystyle\int \left(4x^2 + \frac{6}{x^2} - 4\right) dx = \frac{4}{3}x^3 - \frac{6}{x} - 4x + c.$

9 **Use P to find c**

$4\sqrt{3} = \dfrac{4}{3}(\sqrt{3})^3 - \dfrac{6}{\sqrt{3}} - 4\sqrt{3} + c = 4\sqrt{3} - 2\sqrt{3} - 4\sqrt{3} + c.$

10 **State f(x)**

$c = 6\sqrt{3}$, so $f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}$.

✓ **FINAL ANSWER**

$(a)(i)$ verified.

$(a)(ii)\ x = \dfrac{\sqrt{2}}{2}.$ $\quad (b)\ y - 4\sqrt{3} = -\dfrac{1}{10}(x - \sqrt{3}).$ $\quad (c)\ f(x) = \dfrac{4}{3}x^3 - \dfrac{6}{x} - 4x + 6\sqrt{3}.$

WMA11/01 MAY/JUNE 2026

Question 8

11 marks | Integration

8. A curve C has equation $y = \mathrm{f}(x),\ x > 0$

The point P with x coordinate $\frac{1}{4}$ lies on C.

Given that

- $\mathrm{f}'(x) = 10 + \dfrac{5\sqrt{x} - 4}{2x^2}$
- the tangent to C at P has equation $y = kx + 2$, where k is a constant

(a) (i) find the value of k

(ii) find the equation of the normal to C at P.
Give the answer in the form $y = mx + c$, where m and c are constants.

(6)

(b) Find $\mathrm{f}(x)$.

(5)

(Total for Question 8 is 11 marks)

WORKED SOLUTION

Question 8 · WMA11/01 May/June 2026

1 **(a)(i) Evaluate the tangent gradient**

The tangent gradient at P is $f'\left(\frac{1}{4}\right)$. Using

$$f'(x) = 10 + \frac{5\sqrt{x} - 4}{2x^2},$$

we get

$$f'\left(\frac{1}{4}\right) = 10 + \frac{5\left(\frac{1}{2}\right) - 4}{2\left(\frac{1}{4}\right)^2} = 10 + \frac{-\frac{3}{2}}{\frac{1}{8}} = -2.$$

Therefore $k = -2$.

2 **(a)(ii) Find the point of contact**

The tangent is $y = -2x + 2$. At $x = \frac{1}{4}$,

$$y = -2\left(\frac{1}{4}\right) + 2 = \frac{3}{2},$$

so $P = \left(\frac{1}{4}, \frac{3}{2}\right)$.

3 **(a)(ii) Form the normal**

The tangent gradient is -2, so the normal gradient is $\frac{1}{2}$. Through P,

$$y-\frac{3}{2}=\frac{1}{2}\left(x-\frac{1}{4}\right).$$

Hence

$$y=\frac{1}{2}x+\frac{11}{8}.$$

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4 **(b) Rewrite and integrate**

First split the fractional term:

$$f'(x)=10+\frac{5}{2}x^{-3/2}-2x^{-2}.$$

Integrating term by term gives

$$f(x)=10x-5x^{-1/2}+2x^{-1}+c=10x-\frac{5}{\sqrt{x}}+\frac{2}{x}+c.$$

---

5 **(b) Use the point P to find the constant**

Since $P=\left(\frac{1}{4},\frac{3}{2}\right)$ lies on the curve,

$$\frac{3}{2}=10\left(\frac{1}{4}\right)-5\left(\frac{1}{4}\right)^{-1/2}+2\left(\frac{1}{4}\right)^{-1}+c.$$

Thus

$$\frac{3}{2}=\frac{5}{2}-10+8+c=\frac{1}{2}+c,$$

so $c=1$.

FINAL ANSWER

$(a)(i)\ k=-2.$ $(a)(ii)\ y=\frac{1}{2}x+\frac{11}{8}.$ $(b)\ f(x)=10x-\frac{5}{\sqrt{x}}+\frac{2}{x}+1.$